

Solution Manual for Introduction to Ordinary Differential Equations with Scientific Applications

First Edition

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Solutions for Chapter 1

(1.1)

- (a) Yes, Linear;
- (b) Yes, Separable;
- (c) No (Second-order);
- (d) Yes, Neither;
- (e) Yes, Linear;
- (f) Yes, Neither;
- (g) Yes, Neither;
- (h) Yes, Both.

(1.2)

$$\begin{aligned}\frac{dy}{dx} &= e^{-x}y^3 \\ \int \frac{dy}{y^3} &= \int e^{-x} dx \\ -\frac{1}{2y^2} &= -e^{-x} + C_1 \\ y &= \pm \frac{1}{\sqrt{2e^{-x} + C_2}}\end{aligned}$$

where $C_2 = -2C_1$. We take the positive solution and plug in the I.C. to get $C_2 = -1$ and

$$y = \frac{1}{\sqrt{2e^{-x} - 1}}$$

(1.3)

$$(\csc x) \frac{dy}{dx} = y + 1$$

I

$$\begin{aligned}\int \frac{dy}{y+1} &= \int \sin x dx \\ \ln |y+1| &= -\cos x + C \\ y &= \pm e^{C-\cos x} - 1 = Ae^{-\cos x} - 1\end{aligned}$$

where $A = \pm e^C$. Substituting the I.C. yields $A = 3e$, and therefore the final answer is

$$y = 3e^{1-\cos x} - 1$$

(1.4) The integrating factor is $\exp(\int 2x dx) = e^{x^2}$.

$$\begin{aligned}\frac{dy}{dx} + 2xy &= e^{-x^2} \\ e^{x^2} \left(\frac{dy}{dx} + 2xy \right) &= e^{x^2} e^{-x^2} \\ \frac{d}{dx} (e^{x^2} y) &= 1 \\ e^{x^2} y &= x + C \\ y &= xe^{-x^2} + Ce^{-x^2}\end{aligned}$$

The I.C. requires $C = 1$, hence $y = (x+1)e^{-x^2}$.

(1.5) The integrating factor is $\exp(\int -\frac{1}{2x} dx) = \frac{1}{\sqrt{x}}$.

$$\begin{aligned}\frac{dy}{dx} - \frac{y}{2x} &= 1 \\ \frac{d}{dx} \left(\frac{y}{\sqrt{x}} \right) &= \left(\frac{1}{\sqrt{x}} \frac{dy}{dx} - \frac{y}{2x^{\frac{3}{2}}} \right) = \frac{1}{\sqrt{x}} \\ \frac{y}{\sqrt{x}} &= \int \frac{1}{\sqrt{x}} dx = 2\sqrt{x} + C \\ y &= 2x + C\sqrt{x}\end{aligned}$$

The I.C. requires $C = 1$, hence $y = 2x + \sqrt{x}$.

(1.6) Exactness follows because

$$\frac{\partial}{\partial y} (2x \tan y + 1) = 2x \sec^2 y = \frac{\partial}{\partial x} (x^2 \sec^2 y)$$

Then, we may integrate both parts with respect to x and y to get

$$\int (2x \tan y + 1) dx = x^2 \tan y + x + g(y)$$

$$\int x^2 \sec^2 y \, dy = x^2 \tan y + h(x)$$

Hence, the potential function will be $x^2 \tan y + x = C$. The I.C. indicates that $C = \tan 1 + 1$.

(1.7) Checking exactness:

$$\frac{\partial(y(e^{xy} + y))}{\partial y} = e^{xy} + xy e^{xy} + 2y = \frac{\partial(x(e^{xy} + 2y))}{\partial x}$$

Integration yields

$$\begin{aligned} \int (ye^{xy} + y^2) dx &= e^{xy} + xy^2 + g(y) \\ \int (xe^{xy} + 2xy) dy &= e^{xy} + xy^2 + h(x) \end{aligned}$$

So we can choose the general solution to be

$$e^{xy} + xy^2 = C$$

With the I.C. $y(0) = 1$, $C = 1$ and the desired solution will be $e^{xy} + xy^2 = 1$.

(1.8) We leave it to the readers to check that this ODE is non-exact. Now, notice that

$$\begin{aligned} \frac{\frac{\partial}{\partial x}(2y) - \frac{\partial}{\partial y}(1 - y^2 \tan x)}{-2y} &= \frac{0 - (-2y \tan x)}{-2y} \\ &= -\tan x \end{aligned}$$

which is a function of x exactly. Then, the integrating factor is

$$\begin{aligned} \exp\left(\int -\tan x \, dx\right) &= \exp\left(\int -\frac{\sin x}{\cos x} \, dx\right) \\ &= \exp\left(\int \frac{1}{\cos x} d(\cos x)\right) \\ &= \exp(\ln |\cos x|) = |\cos x| \end{aligned}$$

Upon multiplying this integrating factor (can ignore the absolute value), the ODE becomes exact:

$$(\cos x - y^2 \sin x) dx + 2y \cos x \, dy = 0$$

$$\frac{\partial}{\partial y}(\cos x - y^2 \sin x) = -2y \sin x = \frac{\partial}{\partial x}(2y \cos x)$$

Integration gives

$$\int (\cos x - y^2 \sin x) dx = \sin x + y^2 \cos x + g(y)$$

I

$$\int 2y \cos x \, dy = y^2 \cos x + h(x)$$

So our general solution will be

$$\sin x + y^2 \cos x = C$$

(1.9) Dividing both sides by xy leads to

$$\begin{aligned} \frac{ydx - xdy}{xy} &= x^2 dx \\ d\left(\ln\left(\frac{x}{y}\right)\right) &= d\left(\frac{1}{3}x^3\right) \\ \ln\left(\frac{x}{y}\right) &= \frac{1}{3}x^3 + C \\ \frac{x}{y} &= e^{\frac{1}{3}x^3 + C} \\ y &= xe^{-\left(\frac{1}{3}x^3 + C\right)} = Axe^{-\frac{x^3}{3}} \end{aligned}$$

where we set $A = e^{-C}$. The I.C. $y(1) = 2$ leads to $A = 2e^{\frac{1}{3}}$.

(1.10) The integrating factor is $\exp\left(\int \frac{2}{x} dx\right) = x^2$, and the equation is then solved as

$$\begin{aligned} \frac{dy}{dx} + \frac{2y}{x} &= e^x \\ \frac{d}{dx}(x^2 y) &= x^2 \frac{dy}{dx} + 2xy = x^2 e^x \\ x^2 y &= \int x^2 e^x dx \end{aligned}$$

The integral at the end can be computed by I.B.P. twice, yielding $\int x^2 e^x dx = x^2 e^x - 2xe^x + 2e^x + C$, and so the general solution is

$$y = e^x - \frac{2e^x}{x} + \frac{2e^x}{x^2} + \frac{C}{x^2}$$

The I.C. $y(1) = 1$ requires that $C = 1 - e$ and hence

$$y = e^x - \frac{2e^x}{x} + \frac{2e^x - e + 1}{x^2}$$

(1.11) The integrating factor is $\exp\left(\int \frac{2x}{x^2+1} dx\right) = \exp(\ln(x^2 + 1)) = x^2 + 1$. Hence

$$\begin{aligned} \frac{dy}{dx} + \frac{2x}{x^2+1} y &= \cos(2x) \\ \frac{d}{dx}((x^2+1)y) &= (x^2+1) \frac{dy}{dx} + 2xy = (x^2+1) \cos(2x) \end{aligned}$$

$$(x^2 + 1)y = \int (x^2 + 1) \cos(2x) dx$$

The integral at the end can again be computed by I.B.P. twice. This gives

$$\int (x^2 + 1) \cos(2x) dx = \left(\frac{x^2}{2} + \frac{1}{4}\right) \sin(2x) + \frac{x}{2} \cos(2x) + C$$

Therefore

$$\begin{aligned} (x^2 + 1)y &= \left(\frac{x^2}{2} + \frac{1}{4}\right) \sin(2x) + \frac{x}{2} \cos(2x) + C \\ y &= \frac{1}{x^2 + 1} \left(\frac{x^2}{2} + \frac{1}{4}\right) \sin(2x) + \frac{x}{2(x^2 + 1)} \cos(2x) + \frac{C}{x^2 + 1} \end{aligned}$$

The I.C. $y(0) = 1$ immediately leads to $C = 1$.

(1.12) Rewrite

$$\begin{aligned} \frac{dy}{dx} &= \frac{1 + y^2 \sin(2x)}{2y \cos^2(x)} \\ (1 + y^2 \sin(2x)) dx - 2y \cos^2 x dy &= 0 \end{aligned}$$

This is an exact ODE as

$$\frac{\partial(1 + y^2 \sin(2x))}{\partial y} = 2y \sin(2x) = \frac{\partial(-2y \cos^2 x)}{\partial x}$$

Then

$$\begin{aligned} \int M dx &= \int (1 + y^2 \sin(2x)) dx = x - \frac{1}{2} y^2 \cos(2x) + g(y) \\ \int N dy &= \int -2y \cos^2 x dy = -y^2 \cos^2(x) + h(x) = -\frac{1}{2} y^2 (1 + \cos(2x)) + h(x) \\ &= -\frac{1}{2} y^2 \cos(2x) - \frac{1}{2} y^2 + h(x) \end{aligned}$$

So the potential function can be

$$\begin{aligned} x - \frac{1}{2} y^2 - \frac{1}{2} y^2 \cos(2x) &= C \\ x - y^2 \cos^2 x &= C \end{aligned}$$

With $y(0) = 1$, $C = -1$, and hence the required solution is $x - y^2 \cos^2 x = -1$.

(1.13) The required ODE is

$$\frac{dM}{dt} = 5 - 0.025M$$

a first-order linear ODE. The solution steps are

$$\begin{aligned}\frac{dM}{dt} + 0.025M &= 5 \\ \frac{d}{dt}(e^{0.025t}M) &= e^{0.025t}\left(\frac{dM}{dt} + 0.025M\right) = 5e^{0.025t} \\ e^{0.025t}M &= \int 5e^{0.025t} dt = 200e^{0.025t} + C \\ M &= 200 + Ce^{-0.025t}\end{aligned}$$

where we choose the integrating factor $\mu = \exp(\int 0.025 dt) = e^{0.025t}$. Plugging in the initial conditions $t = 0$, $M = 100$, we have

$$\begin{aligned}100 &= 200 + Ce^{-0.025(0)} = 200 + C \\ C &= -100\end{aligned}$$

So the answer is

$$M = 200 - 100e^{-0.025t} = 100 + 100(1 - e^{-0.025t})$$

The equilibrium mass as $t \rightarrow \infty$ is 200 kg, with the rate of inflow (5 kg/min) balancing that of outflow ($0.025(200) = 5$ kg/min also).

(1.14) The change in the volume of liquid is described by

$$\frac{dM_l}{dt} = 2 - 0.01M_l$$

Similar to the above exercise, the corresponding solution is

$$M_l = 200 - 100e^{-0.01t}$$

On the other hand, the amount of salt is governed by

$$\begin{aligned}\frac{dM_s}{dt} &= 2(0.08) - (0.01M_l)\frac{M_s}{M_l} \\ &= 0.16 - 0.01M_s\end{aligned}$$

and the corresponding solution is

$$M_s = 16 + Ce^{-0.01t}$$

The I.C. would be $M_s(0) = 100(0.035) = 3.5$, so $C = -12.5$, $M_s = 16 - 12.5e^{-0.01t}$, and so the salinity over time will be

$$s = \frac{16 - 12.5e^{-0.01t}}{200 - 100e^{-0.01t}}$$

(1.15) The change in the volume of the balloon is simply $V = 2 - 0.005t$, and hence the radius will be given by

$$\frac{4}{3}\pi r^3 = 2 - 0.005t$$

$$r = \sqrt[3]{\frac{2 - 0.005t}{\frac{4}{3}\pi}}, \quad t \leq 400$$

(1.16) First, differentiate and eliminate to get

$$\frac{dy}{dx} = C\alpha e^{\alpha x} = \alpha y$$

Then, replacing $\frac{dy}{dx}$ by $-\frac{dx}{dy}$ and solve:

$$-\frac{dx}{dy} = \alpha y$$

$$-\int dx = \int \alpha y dy$$

$$A - x = \frac{\alpha}{2}y^2$$

(1.17) Differentiating the given equation against x gives

$$2x = 4C \frac{dy}{dx}$$

The trick is to write

$$4x^2 = 16C^2 \left(\frac{dy}{dx}\right)^2$$

$$x^2 \frac{dx}{dy} = 4C^2 \frac{dy}{dx}$$

by squaring. Then we can eliminate C from the ODE via using the equation at the beginning:

$$x^2 \frac{dx}{dy} + 2xy = 4C^2 \frac{dy}{dx} + 4Cy \frac{dy}{dx} = 4C(C + y) \frac{dy}{dx}$$

$$x^2 \frac{dx}{dy} + 2xy = x^2 \frac{dy}{dx}$$

Note that replacing $\frac{dy}{dx}$ by $-\frac{dx}{dy}$ yields

$$x^2 \left(-\frac{dy}{dx}\right) + 2xy = x^2 \left(-\frac{dx}{dy}\right)$$

I

$$x^2 \frac{dx}{dy} + 2xy = x^2 \frac{dy}{dx}$$

which is unaltered compared to the ODE above. Therefore, the orthogonal trajectories of parabolas with foci at the origin are exactly themselves.

(1.18) Notice that the slope in polar coordinates is given by

$$\begin{aligned} \frac{dy}{dx} &= \frac{d(r \sin \theta)}{d(r \cos \theta)} \\ &= \frac{\frac{dr}{d\theta} \sin \theta + r \cos \theta}{\frac{dr}{d\theta} \cos \theta - r \sin \theta} = \frac{\frac{1}{r} \frac{dr}{d\theta} \sin \theta + \cos \theta}{\frac{1}{r} \frac{dr}{d\theta} \cos \theta - \sin \theta} \end{aligned}$$

The slope of the orthogonal trajectories then should be

$$-\frac{\frac{dr}{d\theta} \cos \theta - r \sin \theta}{\frac{dr}{d\theta} \sin \theta + r \cos \theta} = \frac{r \sin \theta - \frac{dr}{d\theta} \cos \theta}{r \cos \theta + \frac{dr}{d\theta} \sin \theta} = \frac{r \frac{d\theta}{dr} \sin \theta - \cos \theta}{r \frac{d\theta}{dr} \cos \theta + \sin \theta}$$

If we replace $\frac{1}{r} \frac{dr}{d\theta} \rightarrow -r \frac{d\theta}{dr}$, then the first expression will become the second expression as stated.

(1.19) Combining the two given equations:

$$\begin{aligned} \frac{dp}{dz} &= -\rho g = -\frac{p}{R_d T_0} g \\ \int_{p_0}^p \frac{dp}{p} &= -\int_0^z \frac{g}{R_d T_0} dz \\ [\ln |p|]_{p_0}^p &= \ln\left(\frac{p}{p_0}\right) = -\frac{g}{R_d T_0} z \\ p &= p_0 e^{-\frac{g}{R_d T_0} z} \end{aligned}$$

where we have included the initial condition $p(z = 0) = p_0$ at the sea-level surface in the lower limit of integration.

(1.20) The modified hydrostatic equation is

$$\begin{aligned} \frac{dp}{dz} &= -\frac{g}{R_d(T_0 - \Gamma z)} p \\ \frac{dp}{p} &= -\frac{g}{R_d(T_0 - \Gamma z)} dz \end{aligned}$$

Integrating from the surface to some height again, we have

$$\int_{p_0}^p \frac{dp}{p} = -\int_0^z \frac{g}{R_d(T_0 - \Gamma z)} dz$$

$$\begin{aligned}
\ln \left| \frac{p}{p_0} \right| &= \frac{g}{R_d \Gamma} [\ln |T_0 - \Gamma z|]_0^z \\
\ln \left(\frac{p}{p_0} \right) &= \frac{g}{R_d \Gamma} \ln \left(\frac{T_0 - \Gamma z}{T_0} \right) \\
\ln \left(\frac{p}{p_0} \right) &= \ln \left\{ \left(\frac{T_0 - \Gamma z}{T_0} \right)^{\frac{g}{R_d \Gamma}} \right\} \\
p &= p_0 \left(\frac{T_0 - \Gamma z}{T_0} \right)^{\frac{g}{R_d \Gamma}}
\end{aligned}$$

Plugging in all the given values (convert the original units to SI units properly!), the pressure decreases to 30% of the surface value when $z \approx 9.2$ km now, which is lower when compared to $z \approx 10.3$ km for an isothermal atmosphere, because the decreasing temperature going up implies thinner layers.

(1.21)

$$\begin{aligned}
\frac{d[BrO]}{dt} &= -2k[BrO]^2 \\
\int \frac{d[BrO]}{[BrO]^2} &= - \int 2k dt \\
-\frac{1}{[BrO]} &= -2kt + C \\
[BrO] &= \frac{1}{2kt - C}
\end{aligned}$$

Using the initial condition $[BrO] = 1.5 \times 10^{11}$ molecule/cm³ at $t = 0$, we have

$$\begin{aligned}
1.5 \times 10^{11} &= \frac{1}{2k(0) - C} = -\frac{1}{C} \\
C &= -\frac{1}{1.5 \times 10^{11}} = -6.667 \times 10^{-12}
\end{aligned}$$

When the concentration falls below 1% of the initial value, $[BrO] = 1.5 \times 10^9$ molecule/cm³, and

$$\begin{aligned}
1.5 \times 10^9 &= \frac{1}{2(3.0 \times 10^{-12})t + 6.667 \times 10^{-12}} \\
t &= 110.00 \text{ s}
\end{aligned}$$

(1.22) The second case may be solved by partial fractions.

$$\begin{aligned}
\frac{dv}{dt} &= g - \beta v^2 \\
\int \frac{dv}{1 - \frac{\beta}{g}v^2} &= \int g dt
\end{aligned}$$

I

$$\begin{aligned}
\frac{1}{2} \int \left(\frac{1}{1 - \sqrt{\frac{\beta}{g}} v} + \frac{1}{1 + \sqrt{\frac{\beta}{g}} v} \right) dv &= \int g dt \\
\frac{1}{2} \sqrt{\frac{g}{\beta}} \left(-\ln \left(1 - \sqrt{\frac{\beta}{g}} v \right) + \ln \left(1 + \sqrt{\frac{\beta}{g}} v \right) \right) &= gt + C_1 \\
\ln \left(\frac{1 + \sqrt{\frac{\beta}{g}} v}{1 - \sqrt{\frac{\beta}{g}} v} \right) &= \sqrt{\frac{\beta}{g}} (2gt + 2C_1) = 2\sqrt{\beta g} t + C_2 \\
\frac{1 + \sqrt{\frac{\beta}{g}} v}{1 - \sqrt{\frac{\beta}{g}} v} &= e^{2\sqrt{\beta g} t + C_2} = B e^{2\sqrt{\beta g} t}
\end{aligned}$$

where $B = e^{C_2}$, $C_2 = 2\sqrt{\frac{\beta}{g}} C_1$. The same initial condition implies that

$$\begin{aligned}
\frac{1 + \sqrt{\frac{\beta}{g}}(0)}{1 - \sqrt{\frac{\beta}{g}}(0)} &= B e^{2\sqrt{\beta g}(0)} \\
B &= 1
\end{aligned}$$

Therefore,

$$\frac{1 + \sqrt{\frac{\beta}{g}} v}{1 - \sqrt{\frac{\beta}{g}} v} = e^{2\sqrt{\beta g} t}$$

To reach 90% of the terminal speed (which is $\sqrt{\frac{g}{\beta}}$ this time) at $t = 10$ s, we need

$$\begin{aligned}
\frac{1 + \sqrt{\frac{\beta}{g}} \left(0.9 \sqrt{\frac{g}{\beta}} \right)}{1 - \sqrt{\frac{\beta}{g}} \left(0.9 \sqrt{\frac{g}{\beta}} \right)} &= e^{2\sqrt{\beta g}(10)} \\
\frac{1 + 0.9}{1 - 0.9} &= 19 = e^{20\sqrt{\beta g}} \\
20\sqrt{\beta g} &= \ln 19 \\
\beta &= \frac{1}{g} \left(\frac{\ln 19}{20} \right)^2
\end{aligned}$$

(1.23)

$$\frac{dT_b}{dz} = \gamma_a (T_b^{\text{medium}} - T_b)$$

$$\begin{aligned}
\frac{d\tau}{dz} \frac{dT_b}{d\tau} &= \gamma_a \frac{dT_b}{d\tau} = \gamma_a (T_b^{\text{medium}} - T_b) \\
\frac{dT_b}{d\tau} &= T_b^{\text{medium}} - T_b \\
\frac{dT_b}{d\tau} + T_b &= T_b^{\text{medium}} \\
e^\tau \left(\frac{dT_b}{d\tau} + T_b \right) &= \frac{d}{d\tau} (e^\tau T_b) = e^\tau T_b^{\text{medium}} \\
[e^{\tau'} T_b]_0^\tau &= \int_0^\tau e^{\tau'} T_b^{\text{medium}}(\tau') d\tau' \quad (\text{Dummy integration variable}) \\
e^\tau T_b(\tau) - e^{(0)} T_b(0) &= \int_0^\tau e^{\tau'} T_b^{\text{medium}}(\tau') d\tau' \\
T_b(\tau) &= T_b(0) e^{-\tau} + \int_0^\tau T_b^{\text{medium}}(\tau') e^{\tau' - \tau} d\tau'
\end{aligned}$$

(1.24) The strategy is to first move straight towards the sighted location. If the submarine also moves towards the destroyer, then they will surely meet at $r = 1, \theta = 0$. Suppose the submarine instead moves in a direction slightly deflected by $d\theta$, it will be away from the destroyer, indicated by the arc $rd\theta$. To catch up to the submarine, which will then travel by an extra distance of dr , the destroyer may move by a longer distance of $2dr$ so that these three lengths form a right-angled triangle. Therefore,

$$\begin{aligned}
(rd\theta)^2 + dr^2 &= (2dr)^2 \\
\pm rd\theta &= \sqrt{3}dr \\
\int \frac{\sqrt{3}}{r} dr &= \pm \int d\theta \\
\sqrt{3} \ln |r| &= \pm \theta + C \\
r &= A e^{\pm \frac{\theta}{\sqrt{3}}}
\end{aligned}$$

By the above argument, the I.C. is the same as $A = 1$, so the required strategy is to move according to either $r = e^{\pm \frac{\theta}{\sqrt{3}}}$.

Solutions for Chapter 2

(2.1) This is a homogeneous first-order ODE and we let $z = \frac{y}{x}$, with $\frac{dy}{dx} = z + x \frac{dz}{dx}$:

$$\begin{aligned} z + x \frac{dz}{dx} &= 2z^2 - z \\ x \frac{dz}{dx} &= 2z(z - 1) \\ \int \left(\frac{1}{z-1} - \frac{1}{z} \right) dz &= 2 \int \frac{dx}{x} \\ \ln |z-1| - \ln |z| &= 2 \ln |x| + C \\ \frac{z-1}{z} &= Ax^2 \end{aligned}$$

with $A = \pm e^C$. Recovering y gives

$$\begin{aligned} 1 - \frac{x}{y} &= Ax^2 \\ y &= \frac{x}{1 - Ax^2} \end{aligned}$$

The I.C. $y(1) = 2$ indicates that $A = \frac{1}{2}$ and so the full solution is

$$y = \frac{2x}{2 - x^2}$$

(2.2) Again, it is a homogeneous equation and using the same substitution as above gives

$$\begin{aligned} z + x \frac{dz}{dx} &= z - \sin z \\ x \frac{dz}{dx} &= -\sin z \\ \int \csc z \, dz &= -\int \frac{dx}{x} \\ -\ln |\csc z + \cot z| &= -\ln |x| + C \\ \csc z + \cot z &= Ax \end{aligned}$$

with $A = \pm e^C$. Recovering y yields

$$\csc\left(\frac{y}{x}\right) + \cot\left(\frac{y}{x}\right) = Ax$$

(2.3) Same as above again:

$$\begin{aligned} z + x \frac{dz}{dx} &= \frac{\sqrt{z^2 - 1}}{z} + z \\ x \frac{dz}{dx} &= \frac{\sqrt{z^2 - 1}}{z} \\ \int \frac{z dz}{\sqrt{z^2 - 1}} &= \int \frac{dx}{x} \\ \sqrt{z^2 - 1} &= \ln |x| + C \\ \left(\frac{y}{x}\right)^2 - 1 &= (\ln |x| + C)^2 \\ y &= x \sqrt{(\ln |x| + C)^2 + 1} \end{aligned}$$

The I.C. $y(1) = 1$ means that $C = 0$, so the final answer is

$$y = x \sqrt{(\ln x)^2 + 1}$$

(2.4) It can be checked that the ODE is isobaric with $m = \frac{1}{2}$. Setting $y = x^{\frac{1}{2}}z$, then it becomes

$$\frac{dy}{dx} = x^{\frac{1}{2}} \frac{dz}{dx} + \frac{1}{2} x^{-\frac{1}{2}} z$$

and the ODE is transformed into

$$\begin{aligned} x^{\frac{1}{2}} \frac{dz}{dx} + \frac{1}{2} x^{-\frac{1}{2}} z &= \frac{2}{zx^{\frac{1}{2}}} - \frac{2x}{(zx^{\frac{1}{2}})^3} \\ x^{\frac{1}{2}} \frac{dz}{dx} + \frac{1}{2} x^{-\frac{1}{2}} z &= 2z^{-1} x^{-\frac{1}{2}} - 2z^{-3} x^{-\frac{1}{2}} \\ x \frac{dz}{dx} + \frac{1}{2} z &= \frac{2}{z} - \frac{2}{z^3} \\ \frac{dz}{-\frac{1}{2}z + \frac{2}{z} - \frac{2}{z^3}} &= \frac{dx}{x} \\ -\int \frac{2z^3 dz}{z^4 - 4z^2 + 4} &= \int \frac{dx}{x} \\ -\int \frac{z^2 d(z^2 - 2)}{(z^2 - 2)^2} &= \int \frac{dx}{x} \end{aligned}$$

$$-\int \frac{d(z^2 - 2)}{(z^2 - 2)} - \int \frac{2d(z^2 - 2)}{(z^2 - 2)^2} = \ln |x| + C$$

$$-\ln |z^2 - 2| + \frac{2}{z^2 - 2} = \ln |x| + C$$

Putting back $z = \frac{y}{x^{\frac{1}{2}}}$ yields

$$-\ln \left| \frac{y^2}{x} - 2 \right| + \frac{2}{\frac{y^2}{x} - 2} = \ln |x| + C$$

$$\frac{2x}{y^2 - 2x} = \ln |y^2 - 2x| + C$$

The I.C. $y(1) = 1$ implies that $C = -2$ and hence the required answer is

$$\frac{2x}{y^2 - 2x} = \ln(2x - y^2) - 2$$

(2.5) This ODE is isobaric with $m = 2$. Let $y = x^2 z$, and

$$\frac{dy}{dx} = x^2 \frac{dz}{dx} + 2xz$$

Substituting these gives

$$x^2 \frac{dz}{dx} + 2xz = \frac{x^2 z}{x + \sqrt{x^2 z}}$$

$$x \frac{dz}{dx} + 2z = \frac{z}{1 + \sqrt{z}}$$

$$x \frac{dz}{dx} = -z \left(\frac{1 + 2\sqrt{z}}{1 + \sqrt{z}} \right)$$

$$\int \frac{1 + \sqrt{z}}{z(1 + 2\sqrt{z})} dz = \int \left(\frac{1}{z} - \frac{1}{\sqrt{z}(1 + 2\sqrt{z})} \right) dz = -\int \frac{dx}{x}$$

$$\ln |z| - \ln |1 + 2\sqrt{z}| = -\ln |x| + C$$

$$\frac{z}{1 + 2\sqrt{z}} = \frac{A}{x}$$

with $A = \pm e^C$. Recovering y yields

$$\frac{\frac{y}{x^2}}{1 + 2\sqrt{\frac{y}{x^2}}} = \frac{y}{x^2 + 2x\sqrt{y}} = \frac{A}{x}$$

$$\frac{y}{x + 2\sqrt{y}} = A$$

The I.C. $y(0) = 1$ indicates that $A = \frac{1}{2}$. So

$$y = \frac{1}{2}(x + 2\sqrt{y})$$

$$y - \sqrt{y} - \frac{1}{2}x = 0$$

This is a quadratic equation in $\sqrt{y}$, which has a positive root of

$$\sqrt{y} = \frac{1 + \sqrt{1 + 2x}}{2}$$

$$y = \left(\frac{1 + \sqrt{1 + 2x}}{2} \right)^2$$

(2.6) Here $n = 3$. Let $v = y^{1-3} = y^{-2} = \frac{1}{y^2}$, $y = v^{-\frac{1}{2}} = \frac{1}{\sqrt{v}}$, then

$$\frac{dy}{dx} = -\frac{1}{2}v^{-\frac{3}{2}} \frac{dv}{dx} = -\frac{y^3}{2} \frac{dv}{dx}$$

and the ODE becomes first-order linear:

$$-\frac{y^3}{2} \frac{dv}{dx} + xy = x^3 y^3$$

$$-\frac{1}{2} \frac{dv}{dx} + xy^{-2} = x^3$$

$$\frac{dv}{dx} - 2xv = -2x^3$$

The integrating factor is $\exp(\int -2x dx) = e^{-x^2}$:

$$e^{-x^2} \frac{dv}{dx} - 2xe^{-x^2} v = -2x^3 e^{-x^2}$$

$$\frac{d}{dx}(e^{-x^2} v) = -2x^3 e^{-x^2}$$

$$e^{-x^2} v = -2 \int x^3 e^{-x^2} dx = \int x^2 d(e^{-x^2})$$

$$e^{-x^2} v = [x^2 e^{-x^2}] - \int e^{-x^2} d(x^2) = x^2 e^{-x^2} + e^{-x^2} + C$$

$$v = x^2 + 1 + Ce^{x^2}$$

$$y = \frac{1}{\sqrt{x^2 + 1 + Ce^{x^2}}}$$

The I.C. $y(0) = \frac{1}{2}$ implies that $C = 3$, and therefore $y = \frac{1}{\sqrt{x^2+1+3e^{x^2}}}$.

(2.7) This is also a Bernoulli ODE with $n = 2$. Let $v = y^{1-2} = y^{-1} = \frac{1}{y}$, $y = \frac{1}{v}$, so $\frac{dy}{dx} = -\frac{1}{v^2} \frac{dv}{dx}$,

$$\begin{aligned} -\frac{1}{v^2} \frac{dv}{dx} + \frac{2}{vx} &= \frac{x}{v^2} \\ \frac{dv}{dx} - \frac{2v}{x} &= -x \end{aligned}$$

The integrating factor is $\exp\left(\int -\frac{2}{x} dx\right) = \frac{1}{x^2}$:

$$\begin{aligned} \frac{1}{x^2} \frac{dv}{dx} - \frac{2v}{x^3} &= \frac{d}{dx} \left(\frac{1}{x^2} v \right) = -\frac{1}{x} \\ \frac{1}{x^2} v &= \int -\frac{dx}{x} = -\ln|x| + C \\ v &= -x^2 \ln|x| + Cx^2 \\ y &= \frac{1}{-x^2 \ln|x| + Cx^2} \end{aligned}$$

The I.C. $y(1) = 1$ indicates that $C = 1$, hence the required solution is

$$y = \frac{1}{x^2(1 - \ln x)}$$

(2.8) After some trial-and-error you should be able to see that $y_p = x^2$ is a particular solution. Hence we let $y = x^2 + \frac{1}{z}$, which gives

$$\begin{aligned} \frac{d}{dx} \left(x^2 + \frac{1}{z} \right) &= \frac{(x^2 + \frac{1}{z})^2}{x^2} + 3 \frac{(x^2 + \frac{1}{z})}{x} - (x^2 + x) \\ 2x - \frac{1}{z^2} \frac{dz}{dx} &= \frac{x^4 + 2\frac{x^2}{z} + \frac{1}{z^2}}{x^2} + 3x + \frac{3}{xz} - (x^2 + x) \\ -\frac{1}{z^2} \frac{dz}{dx} &= \frac{2}{z} + \frac{1}{x^2 z^2} + \frac{3}{xz} \\ \frac{dz}{dx} &= -2z - \frac{1}{x^2} - \frac{3z}{x} \\ \frac{dz}{dx} + \left(2 + \frac{3}{x} \right) z &= -\frac{1}{x^2} \end{aligned}$$

The integrating factor is

$$\exp\left(\int \left(2 + \frac{3}{x}\right) dx\right) = \exp(2x + 3 \ln|x|) = x^3 e^{2x}$$

So

$$\begin{aligned}\frac{d}{dx}(x^3 e^{2x} z) &= -x e^{2x} \\ x^3 e^{2x} z &= -\int x e^{2x} dx = -\frac{1}{2} x e^{2x} + \frac{1}{4} e^{2x} + C \\ z &= -\frac{1}{2x^2} + \frac{1}{4x^3} + C \frac{e^{-2x}}{x^3}\end{aligned}$$

and the general solution is

$$\begin{aligned}y &= x^2 + \frac{1}{\frac{1}{4x^3} - \frac{1}{2x^2} + C \frac{e^{-2x}}{x^3}} \\ &= x^2 + \frac{x^3}{\frac{1}{4} - \frac{x}{2} + C e^{-2x}} \\ &= \frac{x^3 + x^2(\frac{1}{4} - \frac{x}{2} + C e^{-2x})}{\frac{1}{4} - \frac{x}{2} + C e^{-2x}} \\ &= \frac{\frac{x^3}{2} + \frac{x^2}{4} + C x^2 e^{-2x}}{\frac{1}{4} - \frac{x}{2} + C e^{-2x}} \\ &= x^2 \frac{\frac{x}{2} + \frac{1}{4} + C e^{-2x}}{\frac{1}{4} - \frac{x}{2} + C e^{-2x}}\end{aligned}$$

(2.9) Using the trigonometric identities $\sin(2x) = 2 \sin x \cos x$ and $\cos(2x) = \cos^2 x - \sin^2 x$, we can express the ODE as

$$\begin{aligned}(2 \sin x \cos x)(y')^2 - 2(\cos^2 x - \sin^2 x)yy' - (2 \sin x \cos x)y^2 &= 0 \\ (\sin x \cos x)(y')^2 - (\cos^2 x)yy' + (\sin^2 x)yy' - (\sin x \cos x)y^2 &= 0 \\ (\cos x)y'((\sin x)y' - (\cos x)y) + (\sin x)y((\sin x)y' - (\cos x)y) &= 0 \\ ((\cos x)y' + (\sin x)y)((\sin x)y' - (\cos x)y) &= 0\end{aligned}$$

The solution to the linear ODE dictated by the first factor is computed as

$$\begin{aligned}(\cos x) \frac{dy}{dx} + (\sin x)y &= 0 \\ \frac{dy}{dx} &= -(\tan x)y \\ \int \frac{dy}{y} &= -\int \frac{\sin x}{\cos x} dx \\ \ln |y| &= \int \frac{1}{\cos x} d(\cos x) = \ln |\cos x| + C\end{aligned}$$

$$y = \pm e^C \cos x = A \cos x$$

Similarly, the solution associated with the second factor is $y = A \sin x$, and the full solution is thus

$$(y - A \cos x)(y - A \sin x) = 0$$

(2.10) Rearranging easily shows that the ODE is soluble in x :

$$x = \left(\frac{y}{y'}\right)^2 - \frac{y}{y'} = \left(\frac{y}{p}\right)^2 - \frac{y}{p}$$

For notational clarity, we have written $p = y'$. Using the formula given in the main text, we have

$$\begin{aligned} \frac{1}{p} &= \left(2\frac{y}{p^2} - \frac{1}{p}\right) + \left(-2\frac{y^2}{p^3} + \frac{y}{p^2}\right)\frac{dp}{dy} \\ 0 &= \left(2\frac{y}{p^2} - \frac{2}{p}\right) + \left(-2\frac{y^2}{p^3} + \frac{y}{p^2}\right)\frac{dp}{dy} \\ 0 &= \left(2\frac{y}{p} - 2\right) + \left(-2\frac{y^2}{p^2} + \frac{y}{p}\right)\frac{dp}{dy} \end{aligned}$$

This is a homogeneous ODE and we can let $v = \frac{p}{y}$:

$$\begin{aligned} \left(2\frac{1}{v} - 2\right) + \left(-2\frac{1}{v^2} + \frac{1}{v}\right)\left(v + y\frac{dv}{dy}\right) &= 0 \\ (2v - 2v^2) + (-2 + v)\left(v + y\frac{dv}{dy}\right) &= 0 \\ v + y\frac{dv}{dy} &= \frac{2v^2 - 2v}{v - 2} \\ y\frac{dv}{dy} &= \frac{2v^2 - 2v - v(v - 2)}{v - 2} = \frac{v^2}{v - 2} \\ \int \frac{(v - 2)dv}{v^2} &= \int \frac{dy}{y} \\ \int \left(\frac{1}{v} - \frac{2}{v^2}\right)dv &= \ln |y| + C \\ \ln |v| + \frac{2}{v} &= \ln |y| + C \\ ve^{\frac{2}{v}} &= Ay \\ \left(\frac{p}{y}\right)e^{\frac{2y}{p}} &= Ay \end{aligned}$$

where we set $A = \pm e^C$ as usual. Note that the ODE can also be written as a quadratic equation in p , so by high-school algebra:

$$0 = y^2 - yp - xp^2$$

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$$\Rightarrow p = \frac{-y \pm \sqrt{y^2 + 4xy^2}}{2x}$$

$$\frac{p}{y} = \frac{-1 \pm \sqrt{1 + 4x}}{2x} = \frac{2}{\pm\sqrt{4x+1} + 1}$$

So

$$\frac{2}{\pm\sqrt{4x+1} + 1} e^{\pm\sqrt{4x+1}+1} = Ay$$

$$y = \frac{Be^{\pm\sqrt{4x+1}}}{\sqrt{4x+1} \pm 1}$$

where $B = \frac{2e}{A}$. For the two solution branches, the I.C. $y(1) = 2$ can mean $B = 2(1 + \sqrt{5})e^{-\sqrt{5}}$ and $B = 2(-1 + \sqrt{5})e^{\sqrt{5}}$ correspondingly, i.e.

$$y = \frac{2(1 + \sqrt{5})e^{\sqrt{4x+1}-\sqrt{5}}}{\sqrt{4x+1} + 1} \text{ or } \frac{2(-1 + \sqrt{5})e^{-\sqrt{4x+1}+\sqrt{5}}}{\sqrt{4x+1} - 1}$$

(2.11) This is an equation soluble for y . Differentiating it with respect to x produces (again denoting $p = \frac{dy}{dx}$)

$$e^x p^2 + 2e^x p \frac{dp}{dx} - \frac{dp}{dx} - \frac{p}{2} = 0$$

$$e^x p \left(2 \frac{dp}{dx} + p \right) - \frac{1}{2} \left(2 \frac{dp}{dx} + p \right) = 0$$

$$\left(e^x p - \frac{1}{2} \right) \left(2 \frac{dp}{dx} + p \right) = 0$$

The second factor corresponds to the general solution and is easily seen to have the solution of $p = Ce^{-\frac{x}{2}}$, and hence putting this back into the original equation, we get

$$e^x (Ce^{-\frac{x}{2}})^2 - Ce^{-\frac{x}{2}} - \frac{y}{2} = 0$$

$$C^2 - Ce^{-\frac{x}{2}} - \frac{y}{2} = 0$$

$$y = 2C^2 - 2Ce^{-\frac{x}{2}}$$

Meanwhile, the first factor is the singular solution $p = \frac{1}{2}e^{-x}$, and

$$e^x \left(\frac{1}{2}e^{-x} \right)^2 - \frac{1}{2}e^{-x} - \frac{y}{2} = 0$$

$$y = -\frac{e^{-x}}{2}$$

(2.12) This amounts to enforcing, for

$$\begin{aligned}\frac{ax + by + p}{cx + dy + q} &= \frac{a(\tilde{x} + h) + b(\tilde{y} + k) + p}{c(\tilde{x} + h) + d(\tilde{y} + k) + q} \\ &= \frac{a\tilde{x} + b\tilde{y} + (ah + bk + p)}{c\tilde{x} + d\tilde{y} + (ch + dk + q)}\end{aligned}$$

we have

$$\begin{cases} ah + bk + p = 0 \\ ch + dk + q = 0 \end{cases}$$

so that the argument becomes homogeneous in $\tilde{x}, \tilde{y}$:

$$\begin{aligned}f\left(\frac{ax + by + p}{cx + dy + q}\right) &= f\left(\frac{a\tilde{x} + b\tilde{y} + (ah + bk + p)}{c\tilde{x} + d\tilde{y} + (ch + dk + q)}\right) \\ &= f\left(\frac{a\tilde{x} + b\tilde{y}}{c\tilde{x} + d\tilde{y}}\right) = f\left(\frac{a + b\frac{\tilde{y}}{\tilde{x}}}{c + d\frac{\tilde{y}}{\tilde{x}}}\right)\end{aligned}$$

For the specific problem, we need to satisfy

$$\begin{cases} h + 2k - 1 = 0 \\ 2h - k + 3 = 0 \end{cases}$$

whose solution can be seen to be $h = -1, k = 1$. With this substitution, the ODE becomes

$$\frac{d\tilde{y}}{d\tilde{x}} = \frac{\tilde{x} + 2\tilde{y}}{2\tilde{x} - \tilde{y}} = \frac{1 + 2\frac{\tilde{y}}{\tilde{x}}}{2 - \frac{\tilde{y}}{\tilde{x}}}$$

By a further substitution $\tilde{z} = \frac{\tilde{y}}{\tilde{x}}$, it is changed into and solved as

$$\begin{aligned}\tilde{z} + \tilde{x} \frac{d\tilde{z}}{d\tilde{x}} &= \frac{1 + 2\tilde{z}}{2 - \tilde{z}} \\ \tilde{x} \frac{d\tilde{z}}{d\tilde{x}} &= \frac{1 + 2\tilde{z} - \tilde{z}(2 - \tilde{z})}{2 - \tilde{z}} = \frac{1 + \tilde{z}^2}{2 - \tilde{z}} \\ \int \frac{2 - \tilde{z}}{1 + \tilde{z}^2} d\tilde{z} &= \int \frac{d\tilde{x}}{\tilde{x}} \\ \int \frac{2}{1 + \tilde{z}^2} d\tilde{z} - \int \frac{\tilde{z}}{1 + \tilde{z}^2} d\tilde{z} &= \ln |\tilde{x}| + C \\ 2 \tan^{-1} \tilde{z} - \frac{1}{2} \ln |1 + \tilde{z}^2| &= \ln |\tilde{x}| + C\end{aligned}$$

and

$$2 \tan^{-1} \left(\frac{\tilde{y}}{\tilde{x}} \right) - \frac{1}{2} \ln \left(1 + \left(\frac{\tilde{y}}{\tilde{x}} \right)^2 \right) - \ln |\tilde{x}| = C$$

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$$2 \tan^{-1} \left(\frac{y-1}{x+1} \right) - \frac{1}{2} \ln \left(1 + \left(\frac{y-1}{x+1} \right)^2 \right) - \ln |x+1| = C$$

(2.13) The ODE and its solution are discontinuous at $x = 1$. Hence, information from the initial condition cannot propagate across the discontinuity.

Solutions for Chapter 3

(3.1)

$$\begin{aligned} W[y_1, y_2] &= \begin{vmatrix} e^{-x} \sin(5x) & e^{-x} \cos(5x) \\ \frac{d}{dx}(e^{-x} \sin(5x)) & \frac{d}{dx}(e^{-x} \cos(5x)) \end{vmatrix} \\ &= \begin{vmatrix} e^{-x} \sin(5x) & e^{-x} \cos(5x) \\ -e^{-x} \sin(5x) + 5e^{-x} \cos(5x) & -e^{-x} \cos(5x) - 5e^{-x} \sin(5x) \end{vmatrix} \\ &= e^{-x} \sin(5x)[-e^{-x} \cos(5x) - 5e^{-x} \sin(5x)] - e^{-x} \cos(5x)[-e^{-x} \sin(5x) + 5e^{-x} \cos(5x)] \\ &= -5e^{-2x} \sin^2(5x) - 5e^{-2x} \cos^2(5x) = -5e^{-2x} \neq 0 \end{aligned}$$

Since the Wronskian is never zero, they are linearly independent.

(3.2) The auxiliary equation is $r^2 - 3r - 4 = 0$, which has two real roots $r = -1, 4$. The general solution is hence

$$y = c_1 e^{-x} + c_2 e^{4x}$$

and hence

$$y' = -c_1 e^{-x} + 4c_2 e^{4x}$$

To match the I.C.s, we have

$$\begin{cases} c_1 + c_2 = 2 \\ -c_1 + 4c_2 = 3 \end{cases}$$

It can be checked that $c_1 = 1, c_2 = 1$, so the required answer is $y = e^{-x} + e^{4x}$.

(3.3) The auxiliary equation is $r^2 + 4r + 4 = 0$, which has a repeated real root $r = -2$. The general solution is hence

$$y = c_1 e^{-2x} + c_2 x e^{-2x}$$

and thus

$$y' = (-2c_1 + c_2)e^{-2x} - 2c_2xe^{-2x}$$

To match the I.C.s, we have

$$\begin{cases} c_1 = 2 \\ -2c_1 + c_2 = -5 \end{cases}$$

It is obvious that $c_1 = 2$, $c_2 = -1$, so the required answer is $y = 2e^{-2x} - xe^{-2x}$.

(3.4) The auxiliary equation is simply $r^2 + 12 = 0$ and hence the complex roots are easily seen to be $r_{\pm} = 2\sqrt{3}i$. The two valid complementary solutions are $y_1 = \cos(2\sqrt{3}x)$ and $y_2 = \sin(2\sqrt{3}x)$, so the general solution will take the form of

$$y = c_1 \cos(2\sqrt{3}x) + c_2 \sin(2\sqrt{3}x)$$

and

$$y = -2\sqrt{3}c_1 \sin(2\sqrt{3}x) + 2\sqrt{3}c_2 \cos(2\sqrt{3}x)$$

Substituting the I.C.s readily yields $c_1 = 1$, $c_2 = \frac{1}{\sqrt{3}}$, i.e. $y = \cos(2\sqrt{3}x) + \frac{\sin(2\sqrt{3}x)}{\sqrt{3}}$.

(3.5) The auxiliary equation $r^2 - 6r + 1 = 0$ has two real roots:

$$\begin{aligned} r &= \frac{-(-6) \pm \sqrt{(-6)^2 - 4(1)(1)}}{2(1)} \\ &= \frac{6 \pm \sqrt{32}}{2} = 3 \pm 2\sqrt{2} \end{aligned}$$

So the general solution takes the form of $y = c_1 e^{(3+2\sqrt{2})x} + c_2 e^{(3-2\sqrt{2})x}$. The I.C.s suggest that

$$\begin{cases} c_1 + c_2 = 1 \\ (3 + 2\sqrt{2})c_1 + (3 - 2\sqrt{2})c_2 = 3 \end{cases} \Rightarrow c_1 = \frac{1}{2}, c_2 = \frac{1}{2}$$

and thus $y = \frac{1}{2}e^{(3+2\sqrt{2})x} + \frac{1}{2}e^{(3-2\sqrt{2})x}$.

(3.6) The auxiliary equation $r^2 + 2r + 4 = 0$ has two complex roots as

$$\begin{aligned} r_{\pm} &= \frac{-2 \pm \sqrt{(2)^2 - 4(1)(4)}}{2} \\ &= -1 \pm \sqrt{3}i \end{aligned}$$

So the general solution will be

$$y = c_1 e^{-x} \cos(\sqrt{3}x) + c_2 e^{-x} \sin(\sqrt{3}x)$$

with

$$y = (-c_1 + \sqrt{3}c_2)e^{-x} \cos(\sqrt{3}x) + (-\sqrt{3}c_1 - c_2)e^{-x} \sin(\sqrt{3}x)$$

The I.C.s then indicate

$$\begin{cases} c_1 = 1 \\ -c_1 + \sqrt{3}c_2 = -1 \end{cases}$$

Therefore, $c_1 = 1$, $c_2 = 0$, and the required answer is $y = e^{-x} \cos(\sqrt{3}x)$.

(3.7) The auxiliary equation is solved as

$$\begin{aligned} r^2 - 6r + 9 &= 0 \\ r &= 3 \end{aligned}$$

This is a repeated real root. The general solution is then

$$\begin{aligned} y &= c_1 e^{3x} + c_2 x e^{3x} \\ y' &= (3c_1 + c_2)e^{3x} + 3c_2 x e^{3x} \end{aligned}$$

Substituting the I.C.s $y(0) = 1$, $y'(0) = 4$ gives

$$\begin{cases} c_1 = 1 \\ 3c_1 + c_2 = 4 \end{cases}$$

Therefore $c_1 = 1$, $c_2 = 1$, and $y = e^{3x} + x e^{3x}$.

(3.8) The auxiliary equation is solved as

$$\begin{aligned} r^2 - 2r + 5 &= 0 \\ r &= 1 \pm 2i \end{aligned}$$

This is a case of two complex roots. The solution form is

$$y = c_1 e^x \cos(2x) + c_2 e^x \sin(2x)$$

and

$$y' = (c_1 + 2c_2)e^x \cos(2x) + (-2c_1 + c_2)e^x \sin(2x)$$

Plugging in the I.C.s $y(0) = -1$, $y'(0) = 1$ leads to

$$\begin{cases} c_1 = -1 \\ c_1 + 2c_2 = 1 \end{cases}$$

Hence $c_1 = -1$, $c_2 = 1$, and $y = -e^x \cos(2x) + e^x \sin(2x)$.

(3.9) Direct substitution of either $y_1 = 1$ and $y_2 = x$ into the second-order derivative on the L.H.S. gives 0 so they solve the equation. The more important issue is to verify their linear independence by calculating the Wronskian:

$$\begin{aligned} W[1, x] &= \begin{vmatrix} 1 & x \\ 1' & x' \end{vmatrix} \\ &= \begin{vmatrix} 1 & x \\ 0 & 1 \end{vmatrix} = 1 \neq 0 \end{aligned}$$

which is non-zero. By linearity, the general solution will be $y = Ax + B$ where A, B are some constants. (This can also be deduced by integrating directly twice.)

(3.10) Dividing by L to get:

$$\frac{d^2 q}{dt^2} + \frac{R}{L} \frac{dq}{dt} + \frac{1}{LC} q = 0$$

A direct comparison to the spring-mass system reveals that $\alpha = \frac{R}{L}$, $\omega_0 = \sqrt{\frac{1}{LC}}$, and the critical damping occurs when

$$\begin{aligned} \alpha &= 2\omega_0 \\ \frac{R}{L} &= 2\sqrt{\frac{1}{LC}} \\ \frac{R}{2} &= \sqrt{\frac{L}{C}} \end{aligned}$$

(3.11) The two complementary solutions are clearly $y_1 = \cos x$ and $y_2 = \sin x$. The particular solution can be found by the method of undetermined coefficients. Let $y_p = A \sin(2x) + B \cos(2x)$:

$$\begin{aligned} (-4A \sin(2x) - 4B \cos(2x)) + (A \sin(2x) + B \cos(2x)) &= \sin(2x) \\ -3A \sin(2x) - 3B \cos(2x) &= 0 \end{aligned}$$

Comparing the coefficients shows that $A = -\frac{1}{3}$, $B = 0$, so the full solution looks like

$$y = c_1 \cos x + c_2 \sin x - \frac{1}{3} \sin(2x)$$

and

$$y' = -c_1 \sin x + c_2 \cos x - \frac{2}{3} \cos(2x)$$

Matching the I.C.s, we need

$$\begin{cases} c_1 = 0 \\ c_2 - \frac{2}{3} = 0 \end{cases}$$

So $c_1 = 0$, $c_2 = \frac{2}{3}$, and the final answer is $y = \frac{2}{3} \sin x - \frac{1}{3} \sin(2x)$.

(3.12) It is clear that the complementary solutions are $y_1 = e^x$ and $y_2 = e^{2x}$. The non-homogeneous term $x \sin x$ requires us to propose the particular solution in the form of $y_p = M \sin x + N \cos x + Px \sin x + Qx \cos x$. Its derivatives are

$$\begin{aligned} \frac{dy_p}{dx} &= M \cos x - N \sin x + (Px \cos x + P \sin x) + (-Qx \sin x + Q \cos x) \\ &= (M + Q) \cos x + (P - N) \sin x + Px \cos x - Qx \sin x \\ \frac{d^2 y_p}{dx^2} &= -(M + Q) \sin x + (P - N) \cos x - Px \sin x + P \cos x - Qx \cos x - Q \sin x \\ &= -(M + 2Q) \sin x + (2P - N) \cos x - Px \sin x - Qx \cos x \end{aligned}$$

Substituting these expressions into the ODE leads to

$$\begin{aligned} &[-(M + 2Q) \sin x + (2P - N) \cos x - Px \sin x - Qx \cos x] \\ &- 3[(M + Q) \cos x + (P - N) \sin x + Px \cos x - Qx \sin x] \\ &\quad + 2[M \sin x + N \cos x + Px \sin x + Qx \cos x] = x \sin x \\ &(3N - 3P + M - 2Q) \sin x + (2P + N - 3M - 3Q) \cos x \\ &\quad + (3Q + P)x \sin x + (Q - 3P)x \cos x = x \sin x \end{aligned}$$

Comparing the coefficients gives the system of equations:

$$\begin{cases} 3N - 3P + M - 2Q = 0 \\ 2P + N - 3M - 3Q = 0 \\ 3Q + P = 1 \\ Q - 3P = 0 \end{cases}$$

Solving it yields $P = \frac{1}{10}$, $Q = \frac{3}{10}$, $M = -\frac{3}{25}$, $N = \frac{17}{50}$ (we omit the steps), and hence the general solution is

$$y = c_1 e^x + c_2 e^{2x} - \frac{3}{25} \sin x + \frac{17}{50} \cos x + \frac{1}{10} x \sin x + \frac{3}{10} x \cos x$$

(3.13) The complementary solution consists of $y_1 = e^{2x}$ and $y_2 = xe^{2x}$, so the e^{2x} basis has a multiplicity of 2. This demands us to increase the degree by 2 and guess the form of the particular solution as $y_p = a_3x^3e^{2x} + a_2x^2e^{2x}$. (There is no need for the a_1, a_0 terms, why?) Plugging in this leads to

$$\begin{aligned} & (4a_3x^3e^{2x} + (12a_3 + 4a_2)x^2e^{2x} + (6a_3 + 8a_2)xe^{2x} + 2a_2e^{2x}) \\ & - 4(2a_3x^3e^{2x} + (3a_3 + 2a_2)x^2e^{2x} + 2a_2xe^{2x}) + 4(a_3x^3e^{2x} + a_2x^2e^{2x}) = 3xe^{2x} \\ & 6a_3xe^{2x} + 2a_2e^{2x} = 3xe^{2x} \end{aligned}$$

Comparing the coefficients gives $a_3 = \frac{1}{2}, a_2 = 0$, and so the general solution will look like $y = c_1e^{2x} + c_2xe^{2x} + \frac{1}{2}x^3e^{2x}$. Differentiation yields $y' = (2c_1 + c_2)e^{2x} + 2c_2xe^{2x} + \frac{3}{2}x^2e^{2x} + x^3e^{2x}$, and matching the I.C.s implies $c_1 = 1, c_2 = -1$ with the final answer of $y = e^{2x} - xe^{2x} + \frac{1}{2}x^3e^{2x}$.

(3.14) The auxiliary equation is seen to be $2r^2 - 3r + 1 = 0$ which has the two real root of $r = \frac{1}{2}, 1$. Therefore, the complementary solutions are $y_1 = e^{\frac{x}{2}}$ and $y_2 = e^x$. Note that the leading coefficient of the equation is 2, so we need to normalize it by multiplying by $\frac{1}{2}$, and the source term $f(x)$ will be $\frac{xe^x}{2}$. The linear system to solve is then

$$\begin{aligned} & \begin{cases} v_1'y_1 + v_2'y_2 = 0 \\ v_1'y_1' + v_2'y_2' = \frac{xe^x}{2} \end{cases} \\ \Rightarrow & \begin{cases} v_1'e^{\frac{x}{2}} + v_2'e^x = 0 \\ v_1'\left(\frac{1}{2}e^{\frac{x}{2}}\right) + v_2'e^x = \frac{xe^x}{2} \end{cases} \end{aligned}$$

This is not very hard to solve: $v_1' = -xe^{\frac{x}{2}}$ and $v_2' = x$, and

$$\begin{aligned} v_1 &= -\int xe^{\frac{x}{2}} dx \\ &= -2 \int xd(e^{\frac{x}{2}}) \\ &= -2[xe^{\frac{x}{2}}] + 2 \int e^{\frac{x}{2}} dx = -2xe^{\frac{x}{2}} + 4e^{\frac{x}{2}} \end{aligned}$$

also $v_2 = \frac{x^2}{2}$ easily. The particular solution is hence

$$\begin{aligned} y_p &= v_1y_1 + v_2y_2 \\ &= (-2xe^{\frac{x}{2}} + 4e^{\frac{x}{2}})e^{\frac{x}{2}} + \frac{x^2}{2}e^x \\ &= 4e^x - 2xe^x + \frac{x^2e^x}{2} = -2xe^x + \frac{x^2e^x}{2} \end{aligned}$$

where the $4e^x$ term can be absorbed into y_2 . The general solution will be

$$y = c_1 e^{\frac{x}{2}} + c_2 e^x - 2xe^x + \frac{x^2 e^x}{2}$$

and $y' = \frac{c_1}{2} e^{\frac{x}{2}} + (c_2 - 2)e^x - xe^x + \frac{x^2 e^x}{2}$. Matching the I.C.s reveals that $c_1 = -2$, $c_2 = 3$, and the full solution is therefore

$$y = -2e^{\frac{x}{2}} + \left(\frac{x^2}{2} - 2x + 3\right)e^x$$

This question can also be approached by the previous method of undetermined coefficients.

(3.15) Let's use Variation of Parameters with $y_1 = \cos x$ and $y_2 = \sin x$. The corresponding linear system is

$$\begin{cases} v_1' \cos x + v_2' \sin x = 0 \\ v_1' (-\sin x) + v_2' \cos x = \csc(2x) \end{cases}$$

Then, by Cramer's Rule, we have

$$\begin{aligned} v_1' &= -\sin x \csc(2x) = -\frac{\sec x}{2} \\ v_2' &= \cos x \csc(2x) = \frac{\csc x}{2} \end{aligned}$$

and

$$\begin{aligned} v_1 &= \int -\frac{\sec x}{2} dx = -\frac{1}{2} \ln |\sec x + \tan x| \\ v_2 &= \int \frac{\csc x}{2} dx = -\frac{1}{2} \ln |\csc x + \cot x| \end{aligned}$$

Therefore, the particular solution will be

$$y_p = -\frac{1}{2} \cos x \ln |\sec x + \tan x| - \frac{1}{2} \sin x \ln |\csc x + \cot x|$$

(3.16) The complementary solution takes the form of $c_1 \cos x + c_2 \sin x$. To use undetermined coefficients, assume a particular solution in the form of $y_p = Ax \sin x + Bx \cos x$, then

$$\begin{aligned} y_p' &= A \sin x + Ax \cos x + B \cos x - Bx \sin x \\ &= Ax \cos x - Bx \sin x + B \cos x + A \sin x \\ y_p'' &= A \cos x - Ax \sin x - B \sin x - Bx \cos x - B \sin x + A \cos x \\ &= -Bx \cos x - Ax \sin x + 2A \cos x - 2B \sin x \end{aligned}$$

Plugging this ansatz into the equation, we have

$$(-Bx \cos x - Ax \sin x + 2A \cos x - 2B \sin x) + (Ax \sin x + Bx \cos x) = \sin x$$

$$2A \cos x - 2B \sin x = \sin x$$

and comparing the coefficients we have $A = 0$, $B = -\frac{1}{2}$, and thus $y_p = -\frac{1}{2}x \cos x$. Now the general solution is

$$y = c_1 \sin x + c_2 \cos x - \frac{1}{2}x \cos x$$

The initial conditions require that $c_1 = 1$, $c_2 = 1$ (the detailed computation is skipped), and the final answer is $y = \sin x + \cos x - \frac{1}{2}x \cos x$.

(3.17) This is a simple harmonic oscillator with a frequency of $\omega = \sqrt{\frac{g}{L}}$ and thus the approximated period is $T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{L}{g}}$. This is shorter than the true value, because the magnitude of acceleration in the real case is slower than the approximation, $\sin \theta < \theta$.

(3.18) When $N^2 > 0$, it becomes a simple harmonic oscillator with the solution form of $\cos(Nt - \phi)$, and the motion of the parcel will be bounded and hence stable. On the other hand, if $N^2 < 0$, the solution will be in the form of $e^{\pm Nt}$. The exponentially increasing part e^{Nt} then indicates that the parcel will keep accelerating and diverge, thus unstable. Finally, if $N^2 = 0$, $\frac{d^2 w}{dt^2} = 0$ and there is zero acceleration (neutral).

(3.19) Differentiating the second ODE gives

$$\frac{d^2[B]}{dt^2} = -k_A \frac{d[A]}{dt}$$

Substituting the first ODE into the above leads to

$$\frac{d^2[B]}{dt^2} = -k_A(-k_B[B]) = k_A k_B [B]$$

$$\frac{d^2[B]}{dt^2} - k_A k_B [B] = 0$$

The general solution of this newly acquired second-order ODE is obviously

$$[B] = c_1 e^{-Kt} + c_2 e^{Kt}$$

where $K = \sqrt{k_A k_B} = \sqrt{(0.01)(0.0225)} = 0.015$. The initial conditions are $[B](0) = 20000$, and $[B]'(0) = -k_A[A](0) = -(0.01)(40000) = -400$, and therefore

$$\begin{cases} c_1 + c_2 = 20000 \\ -Kc_1 + Kc_2 = -400 \end{cases}$$

Solving the system gives $c_1 = \frac{70000}{3}$, $c_2 = -\frac{10000}{3}$. So $[B] = \frac{70000}{3}e^{-0.015t} - \frac{10000}{3}e^{0.015t}$. $[B]$ will eventually deplete to zero because of the second term. The time when it happens is found by solving $\frac{70000}{3}e^{-0.015t} - \frac{10000}{3}e^{0.015t} = 0$, which gives $e^{0.03t} = 7$ and $t \approx 64.9$ h. Meanwhile, $[A] = 35000e^{-0.015t} + 5000e^{0.015t}$.

(3.20) The new solution is now $y = c_1y_1(x) + c_2y_2(x) + y_p(x)$ and the new system to be solved is

$$\begin{cases} y(a) = A = c_1y_1(a) + c_2y_2(a) + y_p(a) \\ y'(a) = B = c_1y_1'(a) + c_2y_2'(a) + y_p'(a) \end{cases} \\ \Rightarrow \begin{cases} A - y_p(a) = c_1y_1(a) + c_2y_2(a) + y_p(a) \\ B - y_p'(a) = c_1y_1'(a) + c_2y_2'(a) + y_p'(a) \end{cases}$$

This is the same as before but with A replaced by $A - y_p(a)$ and B by $B - y_p'(a)$, and we immediately conclude the desired statement.

(3.21) Notice that the derivative of a determinant consists of the sum of determinants produced from the original determinant with each of the rows differentiated. In the case of the Wronskian, it is

$$\begin{aligned} \frac{dW}{dx} &= \begin{vmatrix} y_1'(x) & y_2'(x) & y_3'(x) & \cdots & y_n'(x) \\ y_1'(x) & y_2'(x) & y_3'(x) & & y_n'(x) \\ y_1''(x) & y_2''(x) & y_3''(x) & & y_n''(x) \\ \vdots & & & \ddots & \vdots \\ y_1^{(n-1)}(x) & y_2^{(n-1)}(x) & y_3^{(n-1)}(x) & \cdots & y_n^{(n-1)}(x) \end{vmatrix} \\ &+ \begin{vmatrix} y_1(x) & y_2(x) & y_3(x) & \cdots & y_n(x) \\ y_1''(x) & y_2''(x) & y_3''(x) & & y_n''(x) \\ y_1''(x) & y_2''(x) & y_3''(x) & & y_n''(x) \\ \vdots & & & \ddots & \vdots \\ y_1^{(n-1)}(x) & y_2^{(n-1)}(x) & y_3^{(n-1)}(x) & \cdots & y_n^{(n-1)}(x) \end{vmatrix} + \cdots \\ &+ \begin{vmatrix} y_1(x) & y_2(x) & y_3(x) & \cdots & y_n(x) \\ y_1'(x) & y_2'(x) & y_3'(x) & & y_n'(x) \\ y_1''(x) & y_2''(x) & y_3''(x) & & y_n''(x) \\ \vdots & & & \ddots & \vdots \\ y_1^{(n)}(x) & y_2^{(n)}(x) & y_3^{(n)}(x) & \cdots & y_n^{(n)}(x) \end{vmatrix} \end{aligned}$$

All the determinants vanish as there are two identical rows except the last one, which results in

$$\frac{dW}{dx} = \begin{vmatrix} y_1(x) & y_2(x) & y_3(x) & \cdots & y_n(x) \\ y_1'(x) & y_2'(x) & y_3'(x) & & y_n'(x) \\ y_1''(x) & y_2''(x) & y_3''(x) & & y_n''(x) \\ \vdots & & & \ddots & \vdots \\ y_1^{(n)}(x) & y_2^{(n)}(x) & y_3^{(n)}(x) & \cdots & y_n^{(n)}(x) \end{vmatrix}$$

Now, substitute the original ODE $y^{(n)} = -(a_0(x)y + a_1(x)y' + \cdots + a_{n-1}(x)y^{(n-1)})$ into the last row of the determinant. Again, the lower-degree terms disappear due to duplication with the other rows, which leads to just

$$\begin{aligned} \frac{dW}{dx} &= \begin{vmatrix} y_1(x) & y_2(x) & y_3(x) & \cdots & y_n(x) \\ y_1'(x) & y_2'(x) & y_3'(x) & & y_n'(x) \\ y_1''(x) & y_2''(x) & y_3''(x) & & y_n''(x) \\ \vdots & & & \ddots & \vdots \\ -a_{n-1}y_1^{(n-1)}(x) & -a_{n-1}y_2^{(n-1)}(x) & -a_{n-1}y_3^{(n-1)}(x) & \cdots & -a_{n-1}y_n^{(n-1)}(x) \end{vmatrix} \\ &= -a_{n-1} \begin{vmatrix} y_1(x) & y_2(x) & y_3(x) & \cdots & y_n(x) \\ y_1'(x) & y_2'(x) & y_3'(x) & & y_n'(x) \\ y_1''(x) & y_2''(x) & y_3''(x) & & y_n''(x) \\ \vdots & & & \ddots & \vdots \\ y_1^{(n-1)}(x) & y_2^{(n-1)}(x) & y_3^{(n-1)}(x) & \cdots & y_n^{(n-1)}(x) \end{vmatrix} \\ &= -a_{n-1}W \end{aligned}$$

(3.22) This means that

$$\begin{aligned} &\begin{cases} y_1'' + P(x)y_1' + Q(x)y_1 = 0 \\ y_2'' + P(x)y_2' + Q(x)y_2 = 0 \end{cases} \\ \Rightarrow &\begin{cases} P(x)y_1' + Q(x)y_1 = -y_1'' \\ P(x)y_2' + Q(x)y_2 = -y_2'' \end{cases} \end{aligned}$$

Hence, by Cramer's Rule, the desired coefficients are

$$\begin{aligned} P &= \frac{-y_1''y_2 + y_2''y_1}{y_1'y_2 - y_2'y_1} \\ Q &= \frac{-y_1'y_2'' + y_2'y_1''}{y_1'y_2 - y_2'y_1} \end{aligned}$$

Solutions for Chapter 4

(4.1)

By the substitution $z = \ln x$, it will be turned into

$$\begin{aligned}\frac{d^2y}{dz^2} + (3-1)\frac{dy}{dz} + 2y &= 0 \\ \frac{d^2y}{dz^2} + 2\frac{dy}{dz} + 2y &= 0\end{aligned}$$

The auxiliary equation is $r^2 + 2r + 2 = 0$ with roots $r_{\pm} = -1 \pm i$, and so the solution will be in the form of $y(z) = c_1 e^{-z} \cos(z) + c_2 e^{-z} \sin(z)$. After a reverse transformation, it looks like

$$y(x) = c_1 \frac{\cos(\ln x)}{x} + c_2 \frac{\sin(\ln x)}{x}$$

Substituting the I.C.s, it is clear that $c_1 = 1$. And the first derivative is

$$y'(x) = c_1 \frac{-\sin(\ln x) - \cos(\ln x)}{x^2} + c_2 \frac{\cos(\ln x) - \sin(\ln x)}{x^2}$$

so further we have $c_2 = 1$, and the solution should be $y = \frac{\cos(\ln x) + \sin(\ln x)}{x}$.

(4.2) Let $z = \ln x$, then the ODE is transformed into

$$\frac{d^2y}{dz^2} - 4\frac{dy}{dz} + 3y = 0$$

The auxiliary equation is $r^2 - 4r + 3 = 0$ with two roots of $r = 1, 3$ and the general solution is hence

$$\begin{aligned}y &= c_1 e^z + c_2 e^{3z} \\ &= c_1 e^{\ln x} + c_2 e^{3 \ln x} = c_1 x + c_2 x^3\end{aligned}$$

where we have put back $z = \ln x$. It is not hard to see that the I.C.s require that $c_1 = \frac{1}{2}$, $c_2 = \frac{1}{2}$, so the solution will be $y = \frac{1}{2}(x + x^3)$.

(4.3) Again, the change of variable $z = \ln x$ converts the ODE into

$$\frac{d^2y}{dz^2} - 2\frac{dy}{dz} + y = e^z$$

The auxiliary equation is $r^2 - 2r + 1 = (r - 1)^2 = 0$, so $r = 1$ is a repeated real root and the complementary solution is given by $y_c(z) = c_1 e^z + c_2 z e^z$. To deal with the source term, we use the method of undetermined coefficients, but notice that due to the multiplicity of e^z , we need to try $y_p = Az^2 e^z$. Substituting this then yields

$$\begin{aligned} (Az^2 e^z + 4Aze^z + 2Ae^z) - 2(Az^2 e^z + 2Aze^z) + Az^2 e^z &= e^z \\ 2Ae^z &= e^z \\ \therefore A &= \frac{1}{2} \end{aligned}$$

So the general solution will be

$$\begin{aligned} y &= c_1 e^z + c_2 z e^z + \frac{1}{2} z^2 e^z \\ &= c_1 x + c_2 x \ln x + \frac{1}{2} x (\ln x)^2 \end{aligned}$$

(4.4) Dividing by x gives

$$\frac{d^2y}{dx^2} + \frac{2}{x} \frac{dy}{dx} - y = \frac{e^x}{x}$$

Regarding the canonical form, we can identify with $a_1 = \frac{2}{x}$, $a_0 = -1$, and

$$\begin{aligned} u(x) &= \exp\left(-\frac{1}{2} \int a_1(x) dx\right) = \exp\left(-\frac{1}{2} \int \frac{2}{x} dx\right) \\ &= \exp\left(-\int \frac{1}{x} dx\right) = \exp(-\ln x) = \frac{1}{x} \end{aligned}$$

also

$$\begin{aligned} g(x) &= -1 - \frac{1}{4} \left(\frac{2}{x}\right)^2 - \frac{1}{2} \left(-\frac{2}{x^2}\right) = -1 \\ h(x) &= \left(\frac{e^x}{x}\right)(x) = e^x \end{aligned}$$

Therefore, the ODE now has a new form of

$$\frac{d^2v}{dx^2} - v = e^x$$

It is easy to see that the complementary solution consists of $v_1 = e^x$, $v_2 = e^{-x}$. The particular solution is also not hard to derive by the method of undetermined coefficients: $v_p = \frac{xe^x}{2}$. Therefore, the general solution will be

$$y = uv = \frac{1}{x} \left(c_1 e^x + c_2 e^{-x} + \frac{xe^x}{2} \right) = c_1 \frac{e^x}{x} + c_2 \frac{e^{-x}}{x} + \frac{e^x}{2}$$

(4.5) Normalizing the equation first:

$$\frac{d^2 y}{dx^2} + 2 \cot x \frac{dy}{dx} - y = 0$$

Then, $a_1 = 2 \cot x$, $a_0 = -1$, and

$$\begin{aligned} u(x) &= \exp\left(-\frac{1}{2} \int a_1(x) dx\right) = \exp\left(-\frac{1}{2} \int 2 \cot x dx\right) \\ &= \exp\left(-\int \cot x dx\right) = \exp\left(-\int \frac{\cos x}{\sin x} dx\right) \\ &= \exp(-\ln(\sin x)) = \frac{1}{\sin x} \end{aligned}$$

also

$$g(x) = -1 - \frac{1}{4}(2 \cot x)^2 - \frac{1}{2}(-2 \csc^2 x) = 0$$

and $h(x) = 0$ obviously. So the new ODE is just

$$\frac{d^2 v}{dx^2} = 0$$

which has a simple solution of $v = Ax + B$, and thus

$$y = uv = (\csc x)(Ax + B)$$

(4.6) Let $y = uv = xv$, then

$$\begin{aligned} y' &= (xv)' \\ &= xv' + v \\ y'' &= (xv' + v)' \\ &= xv'' + 2v' \end{aligned}$$

and the ODE becomes

$$(x^2 - 1)(xv'' + 2v') - x(xv' + v) + xv = \frac{1}{x}$$

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$$x(x^2 - 1)v'' + (x^2 - 2)v' = \frac{1}{x}$$

Take $w = v'$:

$$\begin{aligned} x(x^2 - 1)w' + (x^2 - 2)w &= \frac{1}{x} \\ w' + \frac{x^2 - 2}{x(x^2 - 1)}w &= \frac{1}{x^2(x^2 - 1)} \\ w' + \left(\frac{2}{x} - \frac{1}{2(x+1)} - \frac{1}{2(x-1)}\right)w &= -\frac{1}{x^2} - \frac{1}{2(x+1)} + \frac{1}{2(x-1)} \end{aligned}$$

by partial fractions. The integrating factor is then

$$\begin{aligned} \exp\left(\int\left(\frac{2}{x} - \frac{1}{2(x+1)} - \frac{1}{2(x-1)}\right)dx\right) &= \exp\left(2\ln x - \frac{1}{2}\ln(x+1) - \frac{1}{2}\ln(x-1)\right) \\ &= \frac{x^2}{\sqrt{(x+1)(x-1)}} = \frac{x^2}{\sqrt{x^2-1}} \end{aligned}$$

Putting this back into the ODE for w :

$$\begin{aligned} \left(\frac{x^2}{\sqrt{x^2-1}}w\right)' &= \frac{1}{x^2(x^2-1)}\left(\frac{x^2}{\sqrt{x^2-1}}\right) \\ \frac{x^2}{\sqrt{x^2-1}}w &= \int \frac{1}{(x^2-1)^{\frac{3}{2}}}dx \end{aligned}$$

The integral on the R.H.S. can be found by trigonometric substitution $x = \sec t$, $dx = \sec t \tan t dt$:

$$\begin{aligned} \int \frac{1}{(x^2-1)^{\frac{3}{2}}}dx &= \int \frac{1}{(\sec^2 t - 1)^{\frac{3}{2}}} \sec t \tan t dt \\ &= \int \frac{1}{(\tan^2 t)^{\frac{3}{2}}} \sec t \tan t dt \\ &= \int \sec t \cot^2 t dt \\ &= \int \csc t \cot t dt \\ &= -\csc t + B = -\csc(\sec^{-1} x) + B = -\frac{x}{\sqrt{x^2-1}} + B \end{aligned}$$

assuming $x > 1$. Hence the ODE is now solved as

$$\begin{aligned} \frac{x^2}{\sqrt{x^2-1}}w &= -\frac{x}{\sqrt{x^2-1}} + B \\ w &= -\frac{1}{x} + B\frac{\sqrt{x^2-1}}{x^2} \end{aligned}$$

and

$$\begin{aligned}
 v &= \int w dx \\
 &= \int \left(-\frac{1}{x} + B \frac{\sqrt{x^2 - 1}}{x^2} \right) dx \\
 &= -\ln x + B \int \frac{\sqrt{x^2 - 1}}{x^2} dx
 \end{aligned}$$

Again, we use a trigonometric substitution of $x = \sec t$, $dx = \sec t \tan t dt$ for the last integral:

$$\begin{aligned}
 \int \frac{\sqrt{x^2 - 1}}{x^2} dx &= \int \frac{\sqrt{\sec^2 t - 1}}{\sec^2 t} \sec t \tan t dt \\
 &= \int \frac{\tan t}{\sec t} \tan t dt \\
 &= \int \sin t \tan t dt \\
 &= - \int \tan t d(\cos t) \\
 &= - [\cos t \tan t] + \int \cos t d(\tan t) \quad (\text{Integrating by Parts}) \\
 &= -\sin t + \int \cos t \sec^2 t dt \\
 &= -\sin t + \int \frac{1}{\cos^2 t} d(\sin t) \\
 &= -\sin t + \int \frac{1}{1 - \sin^2 t} d(\sin t) \\
 &= -\sin t + \tanh^{-1}(\sin t) + A
 \end{aligned}$$

by noting that $\frac{d(\tanh^{-1} u)}{du} = \frac{1}{1-u^2}$. Recovering x then leads to

$$\int \frac{\sqrt{x^2 - 1}}{x^2} dx = -\frac{\sqrt{x^2 - 1}}{x} + \tanh^{-1}\left(\frac{\sqrt{x^2 - 1}}{x}\right) + A$$

and

$$\begin{aligned}
 v &= -\ln x + B \left(-\frac{\sqrt{x^2 - 1}}{x} + \tanh^{-1}\left(\frac{\sqrt{x^2 - 1}}{x}\right) + A \right) \\
 &= -\ln x + A + B \left(\tanh^{-1}\left(\frac{\sqrt{x^2 - 1}}{x}\right) - \frac{\sqrt{x^2 - 1}}{x} \right)
 \end{aligned}$$

(grouping $AB \rightarrow A$), and the final answer is

$$y = xv = -x \ln x + Ax + B \left(x \tanh^{-1}\left(\frac{\sqrt{x^2 - 1}}{x}\right) - \sqrt{x^2 - 1} \right)$$

(4.7) Again, it is not hard to see that $u(x) = x$ is a valid complementary solution. Similar to above, we let $y = xv$ and have

$$\begin{aligned}y' &= xv' + v \\y'' &= xv'' + 2v'\end{aligned}$$

The ODE then becomes

$$\begin{aligned}2(xv'' + 2v') - x(xv' + v) + xv &= 0 \\2xv'' + 4v' - x^2v' &= 0\end{aligned}$$

Denote $w = v'$, then

$$\begin{aligned}2xw' + (4 - x^2)w &= 0 \\w' + \frac{4 - x^2}{2x}w &= 0 \\w' &= \frac{x^2 - 4}{2x}w \\\int \frac{dw}{w} &= \int \frac{x^2 - 4}{2x}dx = \int \left(\frac{x}{2} - \frac{2}{x}\right)dx \\\ln w &= \frac{x^2}{4} - 2 \ln x + \tilde{B} \\w &= B \frac{e^{\frac{x^2}{4}}}{x^2} \qquad (B = e^{\tilde{B}})\end{aligned}$$

Note that

$$\begin{aligned}\int \frac{e^{\frac{x^2}{4}}}{x^2}dx &= -\int e^{\frac{x^2}{4}}d\left(\frac{1}{x}\right) \\&= -\frac{e^{\frac{x^2}{4}}}{x} + \int \frac{1}{x}d(e^{\frac{x^2}{4}}) \\&= -\frac{e^{\frac{x^2}{4}}}{x} + \int \frac{1}{2}e^{\frac{x^2}{4}}dx \\&= -\frac{e^{\frac{x^2}{4}}}{x} + \frac{\sqrt{\pi}}{2}\operatorname{erfi}\left(\frac{x}{2}\right) + A\end{aligned}$$

With suitable replacements for the integration constants, the desired answer will then be

$$\begin{aligned}y = xv &= x\left(B\left(\frac{\sqrt{\pi}}{2}\operatorname{erfi}\left(\frac{x}{2}\right) - \frac{e^{\frac{x^2}{4}}}{x}\right) + A\right) \\&= B\left(\frac{\sqrt{\pi}}{2}x\operatorname{erfi}\left(\frac{x}{2}\right) - e^{\frac{x^2}{4}}\right) + Ax\end{aligned}$$

(4.8) Plugging $p = \frac{dy}{dx}$ and hence $\frac{d^2y}{dx^2} = p \frac{dp}{dy}$ into the equation:

$$\begin{aligned} p - yp \frac{dp}{dy} &= 1 \\ y \frac{dp}{dy} &= \frac{p-1}{p} \\ \int \frac{p}{p-1} dp &= \int \left(1 + \frac{1}{p-1}\right) dp = \int \frac{1}{y} dy \\ p + \ln|p-1| &= \ln|y| + c_1 \\ (p-1)e^p &= Ay \end{aligned}$$

We then can differentiate both sides with respect to x to get

$$\begin{aligned} Ay' &= \frac{\partial}{\partial y'} ((y' - 1)e^{y'}) \frac{dy'}{dx} \\ Ay' &= y' e^{y'} \frac{dy'}{dx} \\ A &= e^{y'} \frac{dy'}{dx} \\ \int e^{y'} dy' &= \int A dx \\ e^{y'} &= Ax + B \\ y' &= \ln(Ax + B) \end{aligned}$$

Substitute this back into the equation relating $p = y'$ and y at the start then produces

$$\begin{aligned} (\ln(Ax + B) - 1)e^{\ln(Ax+B)} &= Ay \\ y &= (\ln(Ax + B) - 1) \left(x + \frac{B}{A}\right) \end{aligned}$$

Also, note that there is a singular solution that corresponds to $p - 1 = 0$, i.e. $y = x + C$.

(4.9) Again, letting $p = \frac{dy}{dx}$ and $\frac{d^2y}{dx^2} = p \frac{dp}{dy}$, it becomes

$$\begin{aligned} 2yp \frac{dp}{dy} + y - p^2 &= 0 \\ \frac{dp}{dy} - \frac{p}{2y} &= -\frac{1}{2p} \end{aligned}$$

This is a Bernoulli ODE with $n = -1$, and thus we shall further let $v = p^2$, $\frac{dv}{dy} = 2p \frac{dp}{dy}$:

$$\frac{1}{2p} \frac{dv}{dy} - \frac{p}{2y} = -\frac{1}{2p}$$

$$\begin{aligned}
\frac{dv}{dy} - \frac{p^2}{y} &= \frac{dv}{dy} - \frac{v}{y} = -1 \\
\frac{1}{y} \frac{dv}{dy} - \frac{v}{y^2} &= \frac{d}{dy} \left(\frac{v}{y} \right) = -\frac{1}{y} \\
\frac{v}{y} &= -\int \frac{1}{y} dy = -\ln |y| + A \\
v &= y(A - \ln |y|) \\
p &= \pm \sqrt{y(A - \ln |y|)}
\end{aligned}$$

Let $y = e^w$, then $p = \frac{dy}{dx} = \frac{dy}{dw} \frac{dw}{dx} = e^w \frac{dw}{dx}$, and

$$\begin{aligned}
e^w \frac{dw}{dx} &= \pm \sqrt{e^w (A - w)} \\
e^{\frac{w}{2}} \frac{dw}{dx} &= \pm \sqrt{A - w}
\end{aligned}$$

Further, let $u = \sqrt{A - w}$, so that $w = A - u^2$ then $\frac{dw}{dx} = \frac{dw}{du} \frac{du}{dx} = -2u \frac{du}{dx}$:

$$\begin{aligned}
-2ue^{\frac{A-u^2}{2}} \frac{du}{dx} &= \pm u \\
e^{\frac{-u^2}{2}} \frac{du}{dx} &= \mp \frac{1}{2} e^{-\frac{A}{2}} \\
\int e^{\frac{-u^2}{2}} du &= \mp \int \frac{1}{2} e^{-\frac{A}{2}} dx \\
\sqrt{\frac{\pi}{2}} \operatorname{erf}\left(\frac{u}{\sqrt{2}}\right) &= \mp \frac{e^{-\frac{A}{2}}}{2} x + B
\end{aligned}$$

Recovering y leads to (absorption into the integration constant) the final answer

$$\operatorname{erf}\left(\frac{\sqrt{A - \ln y}}{\sqrt{2}}\right) = \mp \frac{e^{-\frac{A}{2}}}{\sqrt{2\pi}} x + B$$

(4.10) Using the facts of $\frac{d}{dx} \left(y \frac{dy}{dx} \right) = \left(\frac{dy}{dx} \right)^2 + y \frac{d^2 y}{dx^2}$ and $y \frac{dy}{dx} = \frac{d}{dx} \left(\frac{1}{2} y^2 \right)$ allows us to write

$$\begin{aligned}
\frac{d}{dx} \left(y \frac{dy}{dx} \right) + \frac{d}{dx} \left(\frac{1}{2} y^2 \right) &= x \\
y \frac{dy}{dx} + \frac{1}{2} y^2 &= \int x dx = \frac{1}{2} x^2 + c_1
\end{aligned}$$

Then letting $w = y^2$ and using $y \frac{dy}{dx} = \frac{d}{dx} \left(\frac{1}{2} y^2 \right) = \frac{1}{2} \frac{dw}{dx}$ again produces

$$\frac{1}{2} \frac{dw}{dx} + \frac{1}{2} w = \frac{1}{2} x^2 + c_1$$

$$\begin{aligned}
\frac{dw}{dx} + w &= x^2 + 2c_1 \\
\frac{d}{dx}(e^x w) &= e^x \left(\frac{dw}{dx} + w \right) = x^2 e^x + 2c_1 e^x \\
e^x w &= \int x^2 e^x dx + \int 2c_1 e^x dx \\
e^x w &= e^x (x^2 - 2x + 2) + 2c_1 e^x + c_2 \\
w &= x^2 - 2x + 2 + 2c_1 + c_2 e^{-x} \\
y &= \pm \sqrt{2c_1 + c_2 e^{-x} + x^2 - 2x + 2}
\end{aligned}$$

(4.11) Using $\frac{d}{dx} \left[\left(\frac{dy}{dx} \right)^2 \right] = 2 \frac{dy}{dx} \frac{d^2 y}{dx^2}$ results in

$$\begin{aligned}
\frac{1}{2} \frac{d}{dx} \left(\frac{dy}{dx} \right)^2 + 2x &= 0 \\
\left(\frac{dy}{dx} \right)^2 &= - \int 4x dx = -2x^2 + c_1 \\
\frac{dy}{dx} &= \pm \sqrt{c_1 - 2x^2} \\
y &= \pm \int \sqrt{c_1 - 2x^2} dx
\end{aligned}$$

By trigonometric substitution, we can let $x = \frac{\sqrt{c_1}}{\sqrt{2}} \sin t$ and thus $dx = \frac{\sqrt{c_1}}{\sqrt{2}} \cos t dt$, then

$$\begin{aligned}
y &= \pm \int \sqrt{c_1} \cos t \frac{\sqrt{c_1}}{\sqrt{2}} \cos t dt \\
&= \pm \frac{c_1}{\sqrt{2}} \int \cos^2 t dt \\
&= \pm \frac{c_1}{\sqrt{2}} \int \frac{1 + \cos(2t)}{2} dt \\
&= \pm \frac{c_1}{2\sqrt{2}} \left(t + \frac{1}{2} \sin(2t) + c_2 \right) \\
&= \pm \frac{c_1}{2\sqrt{2}} \left(\sin^{-1} \left(\frac{\sqrt{2}x}{\sqrt{c_1}} \right) + \sin t \cos t + c_2 \right) \\
&= \pm \frac{c_1}{2\sqrt{2}} \left(\sin^{-1} \left(\frac{\sqrt{2}x}{\sqrt{c_1}} \right) + \frac{\sqrt{2}x}{\sqrt{c_1}} \sqrt{1 - \frac{2x^2}{c_1}} + c_2 \right)
\end{aligned}$$

(4.12) The form of the bracketed terms suggests that this equation may be isobaric with $m = 1$, which is indeed true because each additive term has dimensionality $m + 1, m + 1, 2m, m + 1, 2m$

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and are equal to 2 in this case. Now we let $y = zx$ and hence $\frac{dy}{dx} = z + x \frac{dz}{dx}$, $\frac{d^2y}{dx^2} = 2 \frac{dz}{dx} + x \frac{d^2z}{dx^2}$. Plugging these into the ODE gives

$$\begin{aligned} x^3 \left(2 \frac{dz}{dx} + x \frac{d^2z}{dx^2} \right) + (2x^2 + zx^2) \left(z + x \frac{dz}{dx} \right) - (2x^2z + x^2z^2) &= 0 \\ x^4 \frac{d^2z}{dx^2} + (4x^3 + zx^3) \frac{dz}{dx} &= 0 \\ x \frac{d^2z}{dx^2} + (4 + z) \frac{dz}{dx} &= 0 \end{aligned}$$

Now further let $x = e^t$, and

$$\begin{aligned} \frac{dz}{dx} &= \frac{dz}{dt} \frac{dt}{dx} \\ &= \frac{dz}{dt} \left(\frac{1}{\frac{dx}{dt}} \right) = e^{-t} \frac{dz}{dt} \\ \frac{d^2z}{dx^2} &= \frac{d}{dx} \left(\frac{dz}{dx} \right) = \frac{dt}{dx} \frac{d}{dt} \left(e^{-t} \frac{dz}{dt} \right) \\ &= e^{-t} \left(e^{-t} \frac{d^2z}{dt^2} - e^{-t} \frac{dz}{dt} \right) \end{aligned}$$

Substitution into the equation produces

$$\begin{aligned} e^t \left(e^{-t} \left(e^{-t} \frac{d^2z}{dt^2} - e^{-t} \frac{dz}{dt} \right) \right) + (4 + z) \left(e^{-t} \frac{dz}{dt} \right) &= 0 \\ \frac{d^2z}{dt^2} + (3 + z) \frac{dz}{dt} &= 0 \\ \frac{d}{dt} \left(\frac{dz}{dt} \right) + \frac{d}{dt} \left(\frac{1}{2} z^2 \right) + 3 \frac{dz}{dt} &= 0 \end{aligned} \tag{5}$$

by rewriting as differentials. Then

$$\begin{aligned} \frac{dz}{dt} + \frac{1}{2} z^2 + 3z &= c_1 \\ \int \frac{dz}{c_1 - 3z - \frac{1}{2} z^2} &= \int dt \\ \int \frac{dz}{(z + 3)^2 - 2c_1 - 9} &= - \int \frac{1}{2} dt \\ - \frac{1}{\sqrt{2c_1 + 9}} \tanh^{-1} \left(\frac{z + 3}{\sqrt{2c_1 + 9}} \right) &= - \frac{1}{2} t + c_2 \end{aligned}$$

A reverse transformation then finally leads to

$$\frac{z + 3}{\sqrt{2c_1 + 9}} = \tanh \left(\frac{1}{2} \sqrt{2c_1 + 9} t - c_2 \sqrt{2c_1 + 9} \right)$$

$$z = \sqrt{2c_1 + 9} \tanh\left(\frac{1}{2}\sqrt{2c_1 + 9}t - c_2\sqrt{2c_1 + 9}\right) - 3$$

$$y = d_1x \tanh\left(\frac{1}{2}d_1 \ln x - c_2d_1\right) - 3x$$

where we have replaced $\sqrt{2c_1 + 9}$ by d_1 .

(4.13) Integrating twice easily leads to

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{2}x^2 - x + A \\ y &= \frac{1}{6}x^3 - \frac{1}{2}x^2 + Ax + B\end{aligned}$$

The boundary conditions require that

$$\begin{cases} 0 = (0) + A \\ 3 = \frac{1}{6} - \frac{1}{2} + A + B \end{cases}$$

Thus $A = 0$, $B = \frac{10}{3}$, and the solution is $y = \frac{1}{6}x^3 - \frac{1}{2}x^2 + \frac{10}{3}$.

(4.14) The equation itself has two possible solutions: $\sin(\sqrt{\lambda}x)$ and $\cos(\sqrt{\lambda}x)$. The left B.C. $y'(0) = 0$ immediately means that $\sin(\sqrt{\lambda}x)$ is not allowed and only $\cos(\sqrt{\lambda}x)$ persists. The right B.C. $y'(\pi) = 0$ then implies that $\sqrt{\lambda}\pi = n\pi$ where n has to be a non-negative integer. So, the possible values of λ will be just n^2 , $n = 0, 1, 2, \dots$

(4.15) Following the hint, substituting the ansatzes leads to

$$\begin{aligned}fAe^{\alpha z} &= KB\alpha^2e^{\alpha z} \\ fA &= KB\alpha^2\end{aligned}$$

and

$$\begin{aligned}fBe^{\alpha z} &= -KA\alpha^2e^{\alpha z} \\ fB &= -KA\alpha^2\end{aligned}$$

Combining the two equations, we have

$$\begin{aligned}fA &= K\left(-\frac{KA\alpha^2}{f}\right)\alpha^2 \\ f^2 &= -K^2\alpha^4 \\ \alpha^2 &= \pm i\frac{f}{K} \\ \alpha &= \pm(1 \pm i)\sqrt{\frac{f}{2K}}\end{aligned}$$

Therefore,

$$u = u_g + Ae^{\pm(1\pm i)\sqrt{\frac{f}{2K}}z}$$

and hence according to the original equation

$$\begin{aligned} v &= -\frac{K}{f} \frac{\partial^2}{\partial z^2} (Ae^{\pm(1\pm i)\sqrt{\frac{f}{2K}}z}) \\ &= \mp Aie^{\pm(1\pm i)\sqrt{\frac{f}{2K}}z} \end{aligned}$$

Recovering the real part and matching the physical boundary conditions (so that $A = -u_g$ and we take the negative branch), the desired Ekman spiral solution will then be

$$\begin{aligned} u &= u_g - u_g e^{-\frac{z}{d}} \cos\left(\frac{z}{d}\right) = u_g \left(1 - e^{-\frac{z}{d}} \cos\left(\frac{z}{d}\right)\right) \\ v &= u_g e^{-\frac{z}{d}} \sin\left(\frac{z}{d}\right) \end{aligned}$$

where $d = \sqrt{\frac{2K}{f}}$.

Solutions for Chapter 5

(5.1) This ODE is easily seen to have a general solution of $y = ce^{\alpha x}$ obtained from a standard approach. Now, let's put the proposed series to get:

$$\begin{aligned}\sum_{n=1}^{\infty} na_n x^{n-1} &= \alpha \sum_{n=0}^{\infty} a_n x^n \\ \sum_{n=0}^{\infty} (n+1)a_{n+1} x^n &= \sum_{n=0}^{\infty} \alpha a_n x^n\end{aligned}$$

where we have replaced n by $n+1$ in the summation on the L.H.S., and by comparing the coefficients it is not hard to obtain the recurrence relation:

$$a_{n+1} = \frac{\alpha a_n}{n+1}$$

So

$$\begin{aligned}a_1 &= \alpha a_0 \\ a_2 &= \frac{\alpha a_1}{2} = \frac{\alpha^2 a_0}{2} \\ a_3 &= \frac{\alpha a_2}{3} = \frac{\alpha^3 a_0}{6}\end{aligned}$$

and so on. In general, it is not hard to deduce that $a_n = \frac{\alpha^n a_0}{n!}$, and thus the series solution is

$$\begin{aligned}y &= a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots \\ &= a_0 + a_0 \alpha x + \frac{a_0 \alpha^2}{2!} x^2 + \frac{a_0 \alpha^3}{3!} x^3 + \cdots \\ &= a_0 \left(1 + \alpha x + \frac{(\alpha x)^2}{2!} + \frac{(\alpha x)^3}{3!} + \cdots \right) \\ &= a_0 \sum_{n=0}^{\infty} \frac{(\alpha x)^n}{n!} = a_0 e^{\alpha x}\end{aligned}$$

as we recognize that $\sum_{n=0}^{\infty} \frac{(\alpha x)^n}{n!}$ is just the usual Taylor series of $e^{\alpha x}$ (which is analytic), and so the two answers match (only the integration constant is denoted differently).

(5.2) Substituting the series form, in addition to expanding $\frac{1}{1-x}$ into Taylor series, gives

$$\begin{aligned} \sum_{n=1}^{\infty} n a_n x^{n-1} + x \sum_{n=0}^{\infty} a_n x^n &= 1 + x + x^2 + \dots \\ \sum_{n=0}^{\infty} (n+1) a_{n+1} x^n + \sum_{n=0}^{\infty} a_n x^{n+1} &= \sum_{n=0}^{\infty} x^n \\ \sum_{n=0}^{\infty} (n+1) a_{n+1} x^n + \sum_{n=1}^{\infty} a_{n-1} x^n &= \sum_{n=0}^{\infty} x^n \end{aligned}$$

where we have replaced n by $n+1$ and $n-1$ respectively in the two summations on L.H.S. Then by comparing the coefficients, the recurrence relation reads

$$\begin{aligned} (n+1) a_{n+1} + a_{n-1} &= 1 \\ a_{n+1} &= \frac{1 - a_{n-1}}{n+1} \end{aligned}$$

For $n=0$, the second summation does not appear (for convenience we usually assume $a_{-1}=0$), so it simply reduces to $a_1=1$. We subsequently generate some of the remaining terms:

$$\begin{aligned} a_2 &= \frac{1-a_0}{2} = \frac{1}{2} - \frac{a_0}{2} \\ a_3 &= \frac{1-a_1}{3} = 0 \\ a_4 &= \frac{1-a_2}{4} = \frac{1}{4} \left(\frac{1}{2} + \frac{a_0}{2} \right) = \frac{1}{8} + \frac{a_0}{8} \\ a_5 &= \frac{1-a_3}{5} = \frac{1}{5} \\ a_6 &= \frac{1-a_4}{6} = \frac{1}{6} \left(\frac{7}{8} - \frac{a_0}{8} \right) = \frac{7}{48} - \frac{a_0}{48} \\ &\vdots \end{aligned}$$

and hence the series solution is

$$\begin{aligned} y &= a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + a_5 x^5 + a_6 x^6 + \dots \\ &= a_0 + x + \left(\frac{1}{2} - \frac{a_0}{2} \right) x^2 + 0 + \left(\frac{1}{8} + \frac{a_0}{8} \right) x^4 + \frac{1}{5} x^5 + \left(\frac{7}{48} - \frac{a_0}{48} \right) x^6 + \dots \\ &= a_0 \left(1 - \frac{1}{2} x^2 + \frac{1}{8} x^4 - \frac{1}{48} x^6 + \dots \right) + \left(x + \frac{1}{2} x^2 + \frac{1}{8} x^4 + \frac{1}{5} x^5 + \frac{7}{48} x^6 + \dots \right) \\ &= a_0 e^{-\frac{x^2}{2}} + \left(x + \frac{1}{2} x^2 + \frac{1}{8} x^4 + \frac{1}{5} x^5 + \frac{7}{48} x^6 + \dots \right) \end{aligned}$$

(5.3) Again, putting in the series form, we have

$$\sum_{n=1}^{\infty} n a_n x^{n-1} + (x+1) \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} \frac{(-x)^n}{n!}$$

$$\sum_{n=1}^{\infty} n a_n x^{n-1} + \sum_{n=0}^{\infty} a_n x^n + \sum_{n=0}^{\infty} a_n x^{n+1} = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} x^n$$

$$\sum_{n=0}^{\infty} (n+1) a_{n+1} x^n + \sum_{n=0}^{\infty} a_n x^n + \sum_{n=1}^{\infty} a_{n-1} x^n = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} x^n$$

Equating the coefficients of x^n , the recurrence relation is then

$$(n+1)a_{n+1} + a_n + a_{n-1} = \frac{(-1)^n}{n!}$$

$$a_{n+1} = \frac{\frac{(-1)^n}{n!} - a_n - a_{n-1}}{n+1}$$

We now compute the first few terms:

$$a_1 = 1 - a_0$$

$$a_2 = \frac{-1 - a_1 - a_0}{2} = \frac{-1 - (1 - a_0) - a_0}{2} = -1$$

$$a_3 = \frac{\frac{1}{2} - a_2 - a_1}{3} = \frac{\frac{1}{2} - (-1) - (1 - a_0)}{3} = \frac{1}{6} + \frac{a_0}{3}$$

$$a_4 = \frac{-\frac{1}{6} - a_3 - a_2}{4} = \frac{-\frac{1}{6} - (\frac{1}{6} + \frac{a_0}{3}) - (-1)}{4} = \frac{1}{6} - \frac{a_0}{12}$$

$$a_5 = \frac{\frac{1}{24} - a_4 - a_3}{5} = \frac{\frac{1}{24} - (\frac{1}{6} - \frac{a_0}{12}) - (\frac{1}{6} + \frac{a_0}{3})}{5} = -\frac{7}{120} - \frac{a_0}{20}$$

Therefore, the required series solution is

$$y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + a_5 x^5 + \dots$$

$$= a_0 + (1 - a_0)x + (-1)x^2 + \left(\frac{1}{6} + \frac{a_0}{3}\right)x^3 + \left(\frac{1}{6} - \frac{a_0}{12}\right)x^4 + \left(-\frac{7}{120} - \frac{a_0}{20}\right)x^5 + \dots$$

$$= a_0 \left(1 - x + \frac{x^3}{3} - \frac{x^4}{12} - \frac{x^5}{20} + \dots\right) + \left(x - x^2 + \frac{x^3}{6} + \frac{x^4}{6} - \frac{7x^5}{120} + \dots\right)$$

(5.4) Plugging in the series form:

$$\sum_{n=1}^{\infty} n a_n x^{n-1} + (x^2 + 1) \sum_{n=0}^{\infty} a_n x^n = x$$

$$\sum_{n=1}^{\infty} n a_n x^{n-1} + \sum_{n=0}^{\infty} a_n x^n + \sum_{n=0}^{\infty} a_n x^{n+2} = x$$

$$\sum_{n=0}^{\infty} (n+1) a_{n+1} x^n + \sum_{n=0}^{\infty} a_n x^n + \sum_{n=2}^{\infty} a_{n-2} x^n = x$$

The recurrence relation is then, for $n \neq 1$

$$(n+1)a_{n+1} + a_n + a_{n-2} = 0$$

$$a_{n+1} = -\frac{a_n + a_{n-2}}{n+1}$$

First, the case of $n = 0$ easily leads to $a_1 = -a_0$ where we may treat the negative index term as 0. For $n = 1$, the source term comes into play and we have

$$2a_2 + a_1 + a_{-1} = 1$$

$$a_2 = \frac{1 - a_1}{2} = \frac{1 + a_0}{2}$$

The next few terms are

$$a_3 = -\frac{a_2 + a_0}{3} = -\frac{\left(\frac{1}{2} + \frac{a_0}{2}\right) + a_0}{3} = -\frac{1}{6} - \frac{a_0}{2}$$

$$a_4 = -\frac{a_3 + a_1}{4} = -\frac{\left(-\frac{1}{6} - \frac{a_0}{2}\right) + (-a_0)}{4} = \frac{1}{24} + \frac{3}{8}a_0$$

$$a_5 = -\frac{a_4 + a_2}{5} = -\frac{\left(\frac{1}{24} + \frac{3}{8}a_0\right) + \left(\frac{1}{2} + \frac{a_0}{2}\right)}{5} = -\frac{13}{120} - \frac{7}{40}a_0$$

Therefore, the required solution can be written as

$$y = a_0 \left(1 - x + \frac{x^2}{2} - \frac{x^3}{2} + \frac{3x^4}{8} - \frac{7x^5}{40} + \cdots\right) + \left(\frac{x^2}{2} - \frac{x^3}{6} + \frac{x^4}{24} - \frac{13x^5}{120} + \cdots\right)$$

(5.5) Substituting the necessary series expressions into the equation gives

$$\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + 9 \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2} x^n + 9 \sum_{n=0}^{\infty} a_n x^n = 0$$

Comparing coefficients of same-degree terms leads to the recurrence relation of

$$(n+2)(n+1)a_{n+2} + 9a_n = 0$$

$$a_{n+2} = -\frac{9}{(n+2)(n+1)}a_n$$

and hence the patterns of

$$a_2 = -\frac{9}{(2)(1)}a_0 = -\frac{3^2}{2!}a_0$$

$$\begin{aligned}
a_3 &= -\frac{9}{(3)(2)}a_1 = -\frac{3^2}{3!}a_1 \\
a_4 &= -\frac{9}{(4)(3)}a_2 = \frac{9^2}{(4)(3)(2)(1)}a_0 = \frac{3^4}{4!}a_0 \\
a_5 &= -\frac{9}{(5)(4)}a_3 = \frac{9^2}{(5)(4)(3)(2)}a_1 = \frac{3^4}{5!}a_1
\end{aligned}$$

In general, for odd/even terms we have

$$a_{2k} = (-1)^k \frac{3^{2k}}{(2k)!}a_0 \quad a_{2k+1} = (-1)^k \frac{3^{2k}}{(2k+1)!}a_1$$

and the power series solution thus consists of

$$y = a_0 \sum_{k=0}^{\infty} (-1)^k \frac{3^{2k}}{(2k)!} x^{2k} + a_1 \sum_{k=0}^{\infty} (-1)^k \frac{3^{2k}}{(2k+1)!} x^{2k+1}$$

By rearranging, we recognize that the two sums are just $\cos(3x)$ and $\sin(3x)$ as follows.

$$\begin{aligned}
y &= a_0 \sum_{k=0}^{\infty} (-1)^k \frac{(3x)^{2k}}{(2k)!} + \frac{a_1}{3} \sum_{k=0}^{\infty} (-1)^k \frac{(3x)^{2k+1}}{(2k+1)!} \\
&= a_0 \cos(3x) + \frac{a_1}{3} \sin(3x)
\end{aligned}$$

(5.6) Plugging in the proposed series form into the ODE gives

$$\begin{aligned}
\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + x \sum_{n=1}^{\infty} n a_n x^{n-1} + x \sum_{n=0}^{\infty} a_n x^n &= 0 \\
\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + \sum_{n=1}^{\infty} n a_n x^n + \sum_{n=0}^{\infty} a_n x^{n+1} &= 0 \\
\sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2} x^n + \sum_{n=1}^{\infty} n a_n x^n + \sum_{n=1}^{\infty} a_{n-1} x^n &= 0
\end{aligned}$$

So, the recurrence relation is, by comparing the coefficients:

$$\begin{aligned}
(n+2)(n+1)a_{n+2} + n a_n + a_{n-1} &= 0 \\
a_{n+2} &= -\frac{n a_n + a_{n-1}}{(n+2)(n+1)}
\end{aligned}$$

Starting with some a_0 and a_1 , the next few terms are then computed as

$$a_2 = -\frac{(0)a_0}{(2)(1)} = 0$$

$$\begin{aligned}
a_3 &= -\frac{a_1 + a_0}{(3)(2)} = -\frac{a_0}{6} - \frac{a_1}{6} \\
a_4 &= -\frac{2a_2 + a_1}{(4)(3)} = -\frac{2(0) + a_1}{12} = -\frac{a_1}{12} \\
a_5 &= -\frac{3a_3 + a_2}{(5)(4)} = -\frac{3(-\frac{a_0}{6} - \frac{a_1}{6}) + (0)}{20} = \frac{a_0}{40} + \frac{a_1}{40} \\
a_6 &= -\frac{4a_4 + a_3}{(6)(5)} = -\frac{4(-\frac{a_1}{12}) + (-\frac{a_0}{6} - \frac{a_1}{6})}{30} = \frac{a_0}{180} + \frac{a_1}{60}
\end{aligned}$$

Therefore, the required series solution is

$$\begin{aligned}
y &= a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 + \cdots \\
&= a_0 + a_1x + (0) + \left(-\frac{a_0}{6} - \frac{a_1}{6}\right)x^3 + \left(-\frac{a_1}{12}\right)x^4 + \left(\frac{a_0}{40} + \frac{a_1}{40}\right)x^5 + \left(\frac{a_0}{180} + \frac{a_1}{60}\right)x^6 + \cdots \\
&= a_0\left(1 - \frac{x^3}{6} + \frac{x^5}{40} + \frac{x^6}{180} + \cdots\right) + a_1\left(x - \frac{x^3}{6} - \frac{x^4}{12} + \frac{x^5}{40} + \frac{x^6}{60} + \cdots\right)
\end{aligned}$$

(5.7) Similarly, the starting point is

$$\begin{aligned}
\sum_{n=2}^{\infty} n(n-1)a_nx^{n-2} + x \sum_{n=1}^{\infty} na_nx^{n-1} + x^2 \sum_{n=0}^{\infty} a_nx^n &= 0 \\
\sum_{n=2}^{\infty} n(n-1)a_nx^{n-2} + \sum_{n=1}^{\infty} na_nx^n + \sum_{n=0}^{\infty} a_nx^{n+2} &= 0 \\
\sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}x^n + \sum_{n=1}^{\infty} na_nx^n + \sum_{n=2}^{\infty} a_{n-2}x^n &= 0
\end{aligned}$$

The recurrence relation is then

$$\begin{aligned}
(n+2)(n+1)a_{n+2} + na_n + a_{n-2} &= 0 \\
a_{n+2} &= -\frac{na_n + a_{n-2}}{(n+2)(n+1)}
\end{aligned}$$

The three terms involved are either all even or all odd, and hence we can separate the odd/even indices. For the even part, we have

$$a_{2m+2} = -\frac{2ma_{2m} + a_{2m-2}}{(2m+2)(2m+1)}$$

and thus the first few terms are

$$\begin{aligned}
a_2 &= -\frac{(0)a_0}{(2)(1)} = 0 \\
a_4 &= -\frac{2a_2 + a_0}{(4)(3)} = -\frac{2(0) + a_0}{12} = -\frac{a_0}{12}
\end{aligned}$$

$$a_6 = -\frac{4a_4 + a_2}{(6)(5)} = -\frac{4\left(-\frac{a_0}{12}\right) + (0)}{30} = \frac{a_0}{90}$$

Meanwhile, for the odd part, we have

$$a_{2m+3} = -\frac{(2m+1)a_{2m+1} + a_{2m-1}}{(2m+3)(2m+2)}$$

and

$$\begin{aligned} a_3 &= -\frac{(1)a_1}{(3)(2)} = -\frac{a_1}{6} \\ a_5 &= -\frac{3a_3 + a_1}{(5)(4)} = -\frac{3\left(-\frac{a_1}{6}\right) + a_1}{20} = -\frac{a_1}{40} \\ a_7 &= -\frac{5a_5 + a_3}{(7)(6)} = -\frac{5\left(-\frac{a_1}{40}\right) + \left(-\frac{a_1}{6}\right)}{42} = \frac{a_1}{144} \end{aligned}$$

Therefore, the series solution is cleanly

$$y = a_0\left(1 - \frac{x^4}{12} + \frac{x^6}{90} + \cdots\right) + a_1\left(x - \frac{x^3}{6} - \frac{x^5}{40} + \frac{x^7}{144} + \cdots\right)$$

(5.8) Again, substituting the necessary series into the equation:

$$\begin{aligned} &\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} - x^2 \sum_{n=1}^{\infty} na_n x^{n-1} + (x^2 + 1) \sum_{n=0}^{\infty} a_n x^n = 0 \\ &\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} - \sum_{n=1}^{\infty} na_n x^{n+1} + \sum_{n=0}^{\infty} a_n x^n + \sum_{n=0}^{\infty} a_n x^{n+2} = 0 \\ &\sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2} x^n - \sum_{n=2}^{\infty} (n-1)a_{n-1} x^n + \sum_{n=0}^{\infty} a_n x^n + \sum_{n=2}^{\infty} a_{n-2} x^n = 0 \end{aligned}$$

So the recurrence relation is

$$\begin{aligned} (n+2)(n+1)a_{n+2} - (n-1)a_{n-1} + a_n + a_{n-2} &= 0 \\ a_{n+2} &= \frac{(n-1)a_{n-1} - a_n - a_{n-2}}{(n+2)(n+1)} \end{aligned}$$

We now proceed to compute the subsequent terms for fixed a_0, a_1 :

$$\begin{aligned} a_2 &= \frac{-a_0}{(2)(1)} = -\frac{a_0}{2} \\ a_3 &= \frac{-a_1}{(3)(2)} = -\frac{a_1}{6} \\ a_4 &= \frac{a_1 - a_2 - a_0}{(4)(3)} = \frac{a_1 - \left(-\frac{a_0}{2}\right) - a_0}{12} = -\frac{a_0}{24} + \frac{a_1}{12} \end{aligned}$$

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$$a_5 = \frac{2a_2 - a_3 - a_1}{(5)(4)} = \frac{2(-\frac{a_0}{2}) - (-\frac{a_1}{6}) - a_1}{20} = -\frac{a_0}{20} - \frac{a_1}{24}$$

$$a_6 = \frac{3a_3 - a_4 - a_2}{(6)(5)} = \frac{3(-\frac{a_1}{6}) - (-\frac{a_0}{24} + \frac{a_1}{12}) - (-\frac{a_0}{2})}{30} = \frac{13a_0}{720} - \frac{7a_1}{360}$$

Therefore, the series solution will be

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 + \dots$$

$$= a_0 + a_1x + \left(-\frac{a_0}{2}\right)x^2 + \left(-\frac{a_1}{6}\right)x^3 + \left(-\frac{a_0}{24} + \frac{a_1}{12}\right)x^4 + \left(-\frac{a_0}{20} - \frac{a_1}{24}\right)x^5 + \left(\frac{13a_0}{720} - \frac{7a_1}{360}\right)x^6 + \dots$$

$$= a_0\left(1 - \frac{x^2}{2} - \frac{x^4}{24} - \frac{x^5}{20} + \frac{13x^6}{720} + \dots\right) + a_1\left(x - \frac{x^3}{6} + \frac{x^4}{12} - \frac{x^5}{24} - \frac{7x^6}{360} + \dots\right)$$

To match the I.C.s, we simply take the second series, i.e. $a_0 = 0$, $a_1 = 1$.

(5.9) Again, put all the relevant expressions into the equation and rearrange the indices:

$$\sum_{n=2}^{\infty} n(n-1)a_nx^{n-2} + 2 \sum_{n=1}^{\infty} na_nx^{n-1} + \sum_{n=0}^{\infty} a_nx^n = 0$$

$$\sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}x^n + 2 \sum_{n=0}^{\infty} (n+1)a_{n+1}x^n + \sum_{n=0}^{\infty} a_nx^n = 0$$

The recurrence relation is

$$(n+2)(n+1)a_{n+2} + 2(n+1)a_{n+1} + a_n = 0$$

$$a_{n+2} = -\frac{2a_{n+1}}{n+2} - \frac{a_n}{(n+2)(n+1)}$$

As just suggested before, write $a_n = \frac{b_n}{n!}$ to further obtain

$$\frac{b_{n+2}}{(n+2)!} = -\frac{2b_{n+1}}{(n+2)(n+1)!} - \frac{b_n}{(n+2)(n+1)n!}$$

$$b_{n+2} = -2b_{n+1} - b_n$$

The corresponding matrix system is

$$\begin{bmatrix} b_{n+2} \\ b_{n+1} \end{bmatrix} = \begin{bmatrix} -2 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} b_{n+1} \\ b_n \end{bmatrix}$$

The matrix has a repeated (an algebraic multiplicity of 2) eigenvalue of $\lambda = -1$:

$$\begin{vmatrix} -2 - \lambda & -1 \\ 1 & -\lambda \end{vmatrix} = 0$$

$$-(-2 - \lambda)\lambda - (-1) = 0$$

$$\lambda^2 + 2\lambda + 1 = (\lambda + 1)^2 = 0$$

So b_n will contain a $(-1)^n$ term. Apart from this, we need a second series to complete the general solution. Again, with the benefit of hindsight, we may try $n(-1)^n$, inspired by the original repeated root case for the auxiliary equation. Let's see if these really work ($b_n = c_1(-1)^n + c_2n(-1)^n$):

$$\begin{aligned} \text{L.H.S.} &= b_{n+2} \\ &= c_1(-1)^{n+2} + c_2(n+2)(-1)^{n+2} \\ &= c_1(2-1)(-1)^{n+2} + c_2(2(n+1)-n)(-1)^{n+2} \\ &= -2c_1(-1)^{n+1} - c_1(-1)^n - 2c_2(n+1)(-1)^{n+1} - c_2n(-1)^n \\ &= -2[c_1(-1)^{n+1} + c_2(n+1)(-1)^{n+1}] - [c_1(-1)^n + c_2n(-1)^n] \\ &= -2b_{n+1} - b_n = \text{R.H.S.} \end{aligned}$$

This works nicely, hence we may have $a_n = c_1 \frac{(-1)^n}{n!} + c_2 \frac{n(-1)^n}{n!}$, and

$$\begin{aligned} y &= \sum_{n=0}^{\infty} \left(\frac{c_1(-1)^n}{n!} + \frac{c_2n(-1)^n}{n!} \right) x^n \\ &= \sum_{n=0}^{\infty} \frac{c_1(-x)^n}{n!} + \sum_{n=1}^{\infty} \frac{c_2n(-x)^n}{n!} \\ &= c_1 e^{-x} + c_2 \sum_{n=1}^{\infty} \frac{(-x)^n}{(n-1)!} \\ &= c_1 e^{-x} + c_2 \sum_{n=0}^{\infty} \frac{(-x)^{n+1}}{n!} \\ &= c_1 e^{-x} - c_2 x \sum_{n=0}^{\infty} \frac{(-x)^n}{n!} = c_1 e^{-x} - c_2 x e^{-x} \end{aligned}$$

where in the second line we remove the non-existent $n = 0$ term in the second summation, and in the fourth line we replace n by $n + 1$.

(5.10) Rewriting the equation as

$$\frac{d^2 y}{dx^2} + \frac{y}{4x^2} = 0$$

$g(x) = \frac{1}{4x^2} > 0$ obviously, but

$$\begin{aligned} \int_1^{\infty} \frac{1}{4x^2} dx &= \left[-\frac{1}{4x} \right]_1^{\infty} \\ &= 0 - \left(-\frac{1}{4(1)} \right) = \frac{1}{4} \end{aligned}$$

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does not tend to infinity. The solution of the Euler ODE can be found by doing the substitution $z = \ln x$ and the new auxiliary equation is $r^2 - r + \frac{1}{4} = 0$ which has a repeated root of $r = \frac{1}{2}$. So its general solution will be

$$y = c_1 \sqrt{x} + c_2 \sqrt{x} \ln x = \sqrt{x}(c_1 + c_2 \ln x)$$

Assume there is a zero x_0 such that $c_1 + c_2 \ln x_0 = 0$. Since $\ln x$ is always monotonically increasing, there will be no more zeros.

Solutions for Chapter 6

(6.1) The standardized equation is $y'' + \frac{3}{2x}y' - (\frac{1}{2x} + \frac{1}{2x^2})y = 0$ and the corresponding indicial equation for the regular singular point of $x_0 = 0$ is $m^2 + (\frac{3}{2} - 1)m - \frac{1}{2} = m^2 + \frac{1}{2}m - \frac{1}{2} = 0$ which has two roots $m_1 = -1, m_2 = \frac{1}{2}$. Since the two exponents do not differ by an integer, there will be two Frobenius solutions. Substituting the Frobenius form into the ODE leads to

$$\begin{aligned} 2x^2 \sum_{n=0}^{\infty} (m+n)(m+n-1)a_n x^{m+n-2} + 3x \sum_{n=0}^{\infty} (m+n)a_n x^{m+n-1} - (x+1) \sum_{n=0}^{\infty} a_n x^{m+n} &= 0 \\ \sum_{n=0}^{\infty} 2(m+n)(m+n-1)a_n x^{m+n} + \sum_{n=0}^{\infty} 3(m+n)a_n x^{m+n} - \sum_{n=0}^{\infty} a_n x^{m+n} - \sum_{n=0}^{\infty} a_n x^{m+n+1} &= 0 \\ \sum_{n=0}^{\infty} 2(m+n)(m+n-1)a_n x^{m+n} + \sum_{n=0}^{\infty} 3(m+n)a_n x^{m+n} - \sum_{n=0}^{\infty} a_n x^{m+n} - \sum_{n=1}^{\infty} a_{n-1} x^{m+n} &= 0 \end{aligned}$$

where we have shifted the index for the last summation by 1. Comparing the coefficients then leads to the recurrence relation (for $n \geq 1$, while $n = 0$ represents the indicial equation) of

$$\begin{aligned} 2(m+n)(m+n-1)a_n + 3(m+n)a_n - a_n - a_{n-1} &= 0 \\ \Rightarrow a_n &= \frac{a_{n-1}}{2(m+n)(m+n-1) + 3(m+n) - 1} \end{aligned}$$

For the case of $m_1 = -1$, the first few coefficients are generated as

$$\begin{aligned} a_1 &= \frac{a_0}{2(0)(-1) + 3(0) - 1} = -a_0 \\ a_2 &= \frac{a_1}{2(1)(0) + 3(1) - 1} = \frac{a_1}{2} = -\frac{a_0}{2} \\ a_3 &= \frac{a_2}{2(2)(1) + 3(2) - 1} = \frac{a_2}{9} = -\frac{a_0}{18} \\ &\vdots \end{aligned}$$

and similar for the case of $m_2 = \frac{1}{2}$:

$$a_1 = \frac{a_0}{2(\frac{3}{2})(\frac{1}{2}) + 3(\frac{3}{2}) - 1} = \frac{a_0}{5}$$

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$$\begin{aligned} a_2 &= \frac{a_1}{2(\frac{5}{2})(\frac{3}{2}) + 3(\frac{5}{2}) - 1} = \frac{a_1}{14} = \frac{a_0}{70} \\ a_3 &= \frac{a_2}{2(\frac{7}{2})(\frac{5}{2}) + 3(\frac{7}{2}) - 1} = \frac{a_2}{27} = \frac{a_0}{1890} \\ &\vdots \end{aligned}$$

and hence the two Frobenius solutions are $y_1 = \frac{1}{x}(1 - x - \frac{x^2}{2} - \frac{x^3}{18} - \dots)$ and $y_2 = \sqrt{x}(1 + \frac{x}{5} + \frac{x^2}{70} + \frac{x^3}{1890} + \dots)$.

(6.2) The normalized equation will be

$$y'' + y' + \left(\frac{1}{4x} + \frac{3}{16x^2}\right)y = 0$$

with an indicial equation of

$$\begin{aligned} m^2 + (0 - 1)m + \frac{3}{16} &= 0 \\ m^2 - m + \frac{3}{16} &= 0 \\ \left(m - \frac{1}{4}\right)\left(m - \frac{3}{4}\right) &= 0 \end{aligned}$$

and thus the exponents are $m_1 = \frac{1}{4}$, $m_2 = \frac{3}{4}$. Again, they do not differ by an integer and we may seek the two Frobenius solutions. Plugging in the necessary expressions into the ODE leads to

$$\begin{aligned} x^2 \sum_{n=0}^{\infty} (m+n)(m+n-1)a_n x^{m+n-2} + x^2 \sum_{n=0}^{\infty} (m+n)a_n x^{m+n-1} + \left(\frac{x}{4} + \frac{3}{16}\right) \sum_{n=0}^{\infty} a_n x^{m+n} &= 0 \\ \sum_{n=0}^{\infty} (m+n)(m+n-1)a_n x^{m+n} + \sum_{n=0}^{\infty} (m+n)a_n x^{m+n+1} &+ \frac{3}{16} \sum_{n=0}^{\infty} a_n x^{m+n} + \frac{1}{4} \sum_{n=0}^{\infty} a_n x^{m+n+1} = 0 \\ \sum_{n=0}^{\infty} (m+n)(m+n-1)a_n x^{m+n} + \sum_{n=1}^{\infty} (m+n-1)a_{n-1} x^{m+n} &+ \frac{3}{16} \sum_{n=0}^{\infty} a_n x^{m+n} + \frac{1}{4} \sum_{n=1}^{\infty} a_{n-1} x^{m+n} = 0 \end{aligned}$$

The recurrence relation is then

$$(m+n)(m+n-1)a_n + (m+n-1)a_{n-1} + \frac{3}{16}a_n + \frac{1}{4}a_{n-1} = 0$$

$$\begin{aligned}\Rightarrow a_n &= -\frac{(m+n-1+\frac{1}{4})a_{n-1}}{(m+n)(m+n-1)+\frac{3}{16}} = -\frac{(m+n-\frac{3}{4})a_{n-1}}{(m+n-\frac{1}{4})(m+n-\frac{3}{4})} \\ &= -\frac{a_{n-1}}{m+n-\frac{1}{4}}\end{aligned}$$

In the case of $m_1 = \frac{1}{4}$, it is particularly easy:

$$a_n = -\frac{a_{n-1}}{\frac{1}{4} + n - \frac{1}{4}} = -\frac{a_{n-1}}{n}$$

thus we can see that in general $a_n = a_0 \frac{(-1)^n}{n!}$. For the second exponent $m_2 = \frac{3}{4}$, it becomes

$$a_n = -\frac{a_{n-1}}{\frac{3}{4} + n - \frac{1}{4}} = -\frac{a_{n-1}}{n + \frac{1}{2}}$$

Recursively, we have something like

$$a_n = -\frac{a_{n-1}}{n + \frac{1}{2}} = \frac{a_{n-2}(-1)^2}{(n + \frac{1}{2})(n - \frac{1}{2})} = \frac{a_0(-1)^n}{(n + \frac{1}{2})(n - \frac{1}{2}) \cdots (\frac{3}{2})}$$

Therefore, the two Frobenius solutions will then be

$$\begin{aligned}y_1 &= x^{\frac{1}{4}} \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{n!} \\ &= x^{\frac{1}{4}} \sum_{n=0}^{\infty} \frac{(-x)^n}{n!} = x^{\frac{1}{4}} e^{-x}\end{aligned}$$

and

$$\begin{aligned}y_2 &= x^{\frac{3}{4}} \left(1 - \frac{x}{(\frac{3}{2})} + \frac{x^2}{(\frac{5}{2})(\frac{3}{2})} - \frac{x^3}{(\frac{7}{2})(\frac{5}{2})(\frac{3}{2})} + \cdots \right) \\ &= x^{\frac{3}{4}} \left(1 - \frac{2x}{3} + \frac{4x^2}{15} - \frac{8x^3}{105} + \cdots \right)\end{aligned}$$

(6.3) The normalized equation is

$$y'' - \left(\frac{3}{2x} + \frac{1}{2} \right) y' + \left(\frac{3}{2x^2} + \frac{1}{x} \right) y = 0$$

and the indicial equation is

$$\begin{aligned}m^2 + \left(-\frac{3}{2} - 1 \right) m + \frac{3}{2} &= 0 \\ m^2 - \frac{5}{2} m + \frac{3}{2} &= 0\end{aligned}$$

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$$(m-1)\left(m-\frac{3}{2}\right)=0$$

So the exponents are $m_1 = 1$, $m_2 = \frac{3}{2}$. Again, there can be two Frobenius solutions. Now let's substitute the needed expressions into the ODE:

$$\begin{aligned} & 2x^2 \sum_{n=0}^{\infty} (m+n)(m+n-1)a_n x^{m+n-2} - (3x+x^2) \sum_{n=0}^{\infty} (m+n)a_n x^{m+n-1} \\ & \quad + (3+2x) \sum_{n=0}^{\infty} a_n x^{m+n} = 0 \\ & 2 \sum_{n=0}^{\infty} (m+n)(m+n-1)a_n x^{m+n} - 3 \sum_{n=0}^{\infty} (m+n)a_n x^{m+n} - \sum_{n=0}^{\infty} (m+n)a_n x^{m+n+1} \\ & \quad + 3 \sum_{n=0}^{\infty} a_n x^{m+n} + 2 \sum_{n=0}^{\infty} a_n x^{m+n+1} = 0 \\ & 2 \sum_{n=0}^{\infty} (m+n)(m+n-1)a_n x^{m+n} - 3 \sum_{n=0}^{\infty} (m+n)a_n x^{m+n} - \sum_{n=1}^{\infty} (m+n-1)a_{n-1} x^{m+n} \\ & \quad + 3 \sum_{n=0}^{\infty} a_n x^{m+n} + 2 \sum_{n=1}^{\infty} a_{n-1} x^{m+n} = 0 \end{aligned}$$

The recurrence relation is then

$$\begin{aligned} & 2(m+n)(m+n-1)a_n - 3(m+n)a_n - (m+n-1)a_{n-1} + 3a_n + 2a_{n-1} = 0 \\ \Rightarrow a_n &= \frac{(m+n-1-2)a_{n-1}}{2(m+n)(m+n-1)-3(m+n)+3} = \frac{(m+n-3)a_{n-1}}{(m+n-1)(2m+2n-3)} \end{aligned}$$

For $m_1 = 1$, we have

$$\begin{aligned} a_1 &= \frac{(1+1-3)a_0}{(1+1-1)(2(1)+2(1)-3)} = -a_0 \\ a_2 &= \frac{(1+2-3)a_1}{(1+2-1)(2(1)+2(2)-3)} = 0 \end{aligned}$$

and $a_k = 0$ for $k > 2$ so on. Hence the corresponding Frobenius solution is just $y_1 = x(1-x)$.

For $m_2 = \frac{3}{2}$, the first few terms are

$$\begin{aligned} a_1 &= \frac{(\frac{3}{2}+1-3)a_0}{(\frac{3}{2}+1-1)(2(\frac{3}{2})+2(1)-3)} = -\frac{a_0}{6} \\ a_2 &= \frac{(\frac{3}{2}+2-3)a_1}{(\frac{3}{2}+2-1)(2(\frac{3}{2})+2(2)-3)} = \frac{1}{20}\left(-\frac{a_0}{6}\right) = -\frac{a_0}{120} \\ a_3 &= \frac{(\frac{3}{2}+3-3)a_2}{(\frac{3}{2}+3-1)(2(\frac{3}{2})+2(3)-3)} = \frac{1}{14}\left(-\frac{a_0}{120}\right) = -\frac{a_0}{1680} \end{aligned}$$

Therefore, the second Frobenius solution will be

$$y_2 = x^{\frac{3}{2}} \left(1 - \frac{x}{6} - \frac{x^2}{120} - \frac{x^3}{1680} - \cdots \right)$$

(6.4) Normalizing the ODE:

$$y'' - \left(\frac{2}{x} - 1 \right) y' + \left(\frac{2}{x^2} + \frac{1}{x} \right) y = 0$$

The indicial equation is

$$\begin{aligned} m^2 + (-2 - 1)m + 2 &= 0 \\ m^2 - 3m + 2 &= 0 \\ (m - 1)(m - 2) &= 0 \end{aligned}$$

So the exponents are $m_1 = 2, m_2 = 1$. This time they differ by an integer, and it can be shown that there can only be one Frobenius solution and the second solution will involve a logarithm. (We also assign m_1 to be the larger exponent here.) Putting the Frobenius series into the equation, we have

$$\begin{aligned} x^2 \sum_{n=0}^{\infty} (m+n)(m+n-1) a_n x^{m+n-2} - (2x - x^2) \sum_{n=0}^{\infty} (m+n) a_n x^{m+n-1} \\ + (2+x) \sum_{n=0}^{\infty} a_n x^{m+n} = 0 \\ \sum_{n=0}^{\infty} (m+n)(m+n-1) a_n x^{m+n} - 2 \sum_{n=0}^{\infty} (m+n) a_n x^{m+n} + \sum_{n=0}^{\infty} (m+n) a_n x^{m+n+1} \\ + 2 \sum_{n=0}^{\infty} a_n x^{m+n} + \sum_{n=0}^{\infty} a_n x^{m+n+1} = 0 \\ \sum_{n=0}^{\infty} (m+n)(m+n-1) a_n x^{m+n} - 2 \sum_{n=0}^{\infty} (m+n) a_n x^{m+n} + \sum_{n=1}^{\infty} (m+n-1) a_{n-1} x^{m+n} \\ + 2 \sum_{n=0}^{\infty} a_n x^{m+n} + \sum_{n=1}^{\infty} a_{n-1} x^{m+n} = 0 \end{aligned}$$

The recurrence relation will be

$$\begin{aligned} (m+n)(m+n-1) a_n - 2(m+n) a_n + (m+n-1) a_{n-1} + 2a_n + a_{n-1} &= 0 \\ \Rightarrow a_n = -\frac{(m+n-1+1) a_{n-1}}{(m+n)(m+n-1) - 2(m+n) + 2} &= -\frac{(m+n) a_{n-1}}{(m+n-1)(m+n-2)} \end{aligned}$$

For $m_1 = 2$, we have

$$\begin{aligned} a_1 &= -\frac{(2+1)a_0}{(2+1-1)(2+1-2)} = -\frac{3a_0}{2} \\ a_2 &= -\frac{(2+2)a_1}{(2+2-1)(2+2-2)} = -\frac{2}{3}\left(-\frac{3a_0}{2}\right) = a_0 \\ a_3 &= -\frac{(2+3)a_2}{(2+3-1)(2+3-2)} = -\frac{5a_0}{12} \\ a_4 &= -\frac{(2+4)a_3}{(2+4-1)(2+4-2)} = -\frac{3}{10}\left(-\frac{5a_0}{12}\right) = \frac{a_0}{8} \end{aligned}$$

Therefore, the sole Frobenius solution is

$$y_1 = x^2 \left(1 - \frac{3x}{2} + x^2 - \frac{5x^3}{12} + \frac{x^4}{8} - \dots \right)$$

The second solution then will be in the form of

$$\begin{aligned} y_2 &= y_1 \ln x + x^{m_2} \sum_{n=0}^{\infty} c_n x^n \\ &= \sum_{n=0}^{\infty} a_n x^{n+2} \ln x + \sum_{n=0}^{\infty} c_n x^{n+1} \end{aligned}$$

The relevant derivatives are

$$\begin{aligned} y_2' &= \sum_{n=0}^{\infty} (n+2)a_n x^{n+1} \ln x + \sum_{n=0}^{\infty} a_n x^{n+1} + \sum_{n=0}^{\infty} (n+1)c_n x^n \\ y_2'' &= \sum_{n=0}^{\infty} (n+2)(n+1)a_n x^n \ln x + \sum_{n=0}^{\infty} (2n+3)a_n x^n + \sum_{n=0}^{\infty} (n+1)nc_n x^{n-1} \end{aligned}$$

Substituting all the expressions above, we obtain

$$\begin{aligned} &x^2 \left(\sum_{n=0}^{\infty} (n+2)(n+1)a_n x^n \ln x + \sum_{n=0}^{\infty} (2n+3)a_n x^n + \sum_{n=0}^{\infty} (n+1)nc_n x^{n-1} \right) \\ &\quad - (2x - x^2) \left(\sum_{n=0}^{\infty} (n+2)a_n x^{n+1} \ln x + \sum_{n=0}^{\infty} a_n x^{n+1} + \sum_{n=0}^{\infty} (n+1)c_n x^n \right) \\ &\quad + (2+x) \left(\sum_{n=0}^{\infty} a_n x^{n+2} \ln x + \sum_{n=0}^{\infty} c_n x^{n+1} \right) = 0 \end{aligned}$$

It is not hard to see that all the logarithm terms satisfy the previous recurrence relation and will

cancel out, and what remains is

$$\begin{aligned}
& x^2 \left(\sum_{n=0}^{\infty} (2n+3)a_n x^n + \sum_{n=0}^{\infty} (n+1)nc_n x^{n-1} \right) \\
& - (2x - x^2) \left(\sum_{n=0}^{\infty} a_n x^{n+1} + \sum_{n=0}^{\infty} (n+1)c_n x^n \right) + (2+x) \sum_{n=0}^{\infty} c_n x^{n+1} = 0 \\
& \sum_{n=0}^{\infty} (2n+3)a_n x^{n+2} + \sum_{n=0}^{\infty} (n+1)nc_n x^{n+1} \\
& - 2 \sum_{n=0}^{\infty} a_n x^{n+2} - 2 \sum_{n=0}^{\infty} (n+1)c_n x^{n+1} + \sum_{n=0}^{\infty} a_n x^{n+3} + \sum_{n=0}^{\infty} (n+1)c_n x^{n+2} \\
& + 2 \sum_{n=0}^{\infty} c_n x^{n+1} + \sum_{n=0}^{\infty} c_n x^{n+2} = 0 \\
& \sum_{n=1}^{\infty} (2n+1)a_{n-1} x^{n+1} + \sum_{n=0}^{\infty} (n+1)nc_n x^{n+1} \\
& - 2 \sum_{n=1}^{\infty} a_{n-1} x^{n+1} - 2 \sum_{n=0}^{\infty} (n+1)c_n x^{n+1} + \sum_{n=2}^{\infty} a_{n-2} x^{n+1} + \sum_{n=1}^{\infty} nc_{n-1} x^{n+1} \\
& + 2 \sum_{n=0}^{\infty} c_n x^{n+1} + \sum_{n=1}^{\infty} c_{n-1} x^{n+1} = 0
\end{aligned}$$

The recurrence relation is now

$$\begin{aligned}
& (2n+1)a_{n-1} + (n+1)nc_n - 2a_{n-1} - 2(n+1)c_n + a_{n-2} + nc_{n-1} + 2c_n + c_{n-1} = 0 \\
\Rightarrow c_n &= -\frac{((2n+1)-2)a_{n-1} + a_{n-2} + (n+1)c_{n-1}}{(n+1)n - 2(n+1) + 2} = -\frac{(2n-1)a_{n-1} + a_{n-2} + (n+1)c_{n-1}}{n(n-1)}
\end{aligned}$$

Notice that for

$$c_1 = -\frac{a_0 + 2c_0}{(1)(0)}$$

the denominator is zero, which means that for the series to hold, the only possibility is that $c_0 = -\frac{a_0}{2}$. For convenience, we take $a_0 = 1$ and so $c_0 = -\frac{1}{2}$, in addition to $c_1 = 0$. Subsequently,

$$\begin{aligned}
c_2 &= -\frac{3a_1 + a_0 + 3c_1}{(2)(1)} = -\frac{3(-\frac{3}{2}) + 1}{2} = \frac{7}{4} \\
c_3 &= -\frac{5a_2 + a_1 + 4c_2}{(3)(2)} = -\frac{5(1) + (-\frac{3}{2}) + 4(\frac{7}{4})}{6} = -\frac{7}{4}
\end{aligned}$$

And therefore our desired second answer will be

$$y_2 = y_1 \ln x + x \left(-\frac{1}{2} + \frac{7x^2}{4} - \frac{7x^3}{4} + \dots \right)$$

(6.5) As before, we rewrite

$$y'' - \left(\frac{1}{x} + \frac{1}{4}\right)y' + \left(\frac{3}{4x^2} + \frac{1}{4}\right)y = 0$$

The indicial equation is

$$m^2 + (-1 - 1)m + \frac{3}{4} = 0$$

$$m^2 - 2m + \frac{3}{4} = 0$$

$$\left(m - \frac{1}{2}\right)\left(m - \frac{3}{2}\right) = 0$$

So the exponents are $m_1 = \frac{3}{2}$, $m_2 = \frac{1}{2}$. Again, we only have one Frobenius solution for the larger exponent. Substituting the required forms into the ODE to get

$$\begin{aligned} 4x^2 \sum_{n=0}^{\infty} (m+n)(m+n-1)a_n x^{m+n-2} - (4x + x^2) \sum_{n=0}^{\infty} (m+n)a_n x^{m+n-1} \\ + (3 + x^2) \sum_{n=0}^{\infty} a_n x^{m+n} = 0 \\ 4 \sum_{n=0}^{\infty} (m+n)(m+n-1)a_n x^{m+n} - 4 \sum_{n=0}^{\infty} (m+n)a_n x^{m+n} - \sum_{n=0}^{\infty} (m+n)a_n x^{m+n+1} \\ + 3 \sum_{n=0}^{\infty} a_n x^{m+n} + \sum_{n=0}^{\infty} a_n x^{m+n+2} = 0 \\ 4 \sum_{n=0}^{\infty} (m+n)(m+n-1)a_n x^{m+n} - 4 \sum_{n=0}^{\infty} (m+n)a_n x^{m+n} - \sum_{n=1}^{\infty} (m+n-1)a_{n-1} x^{m+n} \\ + 3 \sum_{n=0}^{\infty} a_n x^{m+n} + \sum_{n=2}^{\infty} a_{n-2} x^{m+n} = 0 \end{aligned}$$

The recurrence relation is

$$\begin{aligned} 4(m+n)(m+n-1)a_n - 4(m+n)a_n - (m+n-1)a_{n-1} + 3a_n + a_{n-2} &= 0 \\ \Rightarrow a_n &= \frac{(m+n-1)a_{n-1} - a_{n-2}}{(2m+2n-1)(2m+2n-3)} \end{aligned}$$

Using $m_1 = \frac{3}{2}$, the first few terms associated with the Frobenius solution are

$$\begin{aligned} a_1 &= \frac{\left(\frac{3}{2} + 1 - 1\right)a_0}{(3 + 2(1) - 1)(3 + 2(1) - 3)} = \frac{3a_0}{16} \\ a_2 &= \frac{\left(\frac{3}{2} + 2 - 1\right)a_1 - a_0}{(3 + 2(2) - 1)(3 + 2(2) - 3)} = \frac{\frac{5}{2}\left(\frac{3a_0}{16}\right) - a_0}{24} = -\frac{17a_0}{768} \end{aligned}$$

$$a_3 = \frac{(\frac{3}{2} + 3 - 1)a_2 - a_1}{(3 + 2(3) - 1)(3 + 2(3) - 3)} = \frac{\frac{7}{2}(-\frac{17a_0}{768}) - (\frac{3a_0}{16})}{48} = -\frac{407a_0}{73728}$$

Therefore, the Frobenius solution is

$$y_1 = x^{\frac{3}{2}} \left(1 + \frac{3x}{16} - \frac{17x^2}{768} - \frac{407x^3}{73728} - \dots \right)$$

Just like the previous question, we propose the second solution as

$$\begin{aligned} y_2 &= y_1 \ln x + x^{m_2} \sum_{n=0}^{\infty} c_n x^n \\ &= \sum_{n=0}^{\infty} a_n x^{n+\frac{3}{2}} \ln x + \sum_{n=0}^{\infty} c_n x^{n+\frac{1}{2}} \end{aligned}$$

The relevant derivatives are

$$\begin{aligned} y_2' &= \sum_{n=0}^{\infty} \left(n + \frac{3}{2} \right) a_n x^{n+\frac{1}{2}} \ln x + \sum_{n=0}^{\infty} a_n x^{n+\frac{1}{2}} + \sum_{n=0}^{\infty} \left(n + \frac{1}{2} \right) c_n x^{n-\frac{1}{2}} \\ y_2'' &= \sum_{n=0}^{\infty} \left(n + \frac{3}{2} \right) \left(n + \frac{1}{2} \right) a_n x^{n-\frac{1}{2}} \ln x + \sum_{n=0}^{\infty} (2n+2) a_n x^{n-\frac{1}{2}} + \sum_{n=0}^{\infty} \left(n + \frac{1}{2} \right) \left(n - \frac{1}{2} \right) c_n x^{n-\frac{3}{2}} \end{aligned}$$

Plugging in all these expressions (and we automatically remove the logarithmic terms due to the same reason as the last problem):

$$\begin{aligned} &4x^2 \left(\sum_{n=0}^{\infty} (2n+2) a_n x^{n-\frac{1}{2}} + \sum_{n=0}^{\infty} \left(n + \frac{1}{2} \right) \left(n - \frac{1}{2} \right) c_n x^{n-\frac{3}{2}} \right) \\ &- (4x + x^2) \left(\sum_{n=0}^{\infty} a_n x^{n+\frac{1}{2}} + \sum_{n=0}^{\infty} \left(n + \frac{1}{2} \right) c_n x^{n-\frac{1}{2}} \right) + (3 + x^2) \sum_{n=0}^{\infty} c_n x^{n+\frac{1}{2}} = 0 \\ &4 \sum_{n=0}^{\infty} (2n+2) a_n x^{n+\frac{3}{2}} + 4 \sum_{n=0}^{\infty} \left(n + \frac{1}{2} \right) \left(n - \frac{1}{2} \right) c_n x^{n+\frac{1}{2}} \\ &- 4 \sum_{n=0}^{\infty} a_n x^{n+\frac{3}{2}} - 4 \sum_{n=0}^{\infty} \left(n + \frac{1}{2} \right) c_n x^{n+\frac{1}{2}} - \sum_{n=0}^{\infty} a_n x^{n+\frac{5}{2}} - \sum_{n=0}^{\infty} \left(n + \frac{1}{2} \right) c_n x^{n+\frac{3}{2}} \\ &+ 3 \sum_{n=0}^{\infty} c_n x^{n+\frac{1}{2}} + \sum_{n=0}^{\infty} c_n x^{n+\frac{5}{2}} = 0 \end{aligned}$$

$$\begin{aligned}
& 4 \sum_{n=1}^{\infty} 2na_{n-1}x^{n+\frac{1}{2}} + 4 \sum_{n=0}^{\infty} \left(n + \frac{1}{2}\right) \left(n - \frac{1}{2}\right) c_n x^{n+\frac{1}{2}} \\
& - 4 \sum_{n=1}^{\infty} a_{n-1}x^{n+\frac{1}{2}} - 4 \sum_{n=0}^{\infty} \left(n + \frac{1}{2}\right) c_n x^{n+\frac{1}{2}} - \sum_{n=2}^{\infty} a_{n-2}x^{n+\frac{1}{2}} - \sum_{n=1}^{\infty} \left(n - \frac{1}{2}\right) c_{n-1}x^{n+\frac{1}{2}} \\
& + 3 \sum_{n=0}^{\infty} c_n x^{n+\frac{1}{2}} + \sum_{n=2}^{\infty} c_{n-2}x^{n+\frac{1}{2}} = 0
\end{aligned}$$

The recurrence relation is

$$\begin{aligned}
& 8na_{n-1} + 4\left(n + \frac{1}{2}\right)\left(n - \frac{1}{2}\right)c_n - 4a_{n-1} - 4\left(n + \frac{1}{2}\right)c_n \\
& - a_{n-2} - \left(n - \frac{1}{2}\right)c_{n-1} + 3c_n + c_{n-2} = 0 \\
\Rightarrow c_n = & -\frac{(8n-4)a_{n-1} - a_{n-2} - \left(n - \frac{1}{2}\right)c_{n-1} + c_{n-2}}{4n(n-1)}
\end{aligned}$$

Now we proceed to the computation. However, due to the same reason as in the last question, we need $c_1 = 0$, and

$$\begin{aligned}
(8-4)a_0 - \left(1 - \frac{1}{2}\right)c_0 &= 0 \\
c_0 &= 8a_0
\end{aligned}$$

So we may take $c_0 = 8$ while $a_0 = 1$. The next few terms are

$$\begin{aligned}
c_2 &= -\frac{(8(2)-4)a_1 - a_0 - \left(2 - \frac{1}{2}\right)c_1 + c_0}{4(2)(1)} = -\frac{12\left(\frac{3}{16}\right) - 1 - 0 + 8}{8} = -\frac{37}{32} \\
c_3 &= -\frac{(8(3)-4)a_2 - a_1 - \left(3 - \frac{1}{2}\right)c_2 + c_1}{4(3)(2)} = -\frac{20\left(-\frac{17}{768}\right) - \left(\frac{3}{16}\right) - \left(3 - \frac{1}{2}\right)\left(-\frac{37}{32}\right) + 0}{24} = -\frac{217}{2304}
\end{aligned}$$

So the desired second solution will be

$$y_2 = y_1 \ln x + x^{\frac{1}{2}} \left(8 - \frac{37x^2}{32} - \frac{217x^3}{2304} - \dots \right)$$

(6.6) Any ordinary point is kind of a regular singular point because if $P(x)$ and $Q(x)$ are analytic then $(x - x_0)P(x)$ and $(x - x_0)^2Q(x)$ will also be clearly analytic at $x = x_0$, and this will also be the case for this question. Note that $p_{-1} = 0 = q_{-2}$, and hence the indicial equation is $m^2 - m = m(m - 1) = 0$ with the roots $m_1 = 1, m_2 = 0$, differed by an integer. However, by previous knowledge, we know that such a constant-coefficient ODE always has two series solutions, which are a limiting case of Frobenius solutions. For example, if we are given $y'' + y = 0$, the two complementary solutions are clearly $y_1 = \sin x = x - \frac{x^3}{3!} + \dots$ and $y_2 = \cos x = 1 - \frac{x^2}{2!} + \dots$, which match the exponents 1, 0 respectively from the indicial equation.

(6.7) We can see that for this hypergeometric equation, $a = 1, b = -1$. However, $c = -2$ is an integer, so we have to refer to the special discussion in the main text. The two solutions constitute

$$\begin{aligned} y &= AF(1, -1, -2; x) + Bx^{1-(-2)}F(1 - (-2) + 1, -1 - (-2) + 1, 2 - (-2); x) \\ &= AF(1, -1, -2; x) + Bx^3F(4, 2, 4; x) \end{aligned}$$

A limiting process is needed for the first hypergeometric function $F(1, -1, -2; x)$:

$$\begin{aligned} F(1, -1, -2; x) &= \lim_{c \rightarrow -2} \left(\lim_{b \rightarrow -1} \frac{\Gamma(c)}{\Gamma(1)\Gamma(b)} \sum_{n=0}^{\infty} \frac{\Gamma(1+n)\Gamma(b+n)}{\Gamma(c+n)} \frac{x^n}{n!} \right) \\ &= \lim_{c \rightarrow -2} \left(\Gamma(c) \sum_{n=0}^{\infty} \frac{\Gamma(1+n)((-1)(0) \cdots (-1+n-1))}{\Gamma(c+n)} \frac{x^n}{n!} \right) \\ &= \lim_{c \rightarrow -2} \left(\Gamma(c) \left(\frac{\Gamma(1)}{\Gamma(c)} + \frac{\Gamma(2)(-1)}{\Gamma(c+1)}x + (0) \right) \right) \\ &= 1 + \frac{(1)(-1)}{(-2)}x = 1 + \frac{x}{2} \end{aligned}$$

On the other hand,

$$\begin{aligned} F(4, 2, 4; x) &= 1 + \sum_{n=1}^{\infty} \frac{4(4+1) \cdots (4+n-1)2(2+1) \cdots (2+n-1)}{n!4(4+1) \cdots (4+n-1)} x^n \\ &= 1 + \sum_{n=1}^{\infty} \frac{2(2+1) \cdots (n+1)}{n!} x^n \\ &= 1 + \sum_{n=1}^{\infty} \frac{(n+1)!}{n!} x^n = 1 + \sum_{n=1}^{\infty} (n+1)x^n \\ &= 1 + 2x + 3x^2 + \cdots = \frac{1}{(1-x)^2} \end{aligned}$$

So the desired solution will be

$$y = A \left(1 + \frac{x}{2} \right) + B \frac{x^3}{(1-x)^2}$$

(6.8) This time, $a = 5, b = 1, c = 4$, so the solution will be

$$\begin{aligned} y &= AF(5, 1, 4; x) + Bx^{1-4}F(5 - 4 + 1, 1 - 4 + 1, 2 - 4; x) \\ &= AF(5, 1, 4; x) + Bx^{-3}F(2, -2, -2; x) \end{aligned}$$

The first one reduces to

$$F(5, 1, 4; x) = 1 + \sum_{n=1}^{\infty} \frac{5(5+1) \cdots (5+n-1)1(1+1) \cdots (1+n-1)}{n!4(4+1) \cdots (4+n-1)} x^n$$

I

$$\begin{aligned}
&= 1 + \sum_{n=1}^{\infty} \frac{(n+4)n!}{(4)n!} x^n \\
&= 1 + \sum_{n=1}^{\infty} \left(\frac{n+1}{4} + \frac{3}{4} \right) x^n \\
&= \frac{1}{4(1-x)^2} + \frac{3}{4(1-x)} = \frac{4-3x}{4(1-x)^2}
\end{aligned}$$

The second one requires a limiting process:

$$\begin{aligned}
F(2, -2, -2; x) &= \lim_{c \rightarrow -2} \left(\lim_{b \rightarrow -2} \frac{\Gamma(c)}{\Gamma(2)\Gamma(b)} \sum_{n=0}^{\infty} \frac{\Gamma(2+n)\Gamma(b+n)}{\Gamma(c+n)} \frac{x^n}{n!} \right) \\
&= \lim_{c \rightarrow -2} \left(\Gamma(c) \sum_{n=0}^{\infty} \frac{\Gamma(2+n)((-2)(-1)(0) \cdots (-2+n-1))}{\Gamma(c+n)} \frac{x^n}{n!} \right) \\
&= \lim_{c \rightarrow -2} \left(\Gamma(c) \left(\frac{\Gamma(2)}{\Gamma(c)} + \frac{\Gamma(3)(-2)}{\Gamma(c+1)} x + \frac{\Gamma(4)(-2)(-1)}{\Gamma(c+2)} \frac{x^2}{2} + (0) \right) \right) \\
&= 1 + \frac{(2)(-2)}{(-2)} x + \frac{(6)(-2)(-1)}{(-2)(-1)2} x^2 = 1 + 2x + 3x^2
\end{aligned}$$

So the required solution near $x = 0$ is

$$y = A \frac{4-3x}{4(1-x)^2} + Bx^{-3}(1+2x+3x^2)$$

For the solution near $x = 1$, it should be

$$\begin{aligned}
y &= AF(5, 1, 5+1+1-4; 1-x) + B(1-x)^{4-5-1}F(4-1, 4-5, 4-5-1+1; 1-x) \\
&= AF(5, 1, 3; 1-x) + B(1-x)^{-2}F(3, -1, -1; 1-x)
\end{aligned}$$

For convenience, we write $1-x$ as t below. The first hypergeometric function equals

$$\begin{aligned}
F(5, 1, 3; t) &= 1 + \sum_{n=1}^{\infty} \frac{5(5+1) \cdots (5+n-1)1(1+1) \cdots (1+n-1)}{n!3(3+1) \cdots (3+n-1)} t^n \\
&= 1 + \sum_{n=1}^{\infty} \frac{(n+3)(n+4)n!}{(3)(4)n!} t^n = 1 + \sum_{n=1}^{\infty} \frac{n^2+7n+12}{12} t^n \\
&= 1 + \sum_{n=1}^{\infty} \left(\frac{(n+1)(n+2)}{12} + \frac{4(n+1)}{12} + \frac{1}{2} \right) t^n \\
&= \frac{1}{6(1-t)^3} + \frac{1}{3(1-t)^2} + \frac{1}{2(1-t)} = \frac{3t^2-8t+6}{6(1-t)^3}
\end{aligned}$$

Thus

$$F(5, 1, 3; 1-x) = \frac{3(1-x)^2-8(1-x)+6}{6x^3}$$

Meanwhile, the second one will be, by a limiting procedure again:

$$\begin{aligned}
 F(3, -1, -1; x) &= \lim_{c \rightarrow -1} \left(\lim_{b \rightarrow -1} \frac{\Gamma(c)}{\Gamma(3)\Gamma(b)} \sum_{n=0}^{\infty} \frac{\Gamma(3+n)\Gamma(b+n)}{\Gamma(c+n)} \frac{t^n}{n!} \right) \\
 &= \lim_{c \rightarrow -1} \left(\frac{\Gamma(c)}{2} \sum_{n=0}^{\infty} \frac{\Gamma(3+n)((-1)(0) \cdots (-1+n-1))}{\Gamma(c+n)} \frac{t^n}{n!} \right) \\
 &= \lim_{c \rightarrow -1} \left(\frac{\Gamma(c)}{2} \left(\frac{\Gamma(3)}{\Gamma(c)} + \frac{\Gamma(4)(-1)}{\Gamma(c+1)} t + (0) \right) \right) \\
 &= 1 + 3t
 \end{aligned}$$

Therefore, $F(3, -1, -1; 1-x) = 1 + 3(1-x)$, and the required final answer is

$$y = A \frac{3(1-x)^2 - 8(1-x) + 6}{6x^3} + B \frac{1 + 3(1-x)}{(1-x)^2}$$

However, notice that if we write out all the $(1-x)$ terms, the expression will be equivalent to the one near $x = 0$.

(6.9) This time, $a = 2$, $b = -2$, $c = \frac{3}{2}$, hence

$$\begin{aligned}
 y &= AF\left(2, -2, \frac{3}{2}; x\right) + Bx^{1-\frac{3}{2}}F\left(2 - \frac{3}{2} + 1, -2 - \frac{3}{2} + 1, 2 - \frac{3}{2}; x\right) \\
 &= AF\left(2, -2, \frac{3}{2}; x\right) + Bx^{-\frac{1}{2}}F\left(\frac{3}{2}, -\frac{5}{2}, \frac{1}{2}; x\right)
 \end{aligned}$$

The first term simplifies to

$$\begin{aligned}
 F\left(2, -2, \frac{3}{2}; x\right) &= 1 + \sum_{n=1}^{\infty} \frac{2(2+1) \cdots (2+n-1)(-2)(-2+1)(-2+2) \cdots (-2+n-1)}{n! \left(\frac{3}{2}\right) \left(\frac{3}{2}+1\right) \cdots \left(\frac{3}{2}+n-1\right)} x^n \\
 &= 1 + \frac{(2)(-2)}{1! \left(\frac{3}{2}\right)} x + \frac{(2)(3)(-2)(-1)}{2! \left(\frac{3}{2}\right) \left(\frac{5}{2}\right)} x^2 + (0) \\
 &= 1 - \frac{8x}{3} + \frac{24x^2}{15}
 \end{aligned}$$

Meanwhile, the second hypergeometric function can be written as

$$\begin{aligned}
 F\left(\frac{3}{2}, -\frac{5}{2}, \frac{1}{2}; x\right) &= \frac{\Gamma(\frac{1}{2})}{\Gamma(\frac{3}{2})\Gamma(-\frac{5}{2})} \sum_{n=0}^{\infty} \frac{\Gamma(\frac{3}{2}+n)\Gamma(-\frac{5}{2}+n)}{\Gamma(\frac{1}{2}+n)} \frac{x^n}{n!} \\
 &= \frac{\Gamma(\frac{1}{2})}{\Gamma(\frac{3}{2})\Gamma(-\frac{5}{2})} \sum_{n=0}^{\infty} \left(n + \frac{1}{2}\right) \Gamma\left(-\frac{5}{2}+n\right) \frac{x^n}{n!} \\
 &= \frac{\Gamma(\frac{1}{2})}{\Gamma(\frac{3}{2})\Gamma(-\frac{5}{2})} \sum_{n=0}^{\infty} \left(n + \frac{1}{2}\right) \frac{\pi}{\sin(\pi(-\frac{5}{2}+n))\Gamma(1-(-\frac{5}{2}+n))} \frac{x^n}{n!}
 \end{aligned}$$

(Euler's reflection formula)

$$\begin{aligned}
 &= \frac{\Gamma(\frac{1}{2})}{\Gamma(\frac{3}{2})\Gamma(-\frac{5}{2})\Gamma(\frac{7}{2})} \sum_{n=0}^{\infty} \left(n + \frac{1}{2}\right) \frac{\pi\Gamma(\frac{7}{2})}{(-1)^{n+1}n!\Gamma(\frac{7}{2}-n)} x^n \\
 &= 2 \sum_{n=0}^{\infty} \left(n + \frac{1}{2}\right) \frac{\Gamma(\frac{7}{2})}{n!\Gamma(\frac{7}{2}-n)} (-x)^n
 \end{aligned}$$

Note that differentiation yields

$$\begin{aligned}
 \sum_{n=0}^{\infty} \frac{\Gamma(\frac{7}{2})}{n!\Gamma(\frac{7}{2}-n)} (-x)^n &= (1-x)^{\frac{5}{2}} \\
 \sum_{n=0}^{\infty} \frac{-n\Gamma(\frac{7}{2})}{n!\Gamma(\frac{7}{2}-n)} (-x)^{n-1} &= -\frac{5}{2}(1-x)^{\frac{3}{2}} \\
 \sum_{n=0}^{\infty} \frac{(n+1)\Gamma(\frac{7}{2})}{(n+1)!\Gamma(\frac{7}{2}-(n+1))} (-x)^n &= \frac{5}{2}(1-x)^{\frac{3}{2}} \\
 \sum_{n=0}^{\infty} \frac{(\frac{5}{2}-n)\Gamma(\frac{7}{2})}{n!\Gamma(\frac{7}{2}-n)} (-x)^n &= \frac{5}{2}(1-x)^{\frac{3}{2}}
 \end{aligned}$$

Then, it only takes some attention to see that

$$\begin{aligned}
 F\left(\frac{3}{2}, -\frac{5}{2}, \frac{1}{2}; x\right) &= 2 \sum_{n=0}^{\infty} \left(n - \frac{5}{2} + 3\right) \frac{\Gamma(\frac{7}{2})}{n!\Gamma(\frac{7}{2}-n)} (-x)^n \\
 &= 2 \left(-\frac{5}{2}(1-x)^{\frac{3}{2}} + 3(1-x)^{\frac{5}{2}}\right) \\
 &= (1-6x)(1-x)^{\frac{3}{2}}
 \end{aligned}$$

Hence the desired solution near $x = 0$ is

$$y = A \left(1 - \frac{8x}{3} + \frac{24x^2}{15}\right) + B \frac{(1-6x)(1-x)^{\frac{3}{2}}}{\sqrt{x}}$$

Meanwhile, the solution form at $x = \infty$ is

$$\begin{aligned}
 y &= Ax^{-2}F\left(2, 2 - \frac{3}{2} + 1, 2 - (-2) + 1; \frac{1}{x}\right) + Bx^{-(2)}F\left(-2, -2 - \frac{3}{2} + 1, -2 - 2 + 1; \frac{1}{x}\right) \\
 &= Ax^{-2}F\left(2, \frac{3}{2}, 5; \frac{1}{x}\right) + Bx^2F\left(-2, -\frac{5}{2}, -3; \frac{1}{x}\right)
 \end{aligned}$$

Again, let's write $t = \frac{1}{x}$, the first hypergeometric function (omitting the cumbersome and painful steps) should yield:

$$F\left(2, \frac{3}{2}, 5; \frac{1}{x}\right) = \frac{16}{5}x^2 \left(24x^2\sqrt{1-\frac{1}{x}} - 24x^2 - 28x\sqrt{1-\frac{1}{x}}x + 40x + 4\sqrt{1-\frac{1}{x}} - 15\right)$$

The second one is not hard to get by the limiting process:

$$\begin{aligned}
 F\left(-2, -\frac{5}{2}, -3; t\right) &= \lim_{c \rightarrow -3} \left(\lim_{a \rightarrow -2} \frac{\Gamma(c)}{\Gamma(-\frac{5}{2})\Gamma(a)} \sum_{n=0}^{\infty} \frac{\Gamma(a+n)\Gamma(-\frac{5}{2}+n)}{\Gamma(c+n)} \frac{x^n}{n!} \right) \\
 &= \lim_{c \rightarrow -3} \left(\frac{\Gamma(c)}{\Gamma(-\frac{5}{2})} \sum_{n=0}^{\infty} \frac{(-2)(-1)(0) \cdots (-2+n-1)\Gamma(-\frac{5}{2}+n)}{\Gamma(c+n)} \frac{x^n}{n!} \right) \\
 &= \lim_{c \rightarrow -3} \frac{\Gamma(c)}{\Gamma(-\frac{5}{2})} \left(\frac{\Gamma(-\frac{5}{2})}{\Gamma(c)} + \frac{(-2)\Gamma(-\frac{3}{2})}{\Gamma(c+1)} t + \frac{(-2)(-1)\Gamma(-\frac{1}{2})}{\Gamma(c+2)} \frac{t^2}{2!} + (0) \right) \\
 &= 1 - \frac{5}{3}t + \frac{5}{8}t^2
 \end{aligned}$$

Hence the desired solution at $x = \infty$ is

$$y = \frac{16A}{5} \left(24x^2 \sqrt{1 - \frac{1}{x}} - 24x^2 - 28x \sqrt{1 - \frac{1}{x}} + 40x + 4 \sqrt{1 - \frac{1}{x}} - 15 \right) + B \left(x^2 - \frac{5}{3}x + \frac{5}{8} \right)$$

Just like the last question, the two solutions expanded near $x = 0$ and ∞ can be shown to be equivalent.

(6.10) Note that this is a hypergeometric equation in disguise, seen by making the substitution $x \rightarrow -z$:

$$\begin{aligned}
 z(1 + (-z))y'' - (2 + 3(-z))(-y') - y &= 0 \\
 z(1 - z)y'' + (2 - 3z)y' - y &= 0
 \end{aligned}$$

with $a = b = 1$, $c = 2$, and hence the two linearly independent solutions will consist of

$$\begin{aligned}
 y_1 &= F(1, 1, 2; z) \\
 y_2 &= z^{-1} F(1 - 2 + 1, 1 - 2 + 1, 2 - 2; z) \\
 &= z^{-1} F(0, 0, 0; z) = \frac{1}{z} = -\frac{1}{x}
 \end{aligned}$$

By a direct verification, we can see that $\frac{\ln(1+x)}{x} = -\frac{\ln(1-z)}{z}$ is a solution to the equation above (as we also substitute the corresponding derivatives):

$$\begin{aligned}
 &z(1 - z) \left(-\frac{2 \ln(1 - z)}{z^3} - \frac{1}{(1 - z)z^2} - \frac{1}{z^2(1 - z)} + \frac{1}{z(1 - z)^2} \right) \\
 &+ (2 - 3z) \left(\frac{\ln(1 - z)}{z^2} + \frac{1}{z(1 - z)} \right) - \left(-\frac{\ln(1 - z)}{z} \right) \\
 &= -\frac{(2 - 2z) \ln(1 - z)}{z^2} - \frac{2}{z} + \frac{1}{1 - z} + (2 - 3z) \frac{\ln(1 - z)}{z^2} + \frac{2}{z(1 - z)} - \frac{3}{1 - z} + \frac{z \ln(1 - z)}{z^2} \\
 &= -\frac{2}{z} - \frac{2}{1 - z} + \left(\frac{2}{z} + \frac{2}{1 - z} \right) = 0
 \end{aligned}$$

Therefore, $\frac{\ln(1+x)}{x} = -\frac{\ln(1-z)}{z}$, as a series solution, must coincide with the first solution $F(1, 1, 2; -x) = F(1, 1, 2; z)$ to the equation and we are done.

(6.11) Now $t = 0, 1$ correspond to $x = A, B$ which are expected to remain as the regular singular points. Applying the given transformation should produce

$$\begin{aligned} & ([A + t(B - A)] - A)([A + t(B - A)] - B) \frac{1}{(B - A)^2} y'' \\ & + (C + D[A + t(B - A)]) \frac{1}{B - A} y' + Ey = 0 \\ & \frac{t(B - A)(t - 1)(B - A)}{(B - A)^2} y'' + \left(\frac{C}{B - A} + D \left[\frac{A}{B - A} + t \right] \right) y' + Ey = 0 \\ & t(1 - t)y'' - \left(\frac{C}{B - A} + D \left[\frac{A}{B - A} + t \right] \right) y' - Ey = 0 \end{aligned}$$

This is clearly a hypergeometric equation defined by the relation $E = ab$, $-\frac{C+DA}{B-A} = c$, $D = a+b+1$ in which a, b, c can be inverted. This means that the original ODE may be solved in terms of the hypergeometric function at $x = A, B$ with the appropriate transformation. For the given equation, $A = 1, B = 2, C = 2, D = 0, E = -2$, the transformed version by letting $t = \frac{x-1}{2-1} = x - 1$ should be

$$\begin{aligned} t(1 - t)y'' - \left(\frac{2}{2 - 1} + (0) \right) y' + 2y &= 0 \\ t(1 - t)y'' - 2y' + 2y &= 0 \end{aligned}$$

This has $a = 1, b = -2, c = -2$ as the new parameters. So the solution will be in the form of

$$\begin{aligned} y &= AF(1, -2, -2; t) + Bt^{1-(-2)}F(1 - (-2) + 1, -2 - (-2) + 1, 2 - (-2); t) \\ &= AF(1, -2, -2; t) + Bt^3F(4, 1, 4; t) \end{aligned}$$

The two hypergeometric functions are not hard to compute:

$$\begin{aligned} F(1, -2, -2; t) &= \lim_{c \rightarrow -2} \left(\lim_{b \rightarrow -2} \frac{\Gamma(c)}{\Gamma(1)\Gamma(b)} \sum_{n=0}^{\infty} \frac{\Gamma(1+n)\Gamma(b+n)}{\Gamma(c+n)} \frac{t^n}{n!} \right) \\ &= \lim_{c \rightarrow -2} \left(\Gamma(c) \sum_{n=0}^{\infty} \frac{\Gamma(1+n)(-2)(-1)(0) \cdots t^n}{\Gamma(c+n)} \frac{1}{n!} \right) \\ &= \lim_{c \rightarrow -2} \Gamma(c) \left(\frac{1}{\Gamma(c)} + \frac{\Gamma(2)(-2)}{\Gamma(c+1)} t + \frac{\Gamma(3)(-2)(-1)}{\Gamma(c+2)} \frac{t^2}{2} + (0) \right) \\ &= 1 + t + t^2 \end{aligned}$$

and recalling the fact that the hypergeometric function reduces to the geometric series as $a = c$ are both positive:

$$F(4, 1, 4; t) = \frac{1}{1 - t}$$

Therefore, the desired final answer is

$$\begin{aligned} y &= A(1 + (x-1) + (x-1)^2) + B \frac{1}{1 - (x-1)} \\ &= A(x^2 - x + 1) + \frac{B}{2 - x} \end{aligned}$$

(6.12) Here, $a = 2$, $c = 3$, so the two solutions are

$$y = AM(2, 3; x) + Bx^{1-3}M(2 - 3 + 1, 2 - 3; x) = AM(2, 3; x) + Bx^{-2}M(0, -1; x)$$

The second one is immediate: $M(0, -1; x) = 1$, while the first one is

$$\begin{aligned} M(2, 3; x) &= 1 + \sum_{n=1}^{\infty} \frac{2(2+1) \cdots (2+n-1)}{n!3(3+1) \cdots (3+n-1)} x^n \\ &= 1 + \sum_{n=1}^{\infty} \frac{2}{n!(2+n)} x^n \\ &= 1 + \sum_{n=1}^{\infty} \left(\frac{1}{n!} - \frac{1}{(n-1)!(n+2)} \right) x^n \\ &= 1 + \sum_{n=1}^{\infty} \left(\frac{1}{n!} - \frac{n(n+1)}{(n+2)!} \right) x^n \end{aligned}$$

We consider

$$\begin{aligned} e^x &= 1 + \sum_{n=1}^{\infty} \frac{x^n}{n!} = 1 + \sum_{n=1}^{\infty} \frac{(n+1)(n+2)x^n}{(n+2)!} \\ \frac{e^x}{x} &= \frac{1}{x} + \sum_{n=1}^{\infty} \frac{x^{n-1}}{n!} = \frac{1}{x} + 1 + \sum_{n=1}^{\infty} \frac{x^n}{(n+1)!} = \frac{1}{x} + 1 + \sum_{n=1}^{\infty} \frac{(n+2)x^n}{(n+2)!} \end{aligned}$$

likewise, $\frac{e^x}{x^2} = \frac{1}{x^2} + \frac{1}{x} + \frac{1}{2} + \sum_{n=1}^{\infty} \frac{x^n}{(n+2)!}$, and notice that

$$\frac{n(n+1)}{(n+2)!} = \frac{(n+1)(n+2) - 2(n+2) + 2}{(n+2)!}$$

Therefore,

$$\begin{aligned} M(2, 3; x) &= 1 + \sum_{n=1}^{\infty} \left(\frac{1}{n!} - \frac{(n+1)(n+2) - 2(n+2) + 2}{(n+2)!} \right) x^n \\ &= e^x - \left[(e^x - 1) - 2 \left(\frac{e^x}{x} - \frac{1}{x} - 1 \right) + 2 \left(\frac{e^x}{x^2} - \frac{1}{x^2} - \frac{1}{x} - \frac{1}{2} \right) \right] \\ &= \frac{2(1 - e^x + xe^x)}{x^2} \end{aligned}$$

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and we arrive at the final answer:

$$y = A \frac{2(1 - e^x + xe^x)}{x^2} + \frac{B}{x^2}$$

(6.13) We first obtain the indicial equation, where particularly $p_{-1} = c$:

$$\begin{aligned} m^2 + (c - 1)m &= 0 \\ m(m - (1 - c)) &= 0 \end{aligned}$$

Hence, the roots/exponents will be $m_1 = 0, m_2 = 1 - c$. The first one is reduced to the usual series solution, and we can plug the series form into the equation to get

$$\begin{aligned} x \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + (c-x) \sum_{n=1}^{\infty} na_n x^{n-1} - a \sum_{n=0}^{\infty} a_n x^n &= 0 \\ \sum_{n=2}^{\infty} n(n-1)a_n x^{n-1} + c \sum_{n=1}^{\infty} na_n x^{n-1} - \sum_{n=1}^{\infty} na_n x^n - a \sum_{n=0}^{\infty} a_n x^n &= 0 \\ \sum_{n=1}^{\infty} (n+1)na_{n+1}x^n + c \sum_{n=0}^{\infty} (n+1)a_{n+1}x^n - \sum_{n=1}^{\infty} na_n x^n - a \sum_{n=0}^{\infty} a_n x^n &= 0 \end{aligned}$$

By comparing the coefficients, we obtain the recurrence relation when $n \geq 1$:

$$\begin{aligned} (n+1)na_{n+1} + c(n+1)a_{n+1} - na_n - aa_n &= 0 \\ a_{n+1} &= \frac{(a+n)a_n}{(c+n)(n+1)} \end{aligned}$$

So the general form of the terms is

$$\begin{aligned} a_n &= \frac{(a+n-1)}{(c+n-1)n} a_{n-1} \\ &= \frac{(a+n-1)(a+n-2)}{(c+n-1)(c+n-2)n(n-1)} a_{n-2} \\ &= \frac{(a+n-1)(a+n-2) \cdots a}{(c+n-1)(c+n-2) \cdots cn!} a_0 \end{aligned}$$

and the desired series is (absorbed the a_0 factor)

$$y_1 = 1 + \sum_{n=1}^{\infty} \frac{a(a+1) \cdots (a+n-1)}{c(c+1) \cdots (c+n-1)n!} x^n$$

which is consistent with the form of the confluent hypergeometric function given in the main text. The process for deriving y_2 is similar.

Solutions for Chapter 7

(7.1)

$$(a) \quad 3\left(\frac{5!}{s^6}\right) + \frac{1}{2}\left(\frac{3!}{s^4}\right) - \frac{5}{s + \frac{1}{2}} = \frac{360}{s^6} + \frac{3}{s^4} - \frac{10}{2s + 1};$$

(b) The original expression can be rewritten as $\frac{1}{2}(e^{3t} - e^{-3t}) \cos t$, and the required transform will be $\frac{1}{2}\left(\frac{s - 3}{(s - 3)^2 + 1} - \frac{s + 3}{(s + 3)^2 + 1}\right)$;

(c) The expression simply reduces to $\delta(t - 5)$ and thus its Laplace transform is just e^{-5s} .

(7.2)

(a) Rewrite the given expression into partial fractions: $\frac{1}{s^3 - 1} = \frac{1}{(s - 1)(s^2 + s + 1)} = \frac{1}{3}\left(\frac{1}{s - 1} - \frac{s + 2}{s^2 + s + 1}\right) = \frac{1}{3}\left(\frac{1}{s - 1} - \frac{(s + \frac{1}{2}) + \frac{3}{2}}{(s + \frac{1}{2})^2 + \frac{3}{4}}\right)$ and hence the inverse transform will be $\frac{1}{3}\left(e^t - e^{-\frac{t}{2}} \cos\left(\frac{\sqrt{3}t}{2}\right) - \sqrt{3}e^{-\frac{t}{2}} \sin\left(\frac{\sqrt{3}t}{2}\right)\right)$;

(b) Note that $\frac{e^{-s}}{s^2 - 4s + 5} = \frac{e^{-s}}{(s - 2)^2 + 1}$. Using the two shifting theorems, we have the inverse transform as $H(t - 1)e^{2(t-1)} \sin(t - 1)$;

(c) Note that $\frac{d}{ds}\left(\tan^{-1}\left(\frac{s}{3}\right)\right) = \frac{3}{s^2 + 9}$, so that following the hint, we have

$$\left[\tan^{-1}\left(\frac{s}{3}\right)\right]_{\infty}^s = -\int_s^{\infty} \frac{3}{s^2 + 9} ds = -\mathcal{L}\left[\frac{\sin(3t)}{t}\right]$$

hence the required inverse transform is $-\frac{\sin(3t)}{t}$.

(7.3) Following the same procedure, the inverse Laplace transform is

$$\int_0^t \cos \tau \cos(t - \tau) d\tau = \int_0^t \frac{1}{2} [\cos(\tau + (t - \tau)) + \cos(\tau - (t - \tau))] d\tau$$

$$\begin{aligned}
&= \int_0^t \frac{1}{2} [\cos t + \cos(2\tau - t)] d\tau \\
&= \frac{1}{2} \left[\tau \cos t + \frac{1}{2} \sin(2\tau - t) \right]_0^t \\
&= \frac{1}{2} (t \cos t + \sin t)
\end{aligned}$$

Therefore,

$$\begin{aligned}
\mathcal{L}\left[\frac{1}{2}(t \cos t + \sin t)\right] &= \frac{s^2}{(s^2 + 1)^2} \\
\mathcal{L}[t \cos t] &= \frac{2s^2}{(s^2 + 1)^2} - \frac{1}{s^2 + 1} = \frac{s^2 - 1}{(s^2 + 1)^2}
\end{aligned}$$

(7.4) Applying Laplace transform on both sides:

$$\begin{aligned}
[s^2 Y(s) - sy(0) - y'(0)] - 4[sY(s) - y(0)] + 5Y(s) &= \frac{4s}{s^2 + 1} \\
(s^2 - 4s + 5)Y(s) - \frac{1}{2}s + 2 &= \frac{4s}{s^2 + 1}
\end{aligned}$$

then

$$\begin{aligned}
Y(s) &= \frac{4s}{(s^2 + 1)(s^2 - 4s + 5)} + \frac{1}{2} \left(\frac{s}{s^2 - 4s + 5} \right) - \frac{2}{s^2 - 4s + 5} \\
&= \frac{s - 1}{2(s^2 + 1)} - \frac{s - 5}{2(s^2 - 4s + 5)} + \frac{1}{2} \left(\frac{s}{s^2 - 4s + 5} \right) - \frac{2}{s^2 - 4s + 5} \\
&= \frac{s}{2(s^2 + 1)} - \frac{1}{2(s^2 + 1)} + \frac{1}{2(s^2 - 4s + 5)} = \frac{s}{2(s^2 + 1)} - \frac{1}{2(s^2 + 1)} + \frac{1}{2((s - 2)^2 + 1)}
\end{aligned}$$

So $y = \frac{1}{2} \cos t - \frac{1}{2} \sin t + \frac{1}{2} e^{2t} \sin t$.

(7.5) Again, applying Laplace Transform on both sides and using previous results, we have

$$\begin{aligned}
[s^2 Y(s) - sy(0) - y'(0)] + 2[sY(s) - y(0)] + 2Y(s) &= \frac{1}{(s + 1)^2 + 1} \\
(s^2 + 2s + 2)Y(s) - (s + 3) &= \frac{1}{(s + 1)^2 + 1}
\end{aligned}$$

Rearranging to get

$$\begin{aligned}
((s + 1)^2 + 1)Y(s) &= \frac{1}{(s + 1)^2 + 1} + s + 3 \\
Y(s) &= \frac{1}{((s + 1)^2 + 1)^2} + \frac{s + 1}{(s + 1)^2 + 1} + \frac{2}{(s + 1)^2 + 1}
\end{aligned}$$

as partial fractions. The inverse Laplace transform of the first term can be found via the Convolution Theorem, by noting that $\frac{1}{((s+1)^2+1)^2} = \left(\frac{1}{(s+1)^2+1}\right)\left(\frac{1}{(s+1)^2+1}\right)$ and $\frac{1}{(s+1)^2+1}$ exactly comes from the Laplace transform of $e^{-t} \sin t$. Then,

$$\begin{aligned}\int_0^t e^{-\tau} \sin \tau e^{-(t-\tau)} \sin(t-\tau) d\tau &= \int_0^t e^{-t} \frac{1}{2} [\cos(2\tau - t) - \cos t] d\tau \\ &= \frac{1}{2} e^{-t} \left[\frac{1}{2} \sin(2\tau - t) - \tau \cos t \right]_0^t \\ &= \frac{1}{2} e^{-t} \sin t - \frac{1}{2} t e^{-t} \cos t\end{aligned}$$

Hence the required answer is

$$\begin{aligned}y &= \frac{1}{2} e^{-t} \sin t - \frac{1}{2} t e^{-t} \cos t + e^{-t} \cos t + 2e^{-t} \sin t \\ &= \frac{5}{2} e^{-t} \sin t + \left(1 - \frac{t}{2}\right) e^{-t} \cos t\end{aligned}$$

(7.6) Note that $f(t)$ can be written as $H(t-2) + 3H(t-5)$:

$$\begin{aligned}[s^2 Y(s) - sy(0) - y'(0)] + \frac{9}{4} Y(s) &= \mathcal{L}[H(t-2) + 3H(t-5)] \\ \left(s^2 + \frac{9}{4}\right) Y(s) - s &= \frac{e^{-2s}}{s} + \frac{3e^{-5s}}{s} \\ Y(s) &= \frac{s}{s^2 + \frac{9}{4}} + \frac{e^{-2s}}{s(s^2 + \frac{9}{4})} + \frac{3e^{-5s}}{s(s^2 + \frac{9}{4})}\end{aligned}$$

Since

$$\frac{1}{s(s^2 + \frac{9}{4})} = \frac{4}{9} \left(\frac{1}{s} - \frac{s}{s^2 + \frac{9}{4}} \right)$$

hence

$$Y(s) = \frac{s}{s^2 + \frac{9}{4}} + \frac{4e^{-2s}}{9} \left(\frac{1}{s} - \frac{s}{s^2 + \frac{9}{4}} \right) + \frac{4e^{-5s}}{3} \left(\frac{1}{s} - \frac{s}{s^2 + \frac{9}{4}} \right)$$

and thus

$$y = \cos\left(\frac{3}{2}t\right) + \frac{4}{9}H(t-2) \left[1 - \cos\left(\frac{3}{2}(t-2)\right) \right] + \frac{4}{3}H(t-5) \left[1 - \cos\left(\frac{3}{2}(t-5)\right) \right]$$

(7.7) By definition and according to the given instruction,

$$\mathcal{L}\left[t^{-\frac{1}{2}}\right](s) = \int_0^\infty e^{-st} t^{-\frac{1}{2}} dt$$

$$\begin{aligned}
&= 2 \int_0^\infty e^{-sz^2} dz & (z = \sqrt{t}, dz = t^{-\frac{1}{2}} dt) \\
&= \frac{2}{\sqrt{s}} \int_0^\infty e^{-z'^2} dz' & (z' = z\sqrt{s}) \\
&= \frac{2}{\sqrt{s}} \left(\frac{\sqrt{\pi}}{2} \right) = \sqrt{\frac{\pi}{s}}
\end{aligned}$$

and

$$\begin{aligned}
\mathcal{L}[t^{n-\frac{1}{2}}](s) &= (-1)^n \frac{d^n}{ds^n} (\mathcal{L}[t^{-1/2}](s)) \\
&= (-1)^n \frac{d^n}{ds^n} \sqrt{\frac{\pi}{s}} \\
&= \frac{\sqrt{\pi}(2n-1)!!}{2^n} s^{-\frac{1}{2}-n}
\end{aligned}$$

where $(2n-1)!! = (2n-1)(2n-3)(2n-5) \cdots (3)(1)$.

(7.8) To transform the I.C. into a homogeneous one, the simplest way is through subtracting a suitable polynomial. Note that if $p(x) = x$, then $p(0) = 0$, $p'(0) = 1$, so we can let $w = y - p(x) = y - x$ to be the new dependent variable, and now $w(0) = w'(0) = 0$. The ODE then becomes

$$\begin{aligned}
\frac{d^2(w+x)}{dx^2} - (w+x) &= f(x) \\
\frac{d^2w}{dx^2} - w &= f(x) + x
\end{aligned}$$

so the form of the ODE is the same except that the source term now contains an extra x . Its Green's function is then found by considering $\delta(x-z)$ and applying the technique of Laplace Transform:

$$\begin{aligned}
\frac{d^2}{dx^2} [G(x, z)] - G(x, z) &= \delta(x-z) \\
s^2 \mathcal{L}[G](s, z) - sG(0, z) - G'(0, z) - \mathcal{L}[G](s, z) &= e^{-sz} \\
(s^2 - 1) \mathcal{L}[G](s, z) &= e^{-sz} \\
\mathcal{L}[G](s, z) &= \frac{e^{-sz}}{s^2 - 1}
\end{aligned}$$

Therefore,

$$G(x, z) = H(x-z) \sinh(x-z) = \begin{cases} 0 & \text{when } x < z \\ \sinh(x-z) & \text{when } x > z \end{cases}$$

and the particular solution should be

$$w = \int_0^\infty G(x, z)(f(z) + z) dz$$

$$\begin{aligned}
&= \int_0^x \sinh(x-z)(f(z)+z)dz \\
\Rightarrow y &= \int_0^x \sinh(x-z)(f(z)+z)dz + x
\end{aligned}$$

(7.9) Applying Laplace transform on both sides and using the Convolution Theorem, we have

$$\begin{aligned}
(s^2 Y(s) - sy(0) - y'(0)) + Y(s) &= \frac{1}{s} + Y(s) \left(\frac{1}{s^2 + 1} \right) \\
(s^2 + 1)Y(s) &= \frac{1}{s} + \frac{Y(s)}{s^2 + 1} \\
((s^2 + 1)^2 - 1)Y(s) &= \frac{s^2 + 1}{s} \\
Y(s) &= \frac{s^2 + 1}{s((s^2 + 1)^2 - 1)} = \frac{s^2 + 1}{s^3(s^2 + 2)} \\
&= \frac{1}{2s^3} + \frac{1}{4s} - \frac{s}{4(2 + s^2)}
\end{aligned}$$

Therefore, the recovered solution is

$$y = \frac{t^2}{4} + \frac{1}{4} - \frac{1}{4} \cos(\sqrt{2}t)$$

(7.10)

$$\begin{aligned}
\mathcal{L}[f(t)](s) &= \int_0^\infty e^{-st} f(t) dt \\
&= \int_0^T e^{-st} f(t) dt + \int_T^{2T} e^{-st} f(t) dt + \cdots + \int_{nT}^{(n+1)T} e^{-st} f(t) dt + \cdots \\
&= \sum_{n=0}^\infty \int_{nT}^{(n+1)T} e^{-st} f(t) dt
\end{aligned}$$

Let $t = t' + nT$ and exploiting the periodicity of $f(t)$

$$\begin{aligned}
\int_{nT}^{(n+1)T} e^{-st} f(t) dt &= \int_0^T e^{-s(t'+nT)} f(t' + nT) dt' \\
&= e^{-nsT} \int_0^T e^{-st'} f(t') dt' \\
&= e^{-nsT} \int_0^T e^{-st} f(t) dt
\end{aligned}$$

thus

$$\mathcal{L}[f(t)](s) = \sum_{n=0}^\infty \int_{nT}^{(n+1)T} e^{-st} f(t) dt$$

$$\begin{aligned}
&= \sum_{n=0}^{\infty} \left(e^{-nsT} \int_0^T e^{-st} f(t) dt \right) \\
&= \left(\sum_{n=0}^{\infty} e^{-nsT} \right) \int_0^T e^{-st} f(t) dt
\end{aligned}$$

and

$$\begin{aligned}
\sum_{n=0}^{\infty} e^{-nsT} &= \frac{1}{1 - e^{-sT}} \quad \text{as } |e^{-nsT}| < 1 \\
\therefore \mathcal{L}[f(t)](s) &= \left(\sum_{n=0}^{\infty} e^{-nsT} \right) \int_0^T e^{-st} f(t) dt \\
&= \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt
\end{aligned}$$

Now, as

$$\begin{aligned}
\int_0^2 f(t) e^{-st} dt &= \int_0^1 f(t) e^{-st} dt + \int_1^2 f(t) e^{-st} dt \\
&= 0 + \int_1^2 e^{-st} dt \\
&= \left[-\frac{e^{-st}}{s} \right]_1^2 \\
&= \frac{e^{-s} - e^{-2s}}{s}
\end{aligned}$$

Using all the previous results, we then have

$$\begin{aligned}
\mathcal{L}[f(t)](s) &= \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt \\
&= \frac{1}{1 - e^{-2s}} \int_0^2 e^{-st} f(t) dt \\
&= \frac{1}{1 - e^{-2s}} \left(\frac{e^{-s} - e^{-2s}}{s} \right) = \frac{e^{-s} - e^{-2s}}{s(1 - e^{-2s})}
\end{aligned}$$

Finally, as $\sin t$ has a period of 2π :

$$\begin{aligned}
\mathcal{L}[\sin t](s) &= \frac{1}{s^2 + 1} = \frac{1}{1 - e^{-2\pi s}} \left(\int_0^{2\pi} e^{-st} \sin t dt \right) \\
\int_0^{2\pi} e^{-st} \sin t dt &= \frac{1 - e^{-2\pi s}}{s^2 + 1}
\end{aligned}$$

(7.11) First, the source term on the R.H.S. can be described by

$$F(x) = (H(kx - \pi) - 1) \sin(kx)$$

$$\begin{aligned}
&= H(kx - \pi) \sin(kx - \pi + \pi) - \sin(kx) \\
&= -H(kx - \pi) \sin(kx - \pi) - \sin(kx)
\end{aligned}$$

Applying Laplace Transform on both sides, we have

$$\begin{aligned}
s^2 Y - y'(0) - a^2 Y &= -e^{-\frac{\pi s}{k}} \frac{k}{s^2 + k^2} - \frac{k}{s^2 + k^2} \\
(s^2 - a^2) Y - y'(0) &= -\frac{k}{s^2 + k^2} (1 + e^{-\frac{\pi s}{k}}) \\
Y &= -\frac{k}{(s^2 + k^2)(s^2 - a^2)} (1 + e^{-\frac{\pi s}{k}}) + \frac{y'(0)}{s^2 - a^2}
\end{aligned}$$

where particularly

$$\mathcal{L}[H(kx - \pi) \sin(kx - \pi)] = e^{-\frac{\pi s}{k}} \frac{k}{s^2 + k^2}$$

By partial fractions,

$$-\frac{k}{(s^2 + k^2)(s^2 - a^2)} = \frac{k}{a^2 + k^2} \left(\frac{1}{s^2 + k^2} - \frac{1}{s^2 - a^2} \right)$$

The inverse Laplace Transform of

$$Y = \frac{k}{a^2 + k^2} \left(\frac{1}{s^2 + k^2} - \frac{1}{s^2 - a^2} \right) (1 + e^{-\frac{\pi s}{k}}) + \frac{y'(0)}{s^2 - a^2}$$

is

$$\begin{aligned}
&\frac{k}{a^2 + k^2} \left(\frac{\sin(kx)}{k} - \frac{\sinh(ax)}{a} \right) \\
&+ \frac{k}{a^2 + k^2} \left(\frac{\sin(k(x - \frac{\pi}{k}))}{k} - \frac{\sinh(a(x - \frac{\pi}{k}))}{a} \right) H(kx - \pi) + \frac{y'(0)}{a} \sinh(ax)
\end{aligned}$$

or

$$\begin{aligned}
&\frac{1}{a^2 + k^2} \sin(kx) (1 - H(kx - \pi)) \\
&+ \frac{y'(0) - \frac{k}{a^2 + k^2} \sinh(ax)}{a} - \frac{k}{a(a^2 + k^2)} \sinh(a(x - \frac{\pi}{k})) H(kx - \pi)
\end{aligned}$$

As $x \rightarrow \infty$, the sinusoidal terms evaluate to zero, while the hyperbolic/exponential terms are

$$\begin{aligned}
&\frac{y'(0) - \frac{k}{a^2 + k^2} e^{ax} - e^{-ax}}{a} - \frac{k}{a(a^2 + k^2)} \frac{e^{a(x - \frac{\pi}{k})} - e^{-a(x - \frac{\pi}{k})}}{2} \\
&= \frac{1}{2a} \left[\left(y'(0) - \frac{k}{a^2 + k^2} (1 + e^{-\frac{\pi a}{k}}) \right) e^{ax} - \left(y'(0) - \frac{k}{a^2 + k^2} (1 + e^{\frac{\pi a}{k}}) \right) e^{-ax} \right]
\end{aligned}$$

■

For the solution to be physical, the coefficient for e^{ax} has to be zero. Hence

$$y'(0) = \frac{k}{a^2 + k^2} \left(1 + e^{-\frac{\pi a}{k}}\right)$$

So the solution when $kx \geq \pi$ is

$$\begin{aligned} & -\frac{1}{2a} \left(\frac{k}{a^2 + k^2} (1 + e^{-\frac{\pi a}{k}}) - \frac{k}{a^2 + k^2} (1 + e^{\frac{\pi a}{k}}) \right) e^{-ax} \\ &= \frac{k}{a(a^2 + k^2)} \left(\frac{e^{\frac{\pi a}{k}} - e^{-\frac{\pi a}{k}}}{2} \right) e^{-ax} \\ &= \frac{k}{a(a^2 + k^2)} \sinh\left(\frac{\pi a}{k}\right) e^{-ax} \end{aligned}$$

and when $0 \leq kx < \pi$

$$y = \frac{\sin(kx)}{a^2 + k^2} + \frac{k e^{-\frac{\pi a}{k}}}{a(a^2 + k^2)} \sinh(ax)$$

Solutions for Chapter 8

(8.1) Let's first standardize the hypergeometric equation:

$$y'' + \frac{c - (a + b + 1)x}{x(1 - x)}y' - \frac{ab}{x(1 - x)}y = 0$$

$$y'' + \left(\frac{c - (a + b + 1)}{1 - x} + \frac{c}{x} \right)y' - \frac{ab}{x(1 - x)}y = 0$$

So, the integrating factor will be

$$\begin{aligned} r(x) &= \exp\left(\int \left(\frac{c - (a + b + 1)}{1 - x} + \frac{c}{x} \right) dx\right) \\ &= \exp(-(c - (a + b + 1)) \ln(1 - x) + c \ln x) \\ &= \frac{x^c}{(1 - x)^{c - (a + b + 1)}} \end{aligned}$$

and the desired Sturm-Liouville form is

$$\frac{d}{dx} \left(\frac{x^c}{(1 - x)^{c - (a + b + 1)}} \frac{dy}{dx} \right) - \frac{abx^{c-1}}{(1 - x)^{c - a - b}} y = 0$$

where we identify $p(x) = \frac{x^c}{(1 - x)^{c - (a + b + 1)}}$, $q(x) = 0$, $\lambda = -ab$, $\rho(x) = \frac{x^{c-1}}{(1 - x)^{c - a - b}}$.

(8.2)

$$\begin{aligned} \int_a^b \mathcal{L}[f(x)] \overline{g(x)} dx &= \int_a^b \frac{d}{dx} \left(x \frac{d}{dx} [f(x)] \right) \overline{g(x)} dx \\ &= \left[x \frac{df(x)}{dx} \overline{g(x)} \right]_a^b - \int_a^b x \frac{df(x)}{dx} \frac{d\overline{g(x)}}{dx} dx \\ &= \left[x \frac{df(x)}{dx} \overline{g(x)} \right]_a^b - \left[x f(x) \frac{d\overline{g(x)}}{dx} \right]_a^b + \int_a^b f(x) \frac{d}{dx} \left(x \frac{d\overline{g(x)}}{dx} \right) dx \\ &= \text{Boundary Terms} + \int_a^b f(x) \frac{d}{dx} \left(x \frac{d}{dx} [g(x)] \right) dx \end{aligned}$$

■

so $\mathcal{L}^\dagger[g] = \frac{d}{dx}(x \frac{d}{dx}[g]) = \mathcal{L}[g]$ is self-adjoint under the usual L^2 -inner product. Hermiticity requires that the boundary terms are zero:

$$\left[x \frac{df(x)}{dx} \overline{g(x)} \right]_a^b - \left[x f(x) \frac{d\overline{g(x)}}{dx} \right]_a^b = 0$$

This may be achieved by the boundary conditions that either the pair of $f(x)$, $g(x)$, or the derivatives $f'(x)$, $g'(x)$, vanishes at both end-points. Moreover, if a or $b = 0$, then the boundary condition at this one side will be automatically satisfied.

(8.3) (a) x^2 is even, so only the cosine components remain. We then compute a_m as suggested in the main text, integrating by parts:

$$\begin{aligned} a_m &= \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \cos(mx) dx \\ &= \left[\frac{1}{m\pi} x^2 \sin(mx) \right]_{-\pi}^{\pi} - \frac{2}{m\pi} \int_{-\pi}^{\pi} x \sin(mx) dx \\ &= (0) - \frac{2}{m\pi} \left[-\frac{x}{m} \cos(mx) \right]_{-\pi}^{\pi} + \frac{2}{m^2\pi} \int_{-\pi}^{\pi} (-\cos(mx)) dx \\ &= \frac{2}{m^2\pi} [\pi(-1)^m - (-\pi)(-1)^m] - (0) = \frac{4(-1)^m}{m^2} \end{aligned}$$

Meanwhile, the constant term is just

$$\begin{aligned} \frac{a_0}{2} &= \frac{1}{2\pi} \int_{-\pi}^{\pi} x^2 dx \\ &= \frac{1}{2\pi} \left[\frac{x^3}{3} \right]_{-\pi}^{\pi} \\ &= \frac{1}{2\pi} \left[\frac{\pi^3}{3} - \frac{(-\pi)^3}{3} \right] = \frac{\pi^2}{3} \end{aligned}$$

Therefore, the desired Fourier series is

$$x^2 = \frac{\pi^2}{3} + \sum_{m=1}^{\infty} \frac{4(-1)^m}{m^2} \cos(mx)$$

(b) Similarly, x^3 is odd and we only need to calculate b_m :

$$\begin{aligned} b_m &= \frac{1}{\pi} \int_{-\pi}^{\pi} x^3 \sin(mx) dx \\ &= \left[-\frac{1}{m\pi} x^3 \cos(mx) \right]_{-\pi}^{\pi} + \frac{3}{m\pi} \int_{-\pi}^{\pi} x^2 \cos(mx) dx \\ &= -\frac{1}{m\pi} [\pi^3(-1)^m - (-\pi)^3(-1)^m] + \frac{3}{m^2\pi} [x^2 \sin(mx)]_{-\pi}^{\pi} - \frac{6}{m^2\pi} \int_{-\pi}^{\pi} x \sin(mx) dx \end{aligned}$$

$$\begin{aligned}
&= -\frac{2\pi^2(-1)^m}{m} + (0) + \frac{6}{m^3\pi} [x \cos(mx)]_{-\pi}^{\pi} - \frac{6}{m^3\pi} \int_{-\pi}^{\pi} \cos(mx) dx \\
&= -\frac{2\pi^2(-1)^m}{m} + \frac{6}{m^3\pi} [\pi(-1)^m - (-\pi)(-1)^m] - (0) \\
&= -\frac{2\pi^2(-1)^m}{m} + \frac{12(-1)^m}{m^3}
\end{aligned}$$

So the required Fourier series is

$$x^3 = \sum_{m=1}^{\infty} \left(-\frac{2\pi^2(-1)^m}{m} + \frac{12(-1)^m}{m^3} \right) \sin(mx)$$

For the latter part, to obtain the Fourier series appropriate to the interval $[-a, a]$, we need to make a change of variable $\tilde{x} = \frac{\pi x}{a}$ so that $x = \pm a$ correspond to $\tilde{x} = \pm\pi$. Note that then

$$x^2 = \left(\frac{a}{\pi}\right)^2 \tilde{x}^2 \qquad x^3 = \left(\frac{a}{\pi}\right)^3 \tilde{x}^3$$

Hence, we can just apply the Fourier series found above with an extra scaling factor, and the required forms are

$$\begin{aligned}
x^2 &= \left(\frac{a}{\pi}\right)^2 \left(\frac{\pi^2}{3} + \sum_{m=1}^{\infty} \frac{4(-1)^m}{m^2} \cos\left(m \frac{\pi x}{a}\right) \right) \\
x^3 &= \left(\frac{a}{\pi}\right)^3 \left(\sum_{m=1}^{\infty} \left(-\frac{2\pi^2(-1)^m}{m} + \frac{12(-1)^m}{m^3} \right) \sin\left(m \frac{\pi x}{a}\right) \right)
\end{aligned}$$

(8.4) This is a trick question as the given expression is already a Fourier series.

(8.5) The change of variable needed is simply $\tilde{x} = x - \pi$, so that $e^{-x} = e^{-(\tilde{x}+\pi)} = e^{-\pi} e^{-\tilde{x}}$. So the required Fourier series will be the same as the conventional one, just with an extra $e^{-\pi}$ factor. We now proceed to compute the coefficients. Integrating by parts twice,

$$\begin{aligned}
a_m &= \frac{1}{\pi} \int_{-\pi}^{\pi} e^{-x} \cos(mx) dx \\
&= \frac{1}{\pi} \left(\left[\frac{1}{m} e^{-x} \sin(mx) \right]_{-\pi}^{\pi} - \int_{-\pi}^{\pi} \frac{1}{m} (-e^{-x}) \sin(mx) dx \right) \\
&= \frac{1}{\pi} \left((0) + \int_{-\pi}^{\pi} \frac{1}{m} e^{-x} \sin(mx) dx \right) \\
&= \frac{1}{m\pi} \left(\left[-\frac{1}{m} e^{-x} \cos(mx) \right]_{-\pi}^{\pi} - \int_{-\pi}^{\pi} \frac{1}{m} e^{-x} \cos(mx) dx \right) \\
&= \frac{(-1)^m}{m^2\pi} (e^{\pi} - e^{-\pi}) - \frac{1}{m^2\pi} \int_{-\pi}^{\pi} e^{-x} \cos(mx) dx = \frac{(-1)^m}{m^2\pi} (e^{\pi} - e^{-\pi}) - \frac{a_m}{m^2}
\end{aligned}$$

Therefore,

$$\begin{aligned}\frac{m^2 + 1}{m^2} a_m &= \frac{(-1)^m}{m^2 \pi} (e^\pi - e^{-\pi}) \\ a_m &= \frac{(-1)^m}{(m^2 + 1)\pi} (e^\pi - e^{-\pi}) = \frac{2(-1)^m \sinh(\pi)}{(m^2 + 1)\pi}\end{aligned}$$

The constant term is just

$$\begin{aligned}\frac{a_0}{2} &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-x} dx \\ &= \frac{1}{2\pi} (e^\pi - e^{-\pi}) = \frac{1}{\pi} \sinh(\pi)\end{aligned}$$

Similarly,

$$b_m = \frac{1}{\pi} \int_{-\pi}^{\pi} e^{-x} \sin(mx) dx = \frac{(-1)^m m}{(m^2 + 1)\pi} (e^\pi - e^{-\pi}) = \frac{2(-1)^m m \sinh(\pi)}{(m^2 + 1)\pi}$$

Therefore, the desired Fourier series is

$$\begin{aligned}e^{-x} &= e^{-\pi} e^{-\tilde{x}} \\ &= e^{-\pi} \left(\frac{\sinh(\pi)}{\pi} + \sum_{m=1}^{\infty} \frac{2(-1)^m \sinh(\pi)}{(m^2 + 1)\pi} \cos(m\tilde{x}) + \sum_{m=1}^{\infty} \frac{2(-1)^m m \sinh(\pi)}{(m^2 + 1)\pi} \sin(m\tilde{x}) \right)\end{aligned}$$

Reorganizing the terms and recovering x , the final answer is

$$e^{-x} = \frac{1 - e^{-2\pi}}{2\pi} \left(1 + \sum_{m=1}^{\infty} \frac{2}{m^2 + 1} \cos(mx) + \sum_{m=1}^{\infty} \frac{2m}{m^2 + 1} \sin(mx) \right)$$

(8.6) Finding the desired sine series is equivalent to computing the Fourier series of

$$f_{\text{odd}}(x) = \begin{cases} 1 - x & 0 < x \leq 1 \\ x - 1 & -1 \leq x < 0 \end{cases}$$

The needed coefficients can be found, taking into account the interval scaling and symmetry, as

$$\begin{aligned}b_m &= \int_0^1 (1 - x) \sin(m\pi x) dx + \int_{-1}^0 (x - 1) \sin(m\pi x) dx \\ &= 2 \int_0^1 (1 - x) \sin(m\pi x) dx \\ &= 2 \left(\int_0^1 \sin(m\pi x) dx - \int_0^1 x \sin(m\pi x) dx \right)\end{aligned}$$

$$\begin{aligned}
&= 2 \left(\left[-\frac{1}{m\pi} \cos(m\pi x) \right]_0^1 - \left[-\frac{x}{m\pi} \cos(m\pi x) \right]_0^1 + \int_0^1 -\frac{1}{m\pi} \cos(m\pi x) dx \right) \\
&= 2 \left(-\frac{(-1)^m - 1}{m\pi} + \frac{(-1)^m}{m\pi} - \left[\frac{1}{m^2\pi^2} \sin(m\pi x) \right]_0^1 \right) \\
&= \frac{2}{m\pi} - (0) = \frac{2}{m\pi}
\end{aligned}$$

Therefore, the sine series should be just

$$1 - x = \sum_{m=1}^{\infty} \frac{2}{m\pi} \sin(m\pi x)$$

Meanwhile, the cosine series requires us to calculate, similarly,

$$\begin{aligned}
a_m &= 2 \int_0^1 (1 - x) \cos(m\pi x) dx \\
&= 2 \left(\int_0^1 \cos(m\pi x) dx - \int_0^1 x \cos(m\pi x) dx \right) \\
&= 2 \left(\left[\frac{1}{m\pi} \sin(m\pi x) \right]_0^1 - \left[\frac{x}{m\pi} \sin(m\pi x) \right]_0^1 + \int_0^1 \frac{1}{m\pi} \sin(m\pi x) dx \right) \\
&= 2 \left((0) + \left[-\frac{1}{m^2\pi^2} \cos(m\pi x) \right]_0^1 \right) \\
&= \frac{2(1 - (-1)^m)}{m^2\pi^2}
\end{aligned}$$

and the constant term is just

$$\frac{a_0}{2} = \int_0^1 (1 - x) dx = \frac{1}{2}$$

Hence, the cosine series will be

$$1 - x = \frac{1}{2} + \sum_{m=1}^{\infty} \frac{2(1 - (-1)^m)}{m^2\pi^2} \cos(m\pi x)$$

(8.7) From the exercise before, we know that the Fourier series of x^2 is

$$x^2 = \frac{\pi^2}{3} + \sum_{m=1}^{\infty} \frac{4(-1)^m}{m^2} \cos(mx)$$

If we proceed with the formal differentiation, we can obtain

$$\frac{d}{dx}(x^2) = \frac{d}{dx} \left(\frac{\pi^2}{3} + \sum_{m=1}^{\infty} \frac{4(-1)^m}{m^2} \cos(mx) \right)$$

$$2x = \sum_{m=1}^{\infty} -\frac{4(-1)^m}{m} \sin(mx)$$

which is consistent with the Fourier series of x derived in Example 8.3.

(8.8) An obvious explanation is that the condition of the term-by-term Fourier series differentiation theorem is not satisfied because x is not periodic, so a formal differentiation will fail. Another point to see this is that the corresponding Sturm-Liouville operator $\mathcal{L} = -\frac{d^2}{dx^2}$ is not Hermitian, as the boundary conditions are not satisfied and thus the Spectral Theorem does not work as well.

(8.9) This follows directly from the Cauchy-Schwarz Inequality:

$$\begin{aligned} \int |f(x)g(x)|dx &= \int |f(x)||g(x)|dx = \langle |f|, |g| \rangle \\ &\leq \| |f| \| \| |g| \| \\ &= \left(\int |f(x)|^2 dx \right)^{\frac{1}{2}} \left(\int |g(x)|^2 dx \right)^{\frac{1}{2}} \\ &= \|f\| \|g\| < \infty \end{aligned}$$

(8.10) Similar to the example in the main text, the Green's function must satisfy the homogeneous part, but with the different coefficients in z on the two sides of $x = z$:

$$G(x, z) = \begin{cases} A(z) \sin\left(\frac{x}{2}\right) + B(z) \cos\left(\frac{x}{2}\right) & \text{when } x < z \\ C(z) \sin\left(\frac{x}{2}\right) + D(z) \cos\left(\frac{x}{2}\right) & \text{when } x > z \end{cases}$$

To match the given Neumann B.C.s, we need $A(z) = D(z) = 0$ this time, and thus

$$G(x, z) = \begin{cases} B(z) \cos\left(\frac{x}{2}\right) & \text{when } x < z \\ C(z) \sin\left(\frac{x}{2}\right) & \text{when } x > z \end{cases}$$

The discontinuity needed for $\frac{dG}{dx}$ at $x = z$ is just 1 and $G(x, z)$ is continuous there, hence

$$\begin{cases} C(z) \sin\left(\frac{z}{2}\right) - B(z) \cos\left(\frac{z}{2}\right) = 0 \\ \frac{1}{2}C(z) \cos\left(\frac{z}{2}\right) + \frac{1}{2}B(z) \sin\left(\frac{z}{2}\right) = 1 \end{cases}$$

This requires that $B(z) = 2 \sin\left(\frac{z}{2}\right)$, $C(z) = 2 \cos\left(\frac{z}{2}\right)$. So the Green's function in this context is

$$G(x, z) = \begin{cases} 2 \sin\left(\frac{z}{2}\right) \cos\left(\frac{x}{2}\right) & \text{when } x < z \\ 2 \cos\left(\frac{z}{2}\right) \sin\left(\frac{x}{2}\right) & \text{when } x > z \end{cases}$$

The solution is then

$$\begin{aligned}
y &= \int_0^\pi G(x, z) \sin\left(\frac{z}{2}\right) dz \\
&= \int_0^x \left(2 \cos\left(\frac{z}{2}\right) \sin\left(\frac{x}{2}\right)\right) \sin\left(\frac{z}{2}\right) dz + \int_x^\pi \left(2 \sin\left(\frac{z}{2}\right) \cos\left(\frac{x}{2}\right)\right) \sin\left(\frac{z}{2}\right) dz \\
&= 2 \sin\left(\frac{x}{2}\right) \int_0^x \frac{1}{2} \sin z \, dz + 2 \cos\left(\frac{x}{2}\right) \int_x^\pi \sin^2\left(\frac{z}{2}\right) dz \\
&= \sin\left(\frac{x}{2}\right) [-\cos z]_0^x + \cos\left(\frac{x}{2}\right) \int_x^\pi (1 - \cos z) dz \\
&= -\sin\left(\frac{x}{2}\right) (\cos x - 1) + \cos\left(\frac{x}{2}\right) [z - \sin z]_x^\pi \\
&= -\sin\left(\frac{x}{2}\right) (\cos x - 1) + \cos\left(\frac{x}{2}\right) (\pi - x + \sin x) \\
&= \sin\left(\frac{x}{2}\right) - \sin\left(\frac{x}{2}\right) \cos x + (\pi - x) \cos\left(\frac{x}{2}\right) + \cos\left(\frac{x}{2}\right) \sin x \\
&= \sin\left(\frac{x}{2}\right) + (\pi - x) \cos\left(\frac{x}{2}\right) + \sin\left(\frac{x}{2}\right) = 2 \sin\left(\frac{x}{2}\right) + (\pi - x) \cos\left(\frac{x}{2}\right)
\end{aligned}$$

This should match the solution acquired from the usual approach.

(8.11) The general solution to the associated eigenvalue equation $y'' + \lambda y = 0$, as usual, is

$$y = c_1 \cos(\sqrt{\lambda}x) + c_2 \sin(\sqrt{\lambda}x)$$

The boundary conditions indicate that $c_1 = 0$, and

$$y'(L) + \alpha y(L) = c_2 \sqrt{\lambda} \cos(\sqrt{\lambda}L) + \alpha c_2 \sin(\sqrt{\lambda}L) = c_2 (\sqrt{\lambda} \cos(\sqrt{\lambda}L) + \alpha \sin(\sqrt{\lambda}L)) = 0$$

For the solution to be non-trivial, i.e. $c_2 \neq 0$, this means that

$$\begin{aligned}
\sqrt{\lambda} \cos(\sqrt{\lambda}L) + \alpha \sin(\sqrt{\lambda}L) &= 0 \\
\tan(\sqrt{\lambda}L) &= -\frac{\sqrt{\lambda}}{\alpha}
\end{aligned}$$

It can be shown that there are countably infinite values of λ satisfying this constraint and hence are the eigenvalues. Denote them as $\lambda_1, \lambda_2, \dots$, then the Green's function is written as (be careful about the signs, as the equation forms are slightly different)

$$G(x, z) = \sum_{j=1}^{\infty} \frac{1}{\mu - \lambda_j} \frac{\sin(\sqrt{\lambda_j}x) \sin(\sqrt{\lambda_j}z)}{\int_0^L \sin^2(\sqrt{\lambda_j}x) dx}$$

(8.12) Following the same logic as the previous example/exercise, now the boundary conditions require that $G(-\frac{\pi}{2}, z) = G(\frac{\pi}{2}, z) = 0$, so that $A(z) = C(z) = 0$, and

$$G(x, z) = \begin{cases} B(z) \cos x & \text{when } x < z \\ D(z) \cos x & \text{when } x > z \end{cases}$$

The continuity requirements are then

$$\begin{cases} D(z) \cos z - B(z) \cos z = 0 \\ D(z) \sin z - B(z) \sin z = 1 \end{cases}$$

This is clearly inconsistent, and so the Green's function method does not work here. In the context of constructing the Green's function from eigenfunctions, $\mathcal{L} = \frac{d^2}{dx^2}$ where the eigenfunctions are $\varphi_m = \cos(mx)$ with the eigenvalues of $\lambda_m = -m^2$, $m = 0, 1, 2, \dots$, and $\mu = -1$. When $m = 1$, the denominator in the Green's function formula is $\frac{1}{\lambda_m - \mu} = \frac{1}{(-1) - (-1)} = \frac{1}{0}$, which leads to infinity and hence the Green's function blows up. This phenomenon is known as resonance.

Solutions for Chapter 9

(9.1) Following the instruction, $x = 1 - 2t$, $\frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx} = -\frac{1}{2} \frac{dy}{dt}$ and similarly for $\frac{d^2y}{dx^2}$, the new equation is

$$\begin{aligned} (1 - (1 - 2t)^2) \left(\left(-\frac{1}{2} \right)^2 y'' \right) - 2(1 - 2t) \left(-\frac{1}{2} y' \right) + l(l + 1)y &= 0 \\ t(1 - t)y'' + (1 - 2t)y' + l(l + 1)y &= 0 \end{aligned}$$

Comparison shows that this is a hypergeometric equation with $a = -l$, $b = l + 1$, $c = 1$. By the remark for when l is not an integer, the corresponding hypergeometric function will not converge at $t = 1$ and thus $x = -1$ since $c = 1 = -l + (l + 1) = a + b$, and this also implies the unboundedness at $x = 1$ by symmetry. When l is an integer, the hypergeometric function in the first part of the solution will be a finite polynomial that unsurprisingly corresponds to the Legendre polynomials. The special discussion about the hypergeometric equation when $c = l$ is an integer indicates that the second solution will contain a term consisting of the Legendre polynomial times a logarithm $\ln t$. This logarithm then implies that this second solution will also be unbounded at $t = 0$ and $x = 1$.

(9.2) It will be beneficial to first establish the general values of the integral

$$I_n = \int_{-1}^1 x^n \cos(\pi x) dx$$

It is clear that $I_{2m+1} = 0$ for odd indices as it will be an odd integral. For even indices, we derive the reduction formula:

$$\begin{aligned} I_{2m} &= \int_{-1}^1 x^{2m} \cos(\pi x) dx \\ &= \frac{1}{\pi} [x^{2m} \sin(\pi x)]_{-1}^1 - \frac{2m}{\pi} \int_{-1}^1 x^{2m-1} \sin(\pi x) dx \\ &= (0) + \frac{2m}{\pi^2} [x^{2m-1} \cos(\pi x)]_{-1}^1 - \frac{2m(2m-1)}{\pi^2} \int_{-1}^1 x^{2m-2} \cos(\pi x) dx \\ &= -\frac{4m}{\pi^2} - \frac{2m(2m-1)}{\pi^2} \int_{-1}^1 x^{2m-2} \cos(\pi x) dx \end{aligned}$$

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$$= -\frac{4m}{\pi^2} - \frac{2m(2m-1)}{\pi^2} I_{2m-2}$$

Hence $I_0 = 0$, $I_2 = -\frac{4}{\pi^2}$, $I_4 = -\frac{8}{\pi^2} + \frac{48}{\pi^4}$, and the first five Legendre coefficients of $\cos(\pi x)$ will then be

$$\begin{aligned} a_0 &= \frac{1}{2} \left(\int_{-1}^1 \cos(\pi x) dx \right) = 0 \\ a_1 &= \frac{3}{2} \left(\int_{-1}^1 x \cos(\pi x) dx \right) = 0 \\ a_2 &= \frac{5}{2} \left(\int_{-1}^1 \frac{1}{2} (3x^2 - 1) \cos(\pi x) dx \right) = \frac{5}{4} \left(-\frac{(3)(4)}{\pi^2} - 0 \right) = -\frac{15}{\pi^2} \\ a_3 &= \frac{7}{2} \left(\int_{-1}^1 \frac{1}{2} (5x^3 - 3x) \cos(\pi x) dx \right) = 0 \\ a_4 &= \frac{9}{2} \left(\int_{-1}^1 \frac{1}{8} (35x^4 - 30x^2 + 3) \cos(\pi x) dx \right) \\ &= \frac{9}{16} \left(35 \left(-\frac{8}{\pi^2} + \frac{48}{\pi^4} \right) - 30 \left(-\frac{4}{\pi^2} \right) + 0 \right) \\ &= \frac{945}{\pi^4} - \frac{90}{\pi^2} \end{aligned}$$

and so the desired Legendre series is $\cos(\pi x) = -\frac{15}{\pi^2} P_2 + \left(\frac{945}{\pi^4} - \frac{90}{\pi^2} \right) P_4 + \dots$.

(9.3) To broadcast the given interval into $[-1, 1]$, we simply let $t = 2x$ and then the integral becomes

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} e^x dx = \int_{-1}^1 e^{\frac{t}{2}} d\left(\frac{t}{2}\right) = \frac{1}{2} \int_{-1}^1 e^{\frac{t}{2}} dt$$

and we numerically integrate the integral in t to the right. Using the table in the link provided, we have

$$\begin{aligned} \sum_{i=1}^6 w_i f(x_i) &\approx 0.36076 e^{\frac{0.66121}{2}} + 0.36076 e^{-\frac{0.66121}{2}} \\ &\quad + 0.46791 e^{\frac{0.23862}{2}} + 0.46791 e^{-\frac{0.23862}{2}} \\ &\quad + 0.17132 e^{\frac{0.93247}{2}} + 0.17132 e^{-\frac{0.93247}{2}} \\ &\approx 2.08436 \end{aligned}$$

And hence the required value will be $\frac{1}{2}(2.08436) = 1.04218$, whereas the exact result should be $e^{\frac{1}{2}} - e^{-\frac{1}{2}} \approx 1.04219$.

(9.4) Substituting the series form (with k in place of n) into the Chebyshev equation gives

$$(1 - x^2) \sum_{k=2}^{\infty} k(k-1) a_k x^{k-2} - x \sum_{k=1}^{\infty} k a_k x^{k-1} + n^2 \sum_{k=0}^{\infty} a_k x^k = 0$$

$$\sum_{k=2}^{\infty} k(k-1)a_k x^{k-2} - \sum_{k=2}^{\infty} k(k-1)a_k x^k - \sum_{k=1}^{\infty} k a_k x^k + n^2 \sum_{k=0}^{\infty} a_k x^k = 0$$

$$\sum_{k=0}^{\infty} (k+2)(k+1)a_{k+2} x^k - \sum_{k=2}^{\infty} k(k-1)a_k x^k - \sum_{k=1}^{\infty} k a_k x^k + n^2 \sum_{k=0}^{\infty} a_k x^k = 0$$

Comparing the coefficients, the recurrence relation is

$$(k+2)(k+1)a_{k+2} - k(k-1)a_k - k a_k + n^2 a_k = 0$$

$$\Rightarrow a_{k+2} = -\frac{n^2 - k^2}{(k+2)(k+1)} a_k = -\frac{(n-k)(n+k)}{(k+2)(k+1)} a_k$$

The series will be truncated when $k = n$ and become the Chebyshev polynomial of degree n we want, which is then

$$T_n(x) = x^n - \frac{n(n-1)}{2(2n-2)} x^{n-2} + \frac{n(n-1)(n-2)(n-3)}{(2)(4)(2n-2)(2n-4)} x^{n-4} - \dots$$

$$= \sum_{m=0}^{\lfloor \frac{n}{2} \rfloor} (-1)^m \frac{n!}{(n-2m)!} \frac{(n-m-1)!}{(2^m(n-1)!(2^m m!)} x^{n-2m}$$

$$= \sum_{m=0}^{\lfloor \frac{n}{2} \rfloor} (-1)^m \frac{n C_m^{n-m}}{(n-m)2^{2m}} x^{n-2m}$$

To account for the normalization factor (so that the leading coefficient is 2^{n-1}), it would become

$$T_n(x) = \frac{1}{2} \sum_{m=0}^{\lfloor \frac{n}{2} \rfloor} (-1)^m \frac{n C_m^{n-m}}{n-m} (2x)^{n-2m}$$

Hence by comparing this to the form derived in the main text, we may also obtain the following combinatorial identity:

$$\sum_{l=0}^{\lfloor \frac{n}{2} \rfloor - m} C_{2m+2l}^n C_l^{m+l} = 2^{n-2m-1} \frac{n C_m^{n-m}}{n-m}$$

(9.5) Replacing $x \rightarrow \frac{x}{\sqrt{2}}$ in the original Hermite equation produces (remember to take care of the scaling of the differentials as well)

$$\sqrt{2} \left(e^{-\left(\frac{x}{\sqrt{2}}\right)^2} \sqrt{2} y' \right)' + 2 \sqrt{2} e^{-\left(\frac{x}{\sqrt{2}}\right)^2} y = 0$$

$$2 \left(e^{-\frac{x^2}{2}} y' \right)' + 2 \sqrt{2} e^{-\frac{x^2}{2}} y = 0$$

which matches the new equation. Then, the Rodrigues' formula is also given by the same change of variable:

$$(-1)^n e^{\frac{x^2}{2}} \frac{d^n}{dx^n} (e^{-\frac{x^2}{2}})$$

The factor arising from the derivative here is absorbed so that this always has a leading coefficient of 1.

(9.6) First, we do the normalization step $x = \left(\frac{\hbar^2}{2m}\right)^{\frac{1}{4}} \tilde{x}$ to obtain

$$\begin{aligned} \left(\frac{2m}{\hbar^2}\right)^{\frac{1}{2}} \frac{d^2\psi}{d\tilde{x}^2} + \frac{2m}{\hbar^2} \left(E - \left(\frac{\hbar^2}{2m}\right)^{\frac{1}{2}} \tilde{x}^2\right) \psi &= 0 \\ \frac{d^2\psi}{d\tilde{x}^2} + \left(\left(\frac{2m}{\hbar^2}\right)^{\frac{1}{2}} E - \tilde{x}^2\right) \psi &= 0 \\ \frac{d^2\psi}{d\tilde{x}^2} + (\tilde{E} - \tilde{x}^2) \psi &= 0 \end{aligned}$$

where we assign $\tilde{E} = \left(\frac{2m}{\hbar^2}\right)^{\frac{1}{2}} E$. By the same argument as for the Equatorial wave problem, we convert this parabolic cylinder equation into the Hermite equation by appending the factor $\exp(-\frac{\tilde{x}^2}{2})$, and the required eigenstates will be in the form of

$$\psi(\tilde{x}) = H_n(\tilde{x}) \exp\left(-\frac{\tilde{x}^2}{2}\right)$$

where the eigenvalues follow $\tilde{E} = 2\nu + 1$, $\nu = 0, 1, 2, \dots$

(9.7) We can write down the new first and third starting equations, which are

$$\begin{aligned} -i\omega \hat{u}(y) &= -ik \hat{\Phi}(y) \\ -i\omega \hat{\Phi}(y) &= -gHik \hat{u}(y) \end{aligned}$$

Multiplying them together leads to the dispersion relation of

$$\begin{aligned} -\omega^2 \hat{u}(y) \hat{\Phi}(y) &= -gHk^2 \hat{\Phi}(y) \hat{u}(y) \\ \frac{\omega}{k} &= \sqrt{gH} \end{aligned}$$

Eliminating $\hat{u}$, the second starting equation becomes

$$\begin{aligned} \beta y \left(\frac{k}{\omega} \hat{\Phi}(y)\right) &= -\frac{d\hat{\Phi}(y)}{dy} \\ \frac{d\hat{\Phi}(y)}{dy} + \frac{\beta y}{\sqrt{gH}} \hat{\Phi}(y) &= 0 \end{aligned}$$

Again, making the change of variable $\tilde{y} = \frac{y}{y_0}$, $y_0 = \left(\frac{\sqrt{gH}}{\beta}\right)^{\frac{1}{2}}$, it is reduced to

$$\frac{1}{\left(\frac{\sqrt{gH}}{\beta}\right)^{\frac{1}{2}}} \frac{d\hat{\Phi}(\tilde{y})}{d\tilde{y}} + \frac{\beta\tilde{y}}{\sqrt{gH}} \left(\frac{\sqrt{gH}}{\beta}\right)^{\frac{1}{2}} \hat{\Phi}(\tilde{y}) = 0$$

$$\frac{d\hat{\Phi}(\tilde{y})}{d\tilde{y}} + \tilde{y}\hat{\Phi}(\tilde{y}) = 0$$

This is a simple first-order ODE that can be solved using an integrating factor:

$$e^{\frac{\tilde{y}^2}{2}} \left(\frac{d\hat{\Phi}}{d\tilde{y}} + \tilde{y}\hat{\Phi} \right) = 0$$

$$\frac{d}{d\tilde{y}} \left(e^{\frac{\tilde{y}^2}{2}} \hat{\Phi} \right) = 0$$

$$e^{\frac{\tilde{y}^2}{2}} \hat{\Phi} = A$$

$$\hat{\Phi} = A e^{-\frac{\tilde{y}^2}{2}}$$

The solution for $\hat{u}$ is similar. So in this special solution, the amplitudes of the zonal wind and geopotential will decay at a rate of $\exp\left(-\frac{\tilde{y}^2}{2}\right)$ just like the old solution, without any further complications from the Hermite polynomials. This physically corresponds to the Kelvin wave (Subfigure (f) in the schematic attached). We usually denote this solution by $n = -1$, because H_{-1} is conventionally taken to be 0, and so $\hat{v}$ by the previous formula will also be zero.

(9.8) Differentiating the generating function against t yields

$$\frac{\partial}{\partial t} \left(\frac{-xt}{1-t} \right) \frac{e^{-\frac{xt}{1-t}}}{1-t} + \frac{e^{-\frac{xt}{1-t}}}{(1-t)^2} = \sum_{n=0}^{\infty} n L_n(x) t^{n-1}$$

$$-\frac{x e^{-\frac{xt}{1-t}}}{(1-t)^3} + \frac{e^{-\frac{xt}{1-t}}}{(1-t)^2} = \sum_{n=0}^{\infty} n L_n(x) t^{n-1}$$

$$\frac{(1-x-t) e^{-\frac{xt}{1-t}}}{(1-t)^3} = \sum_{n=0}^{\infty} n L_n(x) t^{n-1}$$

Using the generating function itself and rearranging leads to

$$\frac{1-x-t}{(1-t)^2} \sum_{n=0}^{\infty} L_n(x) t^n = \sum_{n=0}^{\infty} n L_n(x) t^{n-1}$$

$$(1-x-t) \sum_{n=0}^{\infty} L_n(x) t^n = (1-t)^2 \sum_{n=0}^{\infty} n L_n(x) t^{n-1}$$

$$(1-x) \sum_{n=0}^{\infty} L_n(x) t^n - \sum_{n=0}^{\infty} L_n(x) t^{n+1} = \sum_{n=0}^{\infty} n L_n(x) t^{n-1} - 2 \sum_{n=0}^{\infty} n L_n(x) t^n + \sum_{n=0}^{\infty} n L_n(x) t^{n+1}$$

$$(1-x) \sum_{n=0}^{\infty} L_n(x) t^n - \sum_{n=0}^{\infty} L_{n-1}(x) t^n = \sum_{n=0}^{\infty} (n+1) L_{n+1}(x) t^n - 2 \sum_{n=0}^{\infty} n L_n(x) t^n + \sum_{n=0}^{\infty} (n-1) L_{n-1}(x) t^n$$

where we have adjusted the indices accordingly. Equating the coefficients then gives the first relation:

$$(1-x)L_n(x) - L_{n-1}(x) = (n+1)L_{n+1}(x) - 2nL_n(x) + (n-1)L_{n-1}(x)$$

$$(n+1)L_{n+1}(x) = (2n+1-x)L_n(x) - nL_{n-1}(x)$$

The second relation is similarly obtained by differentiating the generating function against x :

$$-\frac{te^{-\frac{xt}{1-t}}}{(1-t)^2} = \sum_{n=0}^{\infty} L'_n(x) t^n$$

$$-t \sum_{n=0}^{\infty} L_n(x) t^n = (1-t) \sum_{n=0}^{\infty} L'_n(x) t^n$$

$$-\sum_{n=0}^{\infty} L_n(x) t^{n+1} = \sum_{n=0}^{\infty} L'_n(x) t^n - \sum_{n=0}^{\infty} L'_n(x) t^{n+1}$$

$$-\sum_{n=0}^{\infty} L_{n-1}(x) t^n = \sum_{n=0}^{\infty} L'_n(x) t^n - \sum_{n=0}^{\infty} L'_{n-1}(x) t^n$$

So

$$L_{n-1}(x) = L'_{n-1}(x) - L'_n(x)$$

Differentiating the first relation with respect to x and using the second relation, we arrive at the third relation:

$$(n+1)L'_{n+1}(x) = (2n+1-x)L'_n(x) - L_n(x) - nL'_{n-1}(x)$$

$$(n+1)(L'_n(x) - L_n(x)) = (n+1-x)L'_n(x) - L_n(x) - nL_{n-1}(x)$$

$$xL'_n(x) = nL_n(x) - nL_{n-1}(x)$$

Differentiating the last relation and applying the second relation, we recover the Laguerre equation:

$$xL''_n(x) + L'_n(x) = nL'_n(x) - nL'_{n-1}(x)$$

$$xL''_n(x) + L'_n(x) = nL'_n(x) - n(L_{n-1}(x) + L'_n(x))$$

$$xL''_n(x) + L'_n(x) = xL'_n(x) - nL_n(x)$$

$$xL_n''(x) + (1-x)L_n'(x) + nL_n(x) = 0$$

(9.9) It is a straightforward use of L'Hôpital's rule on its definition:

$$\begin{aligned} Y_p(x) &= \frac{\frac{\partial}{\partial \nu} [J_\nu(x) \cos(\nu\pi) - J_{-\nu}(x)]_{\nu=p}}{[\frac{\partial}{\partial \nu} \sin(\nu\pi)]_{\nu=p}} \\ &= \frac{[(\frac{\partial}{\partial \nu} J_\nu(x)) \cos(\nu\pi) - \pi J_\nu(x) \sin(\nu\pi) - \frac{\partial}{\partial \nu} J_{-\nu}(x)]_{\nu=p}}{[\pi \cos(\nu\pi)]_{\nu=p}} \\ &= \frac{[\frac{\partial}{\partial \nu} J_\nu(x) - (0) - \frac{1}{\cos(\nu\pi)} \frac{\partial}{\partial \nu} J_{-\nu}(x)]_{\nu=p}}{\pi} \\ &= \frac{1}{\pi} \left[\frac{\partial}{\partial \nu} J_\nu(x) - (-1)^p \frac{\partial}{\partial \nu} J_{-\nu}(x) \right]_{\nu=p} \end{aligned}$$

(9.10) Under the suggested change of variable, the derivatives are now

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{dw} \frac{dw}{dx} = \frac{1}{\sqrt{x}} \frac{dy}{dw} = \frac{2}{w} \frac{dy}{dw} \\ \frac{d^2y}{dx^2} &= \frac{2}{w} \frac{d}{dw} \left(\frac{2}{w} \frac{dy}{dw} \right) \\ &= \frac{2}{w} \left(\frac{2}{w} \frac{d^2y}{dw^2} - \frac{2}{w^2} \frac{dy}{dw} \right) \end{aligned}$$

Hence the equation is transformed into

$$\begin{aligned} \frac{w^2}{4} \left(\frac{2}{w} \left(\frac{2}{w} \frac{d^2y}{dw^2} - \frac{2}{w^2} \frac{dy}{dw} \right) \right) + \frac{2}{w} \frac{dy}{dw} + y &= 0 \\ \frac{d^2y}{dw^2} - \frac{1}{w} \frac{dy}{dw} + \frac{2}{w} \frac{dy}{dw} + y &= 0 \\ w^2 \frac{d^2y}{dw^2} + w \frac{dy}{dw} + w^2 y &= 0 \end{aligned}$$

This is a Bessel equation of order 0, and thus the solution will be

$$y = c_1 J_0(w) + c_2 Y_0(w) = c_1 J_0(2\sqrt{x}) + c_2 Y_0(2\sqrt{x})$$

(9.11) With these changes of variables, the derivatives are now

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{dw} \frac{dw}{dx} = \left(\frac{dv}{dw} w^c + v c w^{c-1} \right) \left(\frac{1}{ab w^{b-1}} \right) = \frac{1}{ab} \left(\frac{dv}{dw} w^{c-b+1} + v c w^{c-b} \right) \\ \frac{d^2y}{dx^2} &= \frac{dw}{dx} \frac{d}{dw} \left(\frac{1}{ab} \left(\frac{dv}{dw} w^{c-b+1} + v c w^{c-b} \right) \right) \end{aligned}$$

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$$\begin{aligned}
&= \frac{1}{abw^{b-1}} \left(\frac{1}{ab} \right) \left(\frac{d^2v}{dw^2} w^{c-b+1} + (c-b+1+c) \frac{dv}{dw} w^{c-b} + c(c-b)v w^{c-b-1} \right) \\
&= \frac{1}{a^2b^2} \left(\frac{d^2v}{dw^2} w^{c-2b+2} + (2c-b+1) \frac{dv}{dw} w^{c-2b+1} + c(c-b)v w^{c-2b} \right)
\end{aligned}$$

Substituting all the transformations into the Bessel equation, we arrive at

$$\begin{aligned}
&\frac{a^2w^{2b}}{a^2b^2} \left(\frac{d^2v}{dw^2} w^{c-2b+2} + (2c-b+1) \frac{dv}{dw} w^{c-2b+1} + c(c-b)v w^{c-2b} \right) \\
&\quad + \frac{aw^b}{ab} \left(\frac{dv}{dw} w^{c-b+1} + vcw^{c-b} \right) + (a^2w^{2b} - v^2)v w^c = 0 \\
&\frac{1}{b^2} \left(\frac{d^2v}{dw^2} w^2 + (2c-b+1) \frac{dv}{dw} w + c(c-b)v \right) + \frac{1}{b} \left(\frac{dv}{dw} w + cv \right) + (a^2w^{2b} - v^2)v = 0 \\
&w^2 \frac{d^2v}{dw^2} + (2c-b+1+b)w \frac{dv}{dw} + (c(c-b) + cb + a^2b^2w^{2b} - b^2v^2)v = 0 \\
&w^2 \frac{d^2v}{dw^2} + (2c+1)w \frac{dv}{dw} + (a^2b^2w^{2b} + (c^2 - b^2v^2))v = 0
\end{aligned}$$

The new form of solution will then be

$$\begin{aligned}
y &= c_1 J_\nu(x) + c_2 Y_\nu(x) \\
vw^c &= c_1 J_\nu(aw^b) + c_2 Y_\nu(aw^b) \\
v &= \frac{1}{w^c} (c_1 J_\nu(aw^b) + c_2 Y_\nu(aw^b))
\end{aligned}$$

(9.12) Multiplying the Airy equation by x^2 :

$$x^2 \frac{d^2y}{dx^2} - x^3 y = 0$$

Since we suppose that $x < 0$, we can let $z = -x$ so that it becomes

$$z^2 \frac{d^2y}{dz^2} + z^3 y = 0$$

and interchanging the roles of z (or x) and w , v and y , recognizing $c = -\frac{1}{2}$, $b = \frac{3}{2}$, $a = \frac{2}{3}$, $\nu = \frac{1}{3}$, we conclude that

$$\begin{aligned}
y &= \frac{1}{z^{-\frac{1}{2}}} \left(c_1 J_{\frac{1}{3}} \left(\frac{2}{3} z^{\frac{3}{2}} \right) + c_2 Y_{\frac{1}{3}} \left(\frac{2}{3} z^{\frac{3}{2}} \right) \right) \\
&= \sqrt{-x} \left(c_1 J_{\frac{1}{3}} \left(\frac{2}{3} (-x)^{\frac{3}{2}} \right) + c_2 Y_{\frac{1}{3}} \left(\frac{2}{3} (-x)^{\frac{3}{2}} \right) \right)
\end{aligned}$$

We simply state the required combination of Bessel functions to construct the Airy function of the first kind here:

$$\text{Ai}(x) = \sqrt{-x} \left(\frac{1}{2} J_{\frac{1}{3}} \left(\frac{2}{3} (-x)^{\frac{3}{2}} \right) - \frac{1}{2\sqrt{3}} Y_{\frac{1}{3}} \left(\frac{2}{3} (-x)^{\frac{3}{2}} \right) \right)$$

(9.13) Multiply it by x first:

$$x^2 y'' + 2xy' + x^2 y = 0$$

so that we can use the result from Exercise 9.11. Again, interchanging the roles of the variables, we identify that $c = \frac{1}{2}$, $b = 1$, $a = 1$, $\nu = \frac{1}{2}$, so the general solution can be written as

$$\begin{aligned} y &= \frac{1}{x^{\frac{1}{2}}} (c_1 J_{\frac{1}{2}}(x) + c_2 J_{-\frac{1}{2}}(x)) \\ &= c_1 \frac{\sin x}{x} + c_2 \frac{\cos x}{x} \end{aligned}$$

where we recall the values of the Bessel function at $\nu = \pm \frac{1}{2}$ and have absorbed the factor.

(9.14) This is a modified Bessel equation where $x \rightarrow \mu r$ and $\nu = 1$. So the two composing solutions are $I_1(\mu r)$ and $K_1(\mu r)$. In the inner core, we need $\hat{\psi}(0) = 0$ so we need to cross out $K_1(\mu r)$ which blows up at $r = 0$ and keep $I_1(\mu r)$. In contrast, at far field, we leave out $I_1(\mu r)$ which diverges at infinity, and keep $K_1(\mu r)$ that decays to zero. The full solution is then formed by patching the two respective solutions over the two regimes while enforcing continuity.

Solutions for Chapter 10

(10.1) If the discriminant $\Delta = B^2 - 4AC > 0$, then the quadratic equation $Av^2 + Bv + C = 0$ has two real roots α, β and can be factorized into $(v - \alpha)(v - \beta) = 0$. Informally substituting $v = \frac{\partial}{\partial t} / \frac{\partial}{\partial x}$ then leads to the statement.

(10.2) We can let $u = v + \gamma x$ so that $u_x(0, t) = u_x(L, t) = \gamma$ becomes homogeneous, $v_x(0, t) = v_x(L, t) = 0$. The heat equation remains the same after replacing u by v :

$$v_t = \kappa v_{xx}$$

and the initial condition becomes $v(x, 0) = (1 - \gamma)x - \frac{L}{2}$. The Neumann B.C.s imply that

$$X = \cos\left(\frac{n\pi}{L}x\right), n = 0, 1, 2, \dots$$

particularly, $n = 0$ stands for the constant term. T has the same form as in the main text, with an extra constant term $T_0 = 1$ for $n = 0$. Therefore,

$$\begin{aligned} v &= A_0 + A_1 \cos\left(\frac{\pi}{L}x\right)e^{-\frac{\pi^2\kappa}{L^2}t} + A_2 \cos\left(\frac{2\pi}{L}x\right)e^{-\frac{4\pi^2\kappa}{L^2}t} + \dots \\ &= \sum_{n=0}^{\infty} A_n \cos\left(\frac{n\pi}{L}x\right)e^{-\frac{n^2\pi^2\kappa}{L^2}t} \end{aligned}$$

Subsequently, we compute the coefficients A_n as a Fourier cosine series:

$$\begin{aligned} A_n &= \frac{\int_0^L \left((1 - \gamma)x - \frac{L}{2}\right) \cos\left(\frac{n\pi}{L}x\right) dx}{\int_0^L \cos^2\left(\frac{n\pi}{L}x\right) dx} \\ &= \frac{\frac{L}{n\pi} \int_0^L \left((1 - \gamma)x - \frac{L}{2}\right) d\left(\sin\left(\frac{n\pi}{L}x\right)\right)}{\frac{L}{2}} \\ &= \frac{2}{n\pi} \left(\left[\left((1 - \gamma)x - \frac{L}{2}\right) \sin\left(\frac{n\pi}{L}x\right) \right]_0^L - (1 - \gamma) \int_0^L \sin\left(\frac{n\pi}{L}x\right) dx \right) \\ &= \frac{2}{n\pi} \left((0) + (1 - \gamma) \left[\frac{L}{n\pi} \cos\left(\frac{n\pi}{L}x\right) \right]_0^L \right) \\ &= \frac{2L(1 - \gamma)((-1)^n - 1)}{n^2\pi^2} \end{aligned}$$

and

$$\begin{aligned}
 A_0 &= \frac{\int_0^L ((1-\gamma)x - \frac{L}{2}) dx}{\int_0^L dx} \\
 &= \frac{[(1-\gamma)\frac{x^2}{2} - \frac{L}{2}x]_0^L}{L} \\
 &= \frac{-\gamma\frac{L^2}{2}}{L} = -\frac{\gamma L}{2}
 \end{aligned}$$

Therefore, the required answer is

$$\begin{aligned}
 v &= -\frac{\gamma L}{2} + \sum_{n=1}^{\infty} \frac{2L(1-\gamma)((-1)^n - 1)}{n^2\pi^2} \cos\left(\frac{n\pi}{L}x\right) e^{-\frac{n^2\pi^2\kappa}{L^2}t} \\
 \Rightarrow u &= \gamma\left(x - \frac{L}{2}\right) + \sum_{n=1}^{\infty} \frac{2L(1-\gamma)((-1)^n - 1)}{n^2\pi^2} \cos\left(\frac{n\pi}{L}x\right) e^{-\frac{n^2\pi^2\kappa}{L^2}t}
 \end{aligned}$$

The Neumann B.C.s represent the flow of heat across the boundary, and here the inflow is equal to the outflow. When $\gamma = 1$, we simply have $u = x - \frac{L}{2} = u_0$, i.e. the temperature profile is in equilibrium to begin with in this case.

(10.3) The thermal diffusivity is

$$\kappa = \frac{k}{\rho c} = \frac{2.5}{(1000)(2500)} = 1 \times 10^{-6} \text{ m}^2 \text{ s}^{-1}$$

The angular frequency of the diurnal oscillation is

$$\omega = \frac{2\pi}{1 \text{ day}} = 7.272 \times 10^{-5} \text{ s}^{-1}$$

The skin depth is therefore

$$L = \sqrt{\frac{2\kappa}{\omega}} = \sqrt{\frac{2(1 \times 10^{-6})}{(7.272 \times 10^{-5})}} = 0.166 \text{ m}$$

For the annual case, we may recalculate ω , or use the proportionality relationship indicated by the skin depth formula. Either way should give 3.17 m ($\sqrt{365} \approx 19.1$ times).

(10.4) The initial condition in this problem can be regarded to be a superposition of the mirroring scenario suggested in the main text:

$$T_1(x, t = 0) = \begin{cases} \frac{T_0}{2} & x > -w \\ -\frac{T_0}{2} & x < -w \end{cases}$$

and

$$T_2(x, t = 0) = \begin{cases} -\frac{T_0}{2} & x > w \\ \frac{T_0}{2} & x < w \end{cases}$$

Therefore, the solution is also a superposition:

$$\begin{aligned} T &= \frac{T_0}{2} \operatorname{erf}\left(\frac{x+w}{\sqrt{4\kappa t}}\right) - \frac{T_0}{2} \operatorname{erf}\left(\frac{x-w}{\sqrt{4\kappa t}}\right) \\ &= \frac{T_0}{2} \left(\operatorname{erf}\left(\frac{w+x}{\sqrt{4\kappa t}}\right) + \operatorname{erf}\left(\frac{w-x}{\sqrt{4\kappa t}}\right) \right) \end{aligned}$$

The temperature at $x = 0$ after a year would be:

$$\begin{aligned} T &= \frac{1000}{2} \left[\operatorname{erf}\left(\frac{1}{\sqrt{4(1 \times 10^{-6})(31536000)}}\right) + \operatorname{erf}\left(\frac{1}{\sqrt{4(1 \times 10^{-6})(31536000)}}\right) \right] \\ &\approx 100.2^\circ \text{C} \end{aligned}$$

(10.5) It is the exact same problem with $x \rightarrow z$, $A \rightarrow u_0$, and $\kappa \rightarrow \nu$. So the new solution is simply

$$u = u_0 \left(1 - \operatorname{erf}\left(\frac{z}{\sqrt{4\nu t}}\right) \right)$$

(10.6) Here, the forcing has a triangular shape. The relevant quantities are

$$\begin{aligned} B_n(t) &= \frac{2}{L} \int_0^L f(\xi, t) \sin\left(\frac{n\pi}{L}\xi\right) d\xi \\ &= \frac{2}{L} \int_0^{\frac{L}{2}} \xi \sin\left(\frac{n\pi}{L}\xi\right) d\xi + \frac{2}{L} \int_{\frac{L}{2}}^L (L - \xi) \sin\left(\frac{n\pi}{L}\xi\right) d\xi \end{aligned}$$

The first term is computed as

$$\begin{aligned} \frac{2}{L} \int_0^{\frac{L}{2}} \xi \sin\left(\frac{n\pi}{L}\xi\right) d\xi &= \frac{2}{L} \left(\left[-\frac{L}{n\pi} \xi \cos\left(\frac{n\pi}{L}\xi\right) \right]_0^{\frac{L}{2}} + \int_0^{\frac{L}{2}} \frac{L}{n\pi} \cos\left(\frac{n\pi}{L}\xi\right) d\xi \right) \\ &= \frac{2}{L} \left(-\frac{L^2}{2n\pi} \cos\left(\frac{n\pi}{2}\right) + \left[\frac{L^2}{n^2\pi^2} \sin\left(\frac{n\pi}{L}\xi\right) \right]_0^{\frac{L}{2}} \right) \\ &= \frac{2}{L} \left(-\frac{L^2}{2n\pi} \cos\left(\frac{n\pi}{2}\right) + \frac{L^2}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right) \right) \end{aligned}$$

Meanwhile, the second term similarly gives

$$\frac{2}{L} \int_{\frac{L}{2}}^L (L - \xi) \sin\left(\frac{n\pi}{L}\xi\right) d\xi = \frac{2}{L} \left(\left[-\frac{L}{n\pi} (L - \xi) \cos\left(\frac{n\pi}{L}\xi\right) \right]_{\frac{L}{2}}^L - \int_{\frac{L}{2}}^L \frac{L}{n\pi} \cos\left(\frac{n\pi}{L}\xi\right) d\xi \right)$$

$$\begin{aligned}
&= \frac{2}{L} \left(\frac{L^2}{2n\pi} \cos\left(\frac{n\pi}{2}\right) - \left[\frac{L^2}{n^2\pi^2} \sin\left(\frac{n\pi}{L}\xi\right) \right]_{\frac{L}{2}}^L \right) \\
&= \frac{2}{L} \left(\frac{L^2}{2n\pi} \cos\left(\frac{n\pi}{2}\right) + \frac{L^2}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right) \right)
\end{aligned}$$

Thus

$$B_n = \frac{4L}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right)$$

and it is clear that $b_n = 0$. Using the variation of parameters formula, we have the final answer as

$$\begin{aligned}
u &= \sum_{n=1}^{\infty} \left(\int_0^t B_n(\tau) e^{-\frac{\kappa n^2 \pi^2}{L^2}(t-\tau)} d\tau \right) \sin\left(\frac{n\pi}{L}x\right) \\
&= \sum_{n=1}^{\infty} \left(\int_0^t \frac{4L}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right) e^{-\frac{\kappa n^2 \pi^2}{L^2}(t-\tau)} d\tau \right) \sin\left(\frac{n\pi}{L}x\right) \\
&= \sum_{n=1}^{\infty} \frac{4L}{n^2\pi^2} \left(\frac{L^2}{\kappa n^2 \pi^2} (1 - e^{-\frac{\kappa n^2 \pi^2}{L^2}t}) \right) \sin\left(\frac{n\pi}{2}\right) \sin\left(\frac{n\pi}{L}x\right) \\
&= \sum_{n=1}^{\infty} \frac{4L^3}{\kappa n^4 \pi^4} (1 - e^{-\frac{\kappa n^2 \pi^2}{L^2}t}) \sin\left(\frac{n\pi}{2}\right) \sin\left(\frac{n\pi}{L}x\right)
\end{aligned}$$

(10.7) To deal with the non-homogeneous Neumann B.C.s, we can let $v = u - x(y - \frac{H}{2})$. The new equation set is then

$$\begin{cases} \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0 \\ v(x, 0) = 1 + \frac{H}{2}x & \text{for } 0 \leq x \leq L \\ v(x, H) = 1 - \frac{H}{2}x & \text{for } 0 \leq x \leq L \\ v_x(0, y) = v_x(L, y) = 0 & \text{for } 0 \leq y \leq H \end{cases}$$

Particularly, the Laplace equation itself does not change. The separation of variables procedure is largely the same except that the sine part is replaced by cosine:

$$v(x, y) = \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi}{L}x\right) \sinh\left(\frac{n\pi}{L}y\right)$$

Now we deal with the new Dirichlet B.C.s. For the top side, we need

$$A_n \sinh\left(\frac{n\pi}{L}H\right) = \frac{2}{L} \int_0^L \left(1 - \frac{H}{2}\xi\right) \cos\left(\frac{n\pi}{L}\xi\right) d\xi$$

$$\begin{aligned}
&= \frac{2}{L} \left(\left[\frac{L}{n\pi} \sin\left(\frac{n\pi}{L}\xi\right) \right]_0^L - \frac{H}{2} \left(\left[\frac{L\xi}{n\pi} \sin\left(\frac{n\pi}{L}\xi\right) \right]_0^L - \int_0^L \frac{L}{n\pi} \sin\left(\frac{n\pi}{L}\xi\right) d\xi \right) \right) \\
&= \frac{2}{L} \left((0) + \frac{HL}{2n\pi} \left[-\frac{L}{n\pi} \cos\left(\frac{n\pi}{L}\xi\right) \right]_0^L \right) = \frac{HL(1 - (-1)^n)}{n^2\pi^2}
\end{aligned}$$

Also, the constant term is $1 - \frac{HL}{4}$, which gives $v = (1 - \frac{HL}{4})(\frac{y}{H})$. Hence,

$$v(x, y) = \sum_{n=1}^{\infty} \frac{HL(1 - (-1)^n)}{n^2\pi^2} \cos\left(\frac{n\pi}{L}x\right) \frac{\sinh\left(\frac{n\pi}{L}y\right)}{\sinh\left(\frac{n\pi}{L}H\right)} + \left(1 - \frac{HL}{4}\right) \frac{y}{H}$$

Similarly, the bottom B.C. gives

$$v(x, y) = \sum_{n=1}^{\infty} -\frac{HL(1 - (-1)^n)}{n^2\pi^2} \cos\left(\frac{n\pi}{L}x\right) \frac{\sinh\left(\frac{n\pi}{L}(H - y)\right)}{\sinh\left(\frac{n\pi}{L}H\right)} + \left(1 + \frac{HL}{4}\right) \left(1 - \frac{y}{H}\right)$$

Therefore, the final answer is

$$\begin{aligned}
v(x, y) &= \sum_{n=1}^{\infty} \frac{HL(1 - (-1)^n)}{n^2\pi^2} \cos\left(\frac{n\pi}{L}x\right) \frac{\sinh\left(\frac{n\pi}{L}y\right) - \sinh\left(\frac{n\pi}{L}(H - y)\right)}{\sinh\left(\frac{n\pi}{L}H\right)} + 1 - \frac{Ly}{2} + \frac{HL}{4} \\
\therefore u(x, y) &= \sum_{n=1}^{\infty} \frac{HL(1 - (-1)^n)}{n^2\pi^2} \cos\left(\frac{n\pi}{L}x\right) \frac{\sinh\left(\frac{n\pi}{L}y\right) - \sinh\left(\frac{n\pi}{L}(H - y)\right)}{\sinh\left(\frac{n\pi}{L}H\right)} \\
&\quad + 1 - \frac{Ly}{2} + \frac{HL}{4} + x\left(y - \frac{H}{2}\right)
\end{aligned}$$

(10.8) Here $a = 3$. In polar coordinates, the B.C. will be

$$u(3, \theta) = 27 \cos^2 \theta \sin \theta = f(\theta)$$

whereas the solution formula is

$$\begin{aligned}
u(r, \theta) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} 27 \cos^2 \xi \sin \xi d\xi + \frac{1}{\pi} \sum_{n=1}^{\infty} \left(\frac{r}{3}\right)^n \left[\left(\int_{-\pi}^{\pi} 27 \cos^2 \xi \sin \xi \cos(n\xi) d\xi \right) \cos(n\theta) \right. \\
&\quad \left. + \left(\int_{-\pi}^{\pi} 27 \cos^2 \xi \sin \xi \sin(n\xi) d\xi \right) \sin(n\theta) \right]
\end{aligned}$$

By trigonometry, we may rewrite

$$27 \cos^2 \xi \sin \xi = \frac{27}{4} (\sin(3\xi) + \sin \xi)$$

So, by the orthogonality of the Fourier basis, we have

$$\int_{-\pi}^{\pi} 27 \cos^2 \xi \sin \xi d\xi = 0$$

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$$\int_{-\pi}^{\pi} 27 \cos^2 \xi \sin \xi \cos(n\xi) d\xi = 0$$

$$\int_{-\pi}^{\pi} 27 \cos^2 \xi \sin \xi \sin(n\xi) d\xi = \begin{cases} \frac{27\pi}{4} & n = 1, 3 \\ 0 & \text{else} \end{cases}$$

And thus the full solution will be

$$u(r, \theta) = \frac{27}{4} \left(\left(\frac{r}{3} \right) \sin \theta + \left(\frac{r}{3} \right)^3 \sin(3\theta) \right)$$

(10.9) The full solution will consist of the superposition of contributions from the two caps. The B.C.s mean that only the $D_{m=2} = \pm V_0$ part remains while other C_m, D_m are zero. The top cap leads to

$$u_{\text{top}}(r, \theta, z) = \sum_{n=1}^{\infty} \frac{2 \int_0^a r V_0 J_2(k_{2n} \frac{r}{a}) dr}{a^2 [J_{2+1}(k_{2n})]^2 \sinh(k_{2n} \frac{L}{a})} \sin(2\theta) J_2\left(k_{2n} \frac{r}{a}\right) \sinh\left(k_{2n} \frac{z}{a}\right)$$

$$= \sum_{n=1}^{\infty} \frac{2V_0(2 - 2J_0(k_{2n}) - k_{2n}J_1(k_{2n}))}{k_{2n}^2 [J_3(k_{2n})]^2 \sinh(k_{2n} \frac{L}{a})} \sin(2\theta) J_2\left(k_{2n} \frac{r}{a}\right) \sinh\left(k_{2n} \frac{z}{a}\right)$$

by recalling the relationships about the Bessel functions to evaluate the integral ($\int x J_2(x) = -2J_0(x) - xJ_1(x)$). Similarly, to deal with the bottom cap, we simply replace $\sinh(k_{2n} \frac{z}{a})$ by $\sinh(k_{2n} \frac{L-z}{a})$, and the result should be

$$u_{\text{bottom}}(r, \theta, z) = - \sum_{n=1}^{\infty} \frac{2V_0(2 - 2J_0(k_{2n}) - k_{2n}J_1(k_{2n}))}{k_{2n}^2 [J_3(k_{2n})]^2 \sinh(k_{2n} \frac{L}{a})} \sin(2\theta) J_2\left(k_{2n} \frac{r}{a}\right) \sinh\left(k_{2n} \frac{L-z}{a}\right)$$

and thus the final answer is

$$u(r, \theta, z)$$

$$= \sum_{n=1}^{\infty} \frac{2V_0(2 - 2J_0(k_{2n}) - k_{2n}J_1(k_{2n}))}{k_{2n}^2 [J_3(k_{2n})]^2 \sinh(k_{2n} \frac{L}{a})} \sin(2\theta) J_2\left(k_{2n} \frac{r}{a}\right) \left(\sinh\left(k_{2n} \frac{z}{a}\right) - \sinh\left(k_{2n} \frac{L-z}{a}\right) \right)$$

(10.10) This time, we need to flip the sign of the separation constants, so that

$$\frac{R''}{R} + \frac{1}{r} \frac{R'}{R} + \frac{1}{r^2} \frac{\Theta''}{\Theta} = k^2$$

$$\frac{Z''}{Z} = -k^2$$

Hence

$$Z'' + k^2 Z = 0$$

$$\Rightarrow Z = E \sin(kz) + F \cos(kz)$$

and we may now match the top/bottom zero B.C.s: we need

$$Z = E \sin(kz) = E \sin\left(\frac{n\pi}{L}z\right), \quad k = \frac{n\pi}{L}, \quad n = 1, 2, \dots$$

Then, we use separation of variables for the first equation: this time

$$r^2 \frac{R''}{R} + r \frac{R'}{R} + \frac{\Theta''}{\Theta} - k^2 r^2 = 0$$

So we may let

$$\begin{aligned} r^2 \frac{R''}{R} + r \frac{R'}{R} - k^2 r^2 &= m^2 \\ \frac{\Theta''}{\Theta} &= -m^2 \end{aligned}$$

It is easy to see that for the azimuthal part, we still have

$$\begin{aligned} \Theta'' + m^2 \Theta &= 0 \\ \Theta &= C \cos(m\theta) + D \sin(m\theta) \end{aligned}$$

Meanwhile, the radial part leads to

$$r^2 R'' + r R' - (k^2 r^2 + m^2) R = 0$$

This becomes a modified Bessel equation with $r \rightarrow kr$ and $\nu = m$, and its general solution will be

$$R = A I_m(kr) + B K_m(kr)$$

To be physical, we leave out the singular K_m part at $r = 0$, so that the required solution form in this scenario is

$$u(r, \theta, z) = \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} I_m\left(\frac{n\pi}{L}r\right) (C_{mn} \cos(m\theta) + D_{mn} \sin(m\theta)) \sin\left(\frac{n\pi}{L}z\right)$$

To comply with the boundary condition along the curved surface, we have $m = 0$ due to azimuthal symmetry, and

$$u(a, \theta, z) = \sum_{n=1}^{\infty} C_{0n} I_0\left(\frac{n\pi}{L}a\right) \sin\left(\frac{n\pi}{L}z\right) = f(z)$$

To compute the coefficients C_{0n} , we simply project the series onto the Fourier basis:

$$C_{0n} = \frac{\frac{2}{L} \int_0^L f(z) \sin\left(\frac{n\pi}{L}z\right) dz}{I_0\left(\frac{n\pi}{L}a\right)}$$

(10.11) Since we care about how the gravity of the Earth as a whole is exerted outside itself, for the original general solution in the main text, we have $A = 0$ instead, so that the gravitational potential at the far field diminishes. The north-south symmetry requires that $l = 2p$ be even. Assume that we adjust the coordinate system so that the straight line $\theta = 0, \pi$ over the Equatorial plane aligns with one of the axes of the ellipsoid, then the ellipsoidal symmetry further means that, $D = 0$ while C only has a non-zero value when $m = 2q$ is also even. So the general solution for the simplified Earth's gravitational potential will be

$$V = -\frac{GM}{r} \left(1 + \sum_{p=1}^{\infty} \sum_{q=0}^p \left(\frac{R}{r} \right)^{2p} C_{2p,2q} \cos(2q\theta) P_{2p}^{2q}(\cos \phi) \right)$$

where $-\frac{GM}{r}$ is the asymptotic expression of the Earth's gravitational potential at a large distance.

(10.12) As suggested in the question,

$$\begin{aligned} u(\mathbf{r}) &= \frac{1}{|\mathbf{r} - \mathbf{r}_0|} \\ &= \frac{1}{\sqrt{\mathbf{r} \cdot \mathbf{r} - 2\mathbf{r} \cdot \mathbf{r}_0 + \mathbf{r}_0 \cdot \mathbf{r}_0}} \\ &= \frac{1}{\sqrt{r^2 - 2rr_0 \cos \alpha + r_0^2}} \end{aligned}$$

is a valid solution to the Laplace equation. We now fix the coordinate system so that $\mathbf{r}_0$ is aligned with the positive z -axis (i.e. $\phi = 0$) so $\alpha = \phi$, and r_0 is scaled to have a length of 1. Now consider the general solution in spherical coordinates, the current setting requires that azimuthal symmetry holds so that $m = 0$ and the associated Legendre functions P_l^m reduce to the usual Legendre polynomials P_l , and for it to be physical, we have

$$u(r, \theta, \phi) = \sum_{l=0}^{\infty} A_l r^l P_l(\cos \phi)$$

Therefore, equating these two expressions:

$$\frac{1}{\sqrt{r^2 - 2r \cos \phi + 1}} = \sum_{l=0}^{\infty} A_l r^l P_l(\cos \phi)$$

Note that we have $A_l = 1$ for all l , because when $\cos \phi = 1$, $P_l(\cos \phi) = P_l(1) = 1$, and

$$\begin{aligned} \sum_{l=0}^{\infty} A_l r^l &= \frac{1}{\sqrt{r^2 - 2r + 1}} \\ &= \frac{1}{1 - r} = 1 + r + r^2 + \dots \end{aligned}$$

Finally, we replace $r \rightarrow t$, $\cos \phi \rightarrow x$, and $l \rightarrow n$ to arrive at the usual form of the generating function:

$$\frac{1}{\sqrt{1-2xt+t^2}} = \sum_{n=0}^{\infty} P_n(x)t^n$$

(10.13) The B.C.s and thus the general form of the solution will be the same as the usual one:

$$u(x, t) = \sum_{n=1}^{\infty} \left(A_n \cos\left(\frac{nc\pi}{L}t\right) + B_n \sin\left(\frac{nc\pi}{L}t\right) \right) \sin\left(\frac{n\pi}{L}x\right)$$

From this, we can infer the frequency of the sinusoidal part oscillating in time as

$$\frac{1}{2\pi} \frac{nc\pi}{L} = \frac{nc}{2L}$$

The $\frac{1}{2\pi}$ factor is needed to convert from angular frequency to linear frequency. The first harmonic then has an eigen-frequency of $\frac{c}{2L}$ when $n = 1$. For it to match 196.00 Hz, we have the required value of $c = \frac{196.00}{2(0.635)} = 248.92 \text{ m s}^{-1}$. Finally, pressing down the string reduces L . To play a note an octave higher, press down the string at its middle exactly.

(10.14) With the new I.C.s, this time $E = 0$, and the velocity is given by

$$z_t(r, \theta, t) = \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} J_m\left(\frac{k_{mn}}{a}r\right) (C_{mn} \cos(m\theta) + D_{mn} \sin(m\theta)) \frac{k_{mn}b}{a} \cos\left(\frac{k_{mn}b}{a}t\right)$$

where we have absorbed the F_{mn} coefficient. Specifically, to match the second I.C., we again assume a decomposition in the form of

$$z_t(r, \theta, 0) = g(r, \theta) = \sum_{m=0}^{\infty} (C_m(r) \cos(m\theta) + D_m(r) \sin(m\theta))$$

Comparing these two expressions then yields

$$C_m(r) = \sum_{n=1}^{\infty} C_{mn} \frac{k_{mn}b}{a} J_m\left(\frac{k_{mn}}{a}r\right)$$

and by the same argument as before

$$\begin{aligned} C_{mn} &= \frac{a}{k_{mn}b} \frac{2 \int_0^a r C_m(r) J_m(k_{mn} \frac{r}{a}) dr}{a^2 [J_{m+1}(k_{mn})]^2} \\ &= \frac{2}{k_{mn}ab} \frac{\int_0^a r C_m(r) J_m(k_{mn} \frac{r}{a}) dr}{[J_{m+1}(k_{mn})]^2} \end{aligned}$$

It is similar for D_m and D_{mn} . Therefore, the corresponding solution will be

$$z(r, \theta, t) = \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} J_m\left(\frac{k_{mn}}{a}r\right) \left[\frac{\int_0^a r C_m(r) J_m(k_{mn} \frac{r}{a}) dr}{[J_{m+1}(k_{mn})]^2} \cos(m\theta) \right. \\ \left. + \frac{\int_0^a r D_m(r) J_m(k_{mn} \frac{r}{a}) dr}{[J_{m+1}(k_{mn})]^2} \sin(m\theta) \right] \frac{2}{k_{mn}ab} \sin\left(\frac{k_{mn}b}{a}t\right)$$

The full solution for the general I.C.s will then be the sum of this and that initially with a displacement only, due to linearity.

(10.15) Substituting the given ansatz into the wave equation, we get

$$\begin{aligned} -\omega^2 R e^{i\omega t} - c^2 \left(R'' e^{i\omega t} + \frac{2}{r} R' e^{i\omega t} \right) &= 0 \\ -\omega^2 R - c^2 \left(R'' + \frac{2}{r} R' \right) &= 0 \\ r^2 R'' + 2r R' + \frac{\omega^2}{c^2} r^2 R &= 0 \end{aligned}$$

Referencing the answer for Exercise 9.13 in view of $r \rightarrow \frac{\omega}{c}r$ and using complex form, we can arrive at

$$R = \frac{e^{\pm i \frac{\omega r}{c}}}{\frac{\omega r}{c}}$$

Therefore, the required wave solution is (taking the outward-propagating part)

$$u = A \frac{e^{-i \frac{\omega r}{c}}}{r} e^{i\omega t} = \frac{A}{r} e^{i\omega(t - \frac{r}{c})}$$

It travels at a speed of c (unsurprising) and the amplitude decays in the order of $\frac{1}{r}$.

(10.16) The proof is from the book Applied Partial Differential Equations by Logan. Following the hint, we have

$$\nabla^2 w = \nabla^2 u + 4\varepsilon = (0) + 4\varepsilon > 0$$

Therefore $\nabla^2 w > 0$ in Ω . If w has a maximum within the interior, by multivariable Calculus we must have $w_{xx}, w_{yy} \leq 0$ at that point, contradicting $\nabla^2 w > 0$ just derived. So the maximum of w will occur at the boundary, let's say at (x_b, y_b) . Subsequently,

$$\begin{aligned} u(x, y) &= w(x, y) - \varepsilon(x^2 + y^2) \\ &< w(x, y) \\ &\leq w(x_b, y_b) \\ &= u(x_b, y_b) + \varepsilon(x_b^2 + y_b^2) \end{aligned}$$

Since ε is arbitrary, we conclude that $u(x, y) \leq u(x_b, y_b)$ for all $(x, y) \in \Omega$. This inequality becomes strict if u is not a constant function.

Solutions for Chapter 11

(11.1) The matrix form of this linear system is $\mathbf{y}' = A\mathbf{y}$ where

$$A = \begin{bmatrix} -1 & 1 & 3 \\ -2 & 3 & 2 \\ -2 & 1 & 4 \end{bmatrix}$$

The eigenvectors of A are $(2, 1, 1)^T, (1, 0, 1)^T, (1, 1, 1)^T$ for the eigenvalues $\lambda = 1, 2, 3$. By substituting $\mathbf{y} = P\mathbf{z}$ where

$$P = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

the system becomes $\mathbf{z}' = (P^{-1}AP)\mathbf{z} = D\mathbf{z}$ with

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

The corresponding solutions are $z_1 = C_1 e^x$, $z_2 = C_2 e^{2x}$, and $z_3 = C_3 e^{3x}$. Computing $\mathbf{y} = P\mathbf{z}$ recovers

$$\begin{cases} y_1 = 2C_1 e^x + C_2 e^{2x} + C_3 e^{3x} \\ y_2 = C_1 e^x + C_3 e^{3x} \\ y_3 = C_1 e^x + C_2 e^{2x} + C_3 e^{3x} \end{cases}$$

The initial conditions imply that

$$\begin{bmatrix} 3 \\ 2 \\ 9 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \\ C_3 \end{bmatrix}$$

whose solutions are $C_1 = -6$, $C_2 = 7$, $C_3 = 8$. Hence, the complete solution is

$$\begin{cases} y_1 = -12e^x + 7e^{2x} + 8e^{3x} \\ y_2 = -6e^x + 8e^{3x} \\ y_3 = -6e^x + 7e^{2x} + 8e^{3x} \end{cases}$$

(11.2) The three ODEs can be written as a matrix system of

$$\mathbf{y}' = A\mathbf{y} = \begin{bmatrix} -k_1 & 0 & 0 \\ k_1 & -k_2 & 0 \\ 0 & k_2 & 0 \end{bmatrix} \mathbf{y}$$

It is easy to see that the eigenvalues of the lower-triangular coefficient matrix A are 0 , $-k_1$, and $-k_2$. We further assume that $k_1 \neq k_2$. The eigenvector for $\lambda = 0$ is clearly $(0, 0, 1)^T$, while that for $\lambda = -k_1$ can be derived from solving

$$\left[\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ k_1 & k_1 - k_2 & 0 & 0 \\ 0 & k_2 & k_1 & 0 \end{array} \right]$$

which leads to an eigenvector of $(k_1 - k_2, -k_1, k_2)^T$. Similarly, the eigenvector for $\lambda = -k_2$ can be checked to be $(0, -1, 1)^T$. The matrix for change of variables will then be

$$P = \begin{bmatrix} k_1 - k_2 & 0 & 0 \\ -k_1 & -1 & 0 \\ k_2 & 1 & 1 \end{bmatrix}$$

and with $\mathbf{y} = P\mathbf{z}$ we have

$$\begin{aligned} \mathbf{z}' &= (P^{-1}AP)\mathbf{z} = D\mathbf{z} \\ &= \begin{bmatrix} -k_1 & 0 & 0 \\ 0 & -k_2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \mathbf{z} \end{aligned}$$

the solution of which is $z_1 = c_1 e^{-k_1 t}$, $z_2 = c_2 e^{-k_2 t}$, $z_3 = c_3$. The initial conditions of $[P] = 1$, $[Q] = [R] = 0$ mean that

$$\begin{aligned} \mathbf{y}(0) &= P\mathbf{z}(0) \\ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} &= \begin{bmatrix} k_1 - k_2 & 0 & 0 \\ -k_1 & -1 & 0 \\ k_2 & 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} \end{aligned}$$

Solving for it yields $c_1 = \frac{1}{k_1 - k_2}$, $c_2 = -\frac{k_1}{k_1 - k_2}$, $c_3 = 1$. Therefore, $z_1 = \frac{1}{k_1 - k_2} e^{-k_1 t}$, $z_2 = -\frac{k_1}{k_1 - k_2} e^{-k_2 t}$, $z_3 = 1$, and

$$\mathbf{y} = \begin{bmatrix} k_1 - k_2 & 0 & 0 \\ -k_1 & -1 & 0 \\ k_2 & 1 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{k_1 - k_2} e^{-k_1 t} \\ -\frac{k_1}{k_1 - k_2} e^{-k_2 t} \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} e^{-k_1 t} \\ -\frac{k_1}{k_1 - k_2} e^{-k_1 t} + \frac{k_1}{k_1 - k_2} e^{-k_2 t} \\ \frac{k_2}{k_1 - k_2} e^{-k_1 t} - \frac{k_1}{k_1 - k_2} e^{-k_2 t} + 1 \end{bmatrix}$$

Hence

$$\begin{aligned} [P] &= e^{-k_1 t} \\ [Q] &= -\frac{k_1}{k_1 - k_2} e^{-k_1 t} + \frac{k_1}{k_1 - k_2} e^{-k_2 t} \\ [R] &= \frac{k_2}{k_1 - k_2} e^{-k_1 t} - \frac{k_1}{k_1 - k_2} e^{-k_2 t} + 1 \end{aligned}$$

The time t_Q where $[Q]$ reaches maximum occurs when $\frac{d[Q]}{dt} = 0$, which leads to

$$\begin{aligned} \frac{d[Q]}{dt} &= k_1[P] - k_2[Q] = 0 \\ k_1 e^{-k_1 t} - k_2 \left(-\frac{k_1}{(k_1 - k_2)} e^{-k_1 t} + \frac{k_1}{(k_1 - k_2)} e^{-k_2 t} \right) &= 0 \\ k_1 e^{-k_1 t} + \frac{k_1 k_2}{(k_1 - k_2)} e^{-k_1 t} - \frac{k_1 k_2}{(k_1 - k_2)} e^{-k_2 t} &= 0 \\ \frac{k_1^2}{(k_1 - k_2)} e^{-k_1 t} - \frac{k_1 k_2}{(k_1 - k_2)} e^{-k_2 t} &= 0 \\ \frac{k_1^2}{(k_1 - k_2)} e^{-k_1 t} &= \frac{k_1 k_2}{(k_1 - k_2)} e^{-k_2 t} \\ e^{-(k_1 - k_2)t} &= \frac{k_2}{k_1} \\ -(k_1 - k_2)t &= \ln\left(\frac{k_2}{k_1}\right) \\ \Rightarrow t_Q &= -\frac{\ln\left(\frac{k_2}{k_1}\right)}{k_1 - k_2} = \frac{\ln k_1 - \ln k_2}{k_1 - k_2} \end{aligned}$$

(11.5) Substituting yields the relation

$$\begin{aligned} \left(k^2 + l^2 - \frac{d^2 W}{dz^2} \right) \omega^2 - N^2 (k^2 + l^2) W &= 0 \\ \frac{d^2 W}{dz^2} + \frac{(N^2 - \omega^2) K^2}{\omega^2} W &= 0 \end{aligned}$$

where $K^2 = k^2 + l^2$ is the total horizontal wavenumber. The WKB solution then has the form $A_0 m^{-\frac{1}{2}} \exp(\pm i \int m(x) dx)$ with $m = \left(\frac{(N^2 - \omega^2) K^2}{\omega^2} \right)^{\frac{1}{2}}$. Locally, the flow behaves like a plane wave with a vertical wavenumber m and its amplitude scaled by $m^{-\frac{1}{2}}$.